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8. ECOLOGY

8.1 Introduction

8.1.1.1 This section presents the ecological baseline conditions within the Assessment Area, and the assessment results of the potential ecological impacts resulting from the construction and operation of the Project. The location plan and description of the project are presented in **Figure 1.1** and **Section 2** respectively.

8.1.1.2 According to the EIAO-TM and Appendix G of the EIA Study Brief (ESB-373/2025), baseline conditions for ecological components of the terrestrial environment were evaluated based on information from available literature and field surveys conducted for the purposes of this EIA. Measures required to mitigate any identified adverse impacts are recommended, where appropriate, and residual impacts, if any, are assessed.

8.2 Environmental Legislation, Standards and Guidelines

8.2.1.1 Reference was made to the following HKSAR Government ordinances, regulations, standards, guidelines and documents when identifying ecological importance of habitats and species, and evaluating and assessing potential impacts of the Project on the ecological resources:

- Country Parks Ordinance (Cap. 208) and its subsidiary legislation;
- Forests and Countryside Ordinance (Cap. 96) and its subsidiary legislation, the Forestry Regulations (Cap. 96A);
- Wild Animals Protection Ordinance (Cap. 170);
- Protection of Endangered Species of Animals and Plants Ordinance (Cap. 586) and its subsidiary legislation;
- Chapter 10 of the Hong Kong Planning Standards and Guidelines;
- Environmental Impact Assessment Ordinance (Cap. 499) and relevant Annexes 8 and 16 of the associated Technical Memorandum (EIAO-TM);
- EIAO Guidance Note No. 6/2010 – Some Observations on Ecological Assessment from the Environmental Impact Assessment Ordinance Perspective;
- EIAO Guidance Note No. 7/2023 – Ecological Baseline Survey for Ecological Assessment;
- EIAO Guidance Note No. 10/2023 – Methodologies for Terrestrial and Freshwater Ecological Baseline Surveys;
- Drainage Services Department Practice Note No. 3/2021 – Guidelines on Design for Revitalisation of River Channel;
- Environment, Transport and Works Bureau Technical Circular (Works) (ETWB TC(W)) No. 5/2005 – Protection of Natural Streams/Rivers from Adverse Impacts Arising from Construction Works;
- New Nature Conservation Policy;
- Hong Kong Biodiversity Strategy and Action Plan (2035);

8.2.1.2 This assessment also made reference to the following international conventions and national legislation:

- The International Union for Conservation of Nature (IUCN) Red List of Threatened Species;
- The List of Wild Plants and Wild Animals Under Special State Protection under the Law of the People's Republic of China on the Protection of Wildlife; and
- The Convention on Biological Diversity (the CBD), and an associated city-level Biodiversity Strategy and Action Plan (BSAP) developed by Agriculture, Fisheries and Conservation Department (AFCD) under the CBD.
- List of Wild Animals under State Protection; and
- List of Wild Plants under State Protection.

8.3 Assessment Methodology

8.3.1 Assessment Area

8.3.1.1 The Assessment Area for the purpose of the terrestrial and aquatic ecological impact assessment included areas within 500 metres from the boundary of the Project and its associated works (**Figure 8.1**).

8.3.2 Ecological Survey Methodology

8.3.2.1 A 12-month ecological field survey covering both wet and dry seasons were conducted to gather updated information on the Assessment Area, and to fill any information gaps identified within the 500 m Assessment Area. The field surveys covered flora, fauna and any other habitats/species of conservation importance. The fauna species were surveyed during the optimal time of the day as stated in Appendix B of the EIAO-TM Annex 16. All field surveys were carried out in ways that do not cause any unnecessary stress or damage to the existing habitats and wildlife.

8.3.2.2 The walk-transects for ecological surveys, freshwater sampling points and locations for camera traps are indicated in **Figure 8.2**. The vantage points for flight line survey are indicated in **Figure 8.3**. Where coverage by walk-transect was not possible (e.g. inaccessible areas within private areas, barrack areas where access is not allowed, hillside areas with limited accessibility and safety concern), high-power binoculars were used to determine the habitat structure, general vegetation composition, at a location as close as possible.

Habitat Mapping and Vegetation Survey

8.3.2.3 Terrestrial and aquatic habitats within the Assessment Area were identified, sized and mapped. Ecological characteristics of each habitat type, including size, vegetation type, species present, dominant species found, species diversity and abundance, community structure, ecological value and inter-dependence of the habitats and species, and presence of any features of ecological importance were defined and characterised. Representative photographs of the habitat types and/or any important ecological features identified were taken. A habitat map of suitable scale (1:1,000 to 1:5,000) showing types and locations of terrestrial and aquatic habitats within the Assessment Area was prepared from digital aerial photographs. The habitat map was checked during ground truthing. For areas that are inaccessible, aerial drones were used to capture photographs and videos to survey ecological features and verify habitats.

8.3.2.4 Vegetation surveys were conducted by direct observation to record the diversity and dominance of plant species present in different habitat types. The locations of any plant species of conservation importance were recorded.

8.3.2.5 Identification of flora species and status in Hong Kong made reference to Xing *et al.* (2000), Hu *et al.* (2003), Lai *et al.* (2008), Hong Kong Herbarium (2012), and Hong Kong Herbarium and South China Botanical Gardens (2007; 2008; 2009; 2011).

Avifauna Survey

8.3.2.6 The presence and abundance of all avifauna species in various habitats were recorded visually and aurally. Avifauna was surveyed quantitatively using the transect count method along the walk-transect during morning, dusk and night times. The location of any avifauna species of conservation importance encountered were recorded, along with notable behaviours (e.g. breeding behaviour such as nesting and the presence of recently fledged juveniles, roosting, and feeding activities).

8.3.2.7 Ornithological nomenclature follows Carey *et al.* (2001), Viney *et al.* (2005), and the most recent list published by the Hong Kong Bird Watching Society.

Roosting Ardeids and Flight Line Survey

8.3.2.8 Flight line surveys were conducted for Sha Po Night Roost and Ko Po Road Night Roost at the vantage points, VP1 and VP2, during the peak activity period of ardeids at night roosts (**Figure 8.3**). The vantage points are indicative in nature and are subject to change due to site condition and accessibility upon the exercise of professional judgement during the course of survey. The flight lines of ardeid individuals were followed by means of high-power binoculars to identify their landing locations, whenever possible. In case the ardeids fly out of sight prior to landing, the locations where they were last seen were recorded. Flight line survey for night roosts was commenced approximately one hour before sunset and continue until nightfall. The flight lines, flight height of the flight lines, the return times of the pre-roosting and night-roosting ardeids, and substrate used at the ardeid night roosts were recorded. The time of sunset on the date of survey made reference to the Hong Kong Observatory.

8.3.2.9 Given the potential presence of roosts for ardeids and other avifauna (e.g. Great Cormorants) within the Assessment Area, the field surveys also focused on confirming the presence of any potential roosting sites, in particular trees adjacent to ponds and wetland habitats. No new roosting sites were identified during the course of the ecological surveys.

Terrestrial Mammal Survey

8.3.2.10 The surveys focused on areas that may potentially be utilised by terrestrial mammals. Field signs including droppings, footprints, diggings or burrows left by larger mammals were searched for. Mammal identification was made as accurately as possible from the field signs encountered. Any mammals directly observed were recorded. Camera traps were deployed at selected locations as indicated in **Figure 8.2**.

8.3.2.11 Bat surveys were undertaken by surveyor(s) equipped with ultrasonic bat detector(s) at potential roosting, commuting, foraging and drinking sites. Bat species was

identified through the detection of echolocation calls and direct observation. The acoustic information recorded were analysed using specialised software.

8.3.2.12 Special attention was paid to the day-roost of Himalayan Leaf-nosed Bats (*Hipposideros armiger*) in one of the structures within the deserted Pok Wai Public School, as indicated as 'Bat Shelter' in **Figure 8.1**.

8.3.2.13 Nomenclature of mammal follows Shek (2006).

Butterfly and Odonate Survey

8.3.2.14 Butterflies and odonates (dragonflies and damselflies) were surveyed along the walk-transects. Attention was paid to their potential habitats. Relative abundance of butterfly, dragonfly and damselfly were recorded, along with any larvae and pupae encountered.

8.3.2.15 Nomenclature of butterfly follows Lo & Hui (2010) and Pun (2024); while odonate follows Tam *et al.* (2011) and Reels (2019).

Herpetofauna (Amphibian and Reptile) Survey

8.3.2.16 Herpetofauna was surveyed along survey transects. Potential microhabitats (e.g. leaf litter, underneath of rotten logs) were searched. All reptiles and amphibians sighted were recorded.

8.3.2.17 Amphibian survey was conducted whenever possible on evenings and night time, focusing on areas suitable for amphibians (e.g. forests, shrublands, grasslands, streams, catchwaters, ponds and marshes). Acoustic records of amphibians constitute the majority of data collected, supplemented by visual observation of eggs, tadpoles and frogs / toads whenever possible.

8.3.2.18 During reptile surveys, careful searches were conducted at the preferrable microhabitats and refugia of reptiles (e.g. stones, pond bunds, crevices, leaf litter/debris, rotten log). All reptiles encountered were recorded. In addition to active searching, observation of exposed, basking and foraging reptiles were made.

8.3.2.19 Nomenclature of amphibian and reptile follows Chan *et al.* (2005) and Chan *et al.* (2006), respectively.

Freshwater Invertebrates and Fish Survey

8.3.2.20 Freshwater invertebrates and fish survey was conducted through active searching and/or direct observation at representative freshwater sampling points, as shown in **Figure 8.2** (FS1 to FS12). The sampling points are selected based on the likelihood of direct and indirect impacts on the watercourses.

8.3.2.21 Freshwater invertebrates survey was conducted during daytime; whereas freshwater fish survey was conducted during day-time for diurnal species, and night-time for nocturnal species.

8.3.2.22 To avoid driving organisms (e.g. fish and shrimps) away and disturbing the bottom substrate, direct observation from a suitable distance was conducted before active searching and kick sampling. Boulders within the watercourses were turned over to locate any aquatic animals beneath. Hand net was used to collect organisms

along the watercourses. Organisms encountered were recorded and identified to the lowest possible taxon level.

8.3.2.23 Nomenclature of aquatic fish and invertebrate communities follows Lee *et al.* (2004) and Dudgeon (2003) respectively.

Firefly Survey

8.3.2.24 Fireflies were surveyed along the walk-transects. Dusk and night-time surveys were conducted immediately after sunset for at least 2 hours to cover the active periods of both crepuscular and nocturnal species. The time of sunset on the date of survey made reference to the Hong Kong Observatory. Particular attention was paid to potential habitats for fireflies and locations with less artificial lighting along the transects. Where site situation permitted (e.g. under safe conditions), lighting devices were switched off during survey to allow detection of fireflies before progressing along the transects.

8.3.2.25 During the survey, any adult and larva firefly observed were identified to the species level, where possible. The abundance and distribution of fireflies were recorded. Any sighting of mass occurrence and/or breeding behaviour of fireflies, along with the associated locations and habitats, were recorded. Diurnal fireflies found during other day-time surveys were also recorded.

8.3.2.26 Reference for nomenclature and conservation status of fireflies (e.g. endemic to Hong Kong) made to Yiu (2023).

8.3.3 Ecological Survey Programme

8.3.3.1 The schedule for ecological surveys is shown in **Table 8.1**. Reference is made to Appendix B of the EIAO-TM Annex 16, which outlines the optimal time of year, minimum frequency, and optimal time of the day for conducting surveys for major faunal and floral groups.

Table 8.1 Schedule of Ecological Surveys (12-month Survey)

Ecological Survey	Dry Season	Wet Season							Dry Season			
	Mar 2025	Apr 2025	May 2025	Jun 2025	Jul 2025	Aug 2025	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026
Habitat survey (Daytime)	✓						✓					
Vegetation survey (Daytime)	✓						✓					
Terrestrial mammal survey (Daytime; Night-time; Dusk)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Avifauna survey (Early Morning; Dusk; Night-time)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Reptile survey (Daytime; Night-time)		✓		✓		✓		✓				

Ecological Survey	Dry Season	Wet Season							Dry Season			
	Mar 2025	Apr 2025	May 2025	Jun 2025	Jul 2025	Aug 2025	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026
Amphibian survey (Night-time)	✓		✓		✓		✓					
Butterfly survey (Daytime)	✓		✓		✓		✓		✓			
Odonate survey (Daytime)		✓		✓		✓		✓				
Fireflies Survey (Dusk; Night-time)		✓	✓	✓				✓	✓	✓		
Freshwater Fish survey (Daytime; Night-time)	✓		✓		✓		✓		✓			
Freshwater Invertebrates survey (Daytime)	✓		✓		✓		✓		✓			
Flight Line Survey for Night Roosts	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Notes:
✓ Completed surveys
Optimal time of the year as stated in Appendix B of EIAO-TM Annex 16

8.4 Description of the Environment

8.4.1 Recognised Sites of Conservation Importance

Deep Bay Wetland outside Ramsar Site Priority Site for Enhanced Conservation

8.4.1.1 Deep Bay Wetland outside Ramsar Site is a Priority Site of Enhanced Conservation under the New Nature Conservation Policy, comprising mainly fishponds. The Deep Bay Wetland outside Ramsar Site supports species of conservation importance such as Black-faced spoonbill (*Platalea minor*). A corner of this Priority Site of Enhanced Conservation is overlapping with the Project Boundary to the west of Sha Po.

Lam Tsuen Country Park (LTCP)

8.4.1.2 LTCP was designated in 1979 and spans across Tai Po, Fanling and Yuen Long. The LTCP occupies 1,520 ha of land in the Northern New Territories (AFCD, 2023a). Plantation stands of exotic tree species such as Taiwan Acacia (*Acacia confusa*) and Brisbane Box (*Lophostemon confertus*), as well as native *Machilus* species were established throughout the territory of LTCP. Faunal species of importance such as Chinese Porcupine (*Hystrix brachyura*), Chinese Ferret Badger (*Melogale moschata*) and Chinese Pangolin (*Manis pentadactyla*) could be found within the LTCP. The LTCP is located adjacent to the Project Boundary to the south of NTM area.

"Conservation Area" Zone

8.4.1.3 Conservation Area (CA) are identified within the Assessment Area under five Outline Zoning Plans (OZPs), i.e. S/YL-NTM/15 - Ngau Tam Mei, S/YL-KTN/11 -

Kam Tin North, S/STT/2 – San Tin Technopole, S/YL-NSW/11 – Nam Sang Wai and S/YL-KTS/15 – Kam Tin South. S/YL-KTN/11 - Kam Tin North, S/YL-NSW/11 – Nam Sang Wai and S/YL-KTS/15 – Kam Tin South are located in approximately 250 m, 25 m and 100 m respectively, while S/YL-NTM/15 - Ngau Tam Mei and S/STT/2 – San Tin Technopole are within the Project Boundary.

8.4.1.4 The CAs gazetted under S/YL-NTM/15- Ngau Tam Mei, S/YL-KTS/15 – Kam Tin South, S/YL-KTN/11 - Kam Tin North and S/STT/2 – San Tin Technopole are intended to protect and retain the existing natural landscape, ecological or topographical features of the area for conservation, educational and research purposes and to separate sensitive natural environment such as Country Park from the adverse effects of development. While the CA gazetted under S/YL-NSW/11 – Nam Sang Wai aims to conserve the ecological value of wetland and fishponds which form an integral part of the wetland ecosystem in the Deep Bay Area.

Wetland Conservation Area

8.4.1.5 Wetland Conservation Area (WCA) comprises existing and contiguous, active or abandoned fishponds in the Deep Bay Area, along the landward side of the Mai Po Inner Deep Bay Ramsar Site. The planning intention is to conserve the ecological value of the fishponds which form an integral part of the wetland ecosystem in the Deep Bay Area. The WCA lies directly adjacent to the Project Boundary to the west of Sha Po.

Wetland Buffer Area

8.4.1.6 Wetland Buffer Area (WBA) is located along the landward periphery of the WCA. It is a designated buffer area of about 500 m along the landward boundary of the WCA. It is intended to protect the ecological integrity of the fishponds and wetland within the WCA, and to prevent development that would have a negative offsite disturbance impact on the ecological value of fishponds. Part of the WBA is situated within the Project Boundary along the San Tin Highway.

8.4.2 Other Ecologically Sensitive Resources

Inner Deep Bay and Shenzhen River Catchment Important Bird Area (IBA)

8.4.2.1 Mai Po Inner Deep Bay area and the vicinity around Shenzhen River Catchment was recognised as an Important Bird Area (IBA) by BirdLife International, for its international significance in terms of the size and trend of bird population. The recognition was done as an effort to secure long-term conservation of areas that are important for birds and biodiversity. A mosaic of wetland habitats (total 3,150 ha) was recognised for its global importance, and support to large number of passage and wintering waterbirds, including species of conservation importance and threatened species.

Roosting Sites

8.4.2.2 Two night roosts are located within the Assessment Area, namely Sha Po Night Roost and Ko Po Road Night Roost.

8.4.2.3 For Sha Po Night Roost, ardeids such as Great Egret (*Ardea alba*), Grey Heron (*Ardea cinerea*), Chinese Pond Heron (*Ardeola bacchus*) and Little Egret (*Egretta garzetta*) were observed utilizing trees located at the eastern bank of Kam Tin River (MTRCL, 2024). Roosting substratum included White Popinac (*Leucaena*

leucocephala), Taiwan Acacia and *Sonneratia caseolaris*. Sha Po Night Roost lies within the Project Boundary.

8.4.2.4 For Ko Po Road Night Roost, ardeids such as Eastern Cattle Egret (*Bubulcus coromandus*), Chinese Pond Heron (*Ardeola bacchus*) and Little Egret (*Egretta garzetta*) were observed utilising trees along the southern bank of Kam Tin River next to Ko Po Road (MTRCL, 2024). Roosting substratum included Chinese Banyan (*Ficus macrocarpa*) and White Popinac. Ko Po Night Roost is located in approximately 380 m away from the Project Boundary.

Kam Tin Main Drainage Channel (KTMDC)

8.4.2.5 KTMDC is a channelized section of Kam Tin River. The KTMDC is tidal-influenced and utilized by various wetland associated bird species such as egrets, sandpipers, kingfishers and plovers. Winter visitor and migrant, such as Grey-headed Lapwing (*Vanellus cinereus*), was often found utilising the channel (MTRCL, 2023). Part of the KTMDC is located within the Project Boundary.

Sha Po Marsh

8.4.2.6 The Sha Po Marsh is currently managed by a private estate developer, Sun Hung Kai Properties Limited, as an Ecological Enhancement Area. It is a restored brackish marsh, where 180 wildlife species were *recorded* (SHKPL, 2017). The marsh was reported to support a Near Threatened damselfly species Four-spot Midget (*Mortonagrion hirosei*). The Sha Po Marsh is located directly adjacent to the Project Boundary.

Yuen Long Bypass Floodway Engineered Wetland

8.4.2.7 Yuen Long Bypass Floodway Engineered Wetland (YLBFEW) is located near the downstream of Yuen Long Bypass Floodway and managed by AFCD since 2006 (DSD, 2019). This *wetland* was constructed as part of the ecological mitigation measures for Yuen Long Bypass Floodway Project. The engineered wetland consists of three main water ponds, a seasonal shallow pond and a permanent shallow pond. There are also crushed brick and oyster shell ponds, reed beds and a deep-water pond within the wetland. The dry weather flow of the Bypass Floodway, after purified by passing through the oyster shells, crushed bricks and reed bed, flows into the engineered wetland to offer an ecologically enriched habitat for the wild birds, amphibians and dragonflies (DSD, 2019). The YLBFEW is situated in approximately 25 m away from the Project Boundary.

West Rail Compensatory Wetland

8.4.2.8 Part of the West Rail Compensatory Wetland lies within the 500 m Assessment Area of the Project and is located in approximately 325 m to the east of the Project Boundary in Kam Tin. This compensatory wetland was created since 2022 to mitigate for the loss of wetlands in Kam Tin Valley due to the construction of West Rail in accordance with Environmental Permit No. FEP-24/004/1998/J. The West Rail Compensatory Wetland is currently maintained and managed by MTR Corporation.

Planned Bat Shelter in Pok Wai

8.4.2.9 The approved EIA for the Northern Link (Register No.: AEIAR-259/2024) reported that the day-roost of Himalayan Leaf-nosed Bats (*Hipposideros armiger*), recorded

in the deserted Pok Wai *Public* School, would be directly impacted and lost. As a compensation measure, a purpose-built bat shelter was proposed under the EIA to provide alternative roosting opportunities for the bat species. The planned bat shelter in Pok Wai is located within the Project Boundary.

Planned Wetland Compensation Site in Kam Tin

8.4.2.10 Under the approved EIA for the Northern Link (Register No.: AEIAR-259/2024), a wetland compensation site of about 1.9 ha is identified in Kam Tin, adjacent to the future Ecological Enhancement Area (EEA) of the Park Yoho Phases 2 and 3 Development and the West Rail Compensatory Wetlands. This site was proposed to compensate for the loss of marsh/reed habitat due to the project. The planned compensation wetland for the Northern Link is expected to locate 350 m away from the Project Boundary.

Planned Compensation Wetland for Ngau Tam Mei (NTM) Development Area

8.4.2.11 Under the approved EIA for the NTM Development Area (Register No.: AEIAR-271/2025, CEDD, 2025), a wetland compensation site of about 3.5 ha has been identified adjacent to Kam Tin River, between Tsing Long Highway and San Tam Road at Sha Po area. This site was proposed to accommodate the wetland compensation requirement of about 2.55 ha under Ngau Tam Mei development, and is currently a developed area/wasteland habitat overgrown with herbs/shrubs and exotic/pioneer tree species (e.g. *Leucaena leucocephala)*, considered of low ecological value. Despite part of the wetland compensation site is overlapping with the proposed NMH alignment, based on latest project coordination during the preparation of this EIA Report, “no work zone” is incorporated into the Project design to delineate areas where no construction activities will be undertaken, direct and indirect impacts to this planned compensation wetland during the establishment period is avoided and minimised (**Figure 8.9.1 to 8.9.4** refer). The planned compensation wetland for Ngau Tam Mei development is expected to lie directly adjacent to the NMH alignment and within the Project Boundary.

Proposed Nam Sang Wai Wetland Conservation Park

8.4.2.12 As recommended in the Study Report of the Strategic Feasibility Study on the Development of the Wetland Conservation Parks System released by AFCD in October 2024, the proposed Nam Sang Wai Wetland Conservation Park (WCP) is part of the proposed WCPs System which serve to conserve the ecological value of the wetlands and safeguard the integrity of the wetland system; develop modernised aquaculture industry to create more job opportunities for the agriculture and fisheries industries; promote scientific research on aquaculture to facilitate the upgrade and transformation of the agriculture and fisheries industries; and provide ecological education and recreational facilities for the public (AFCD, 2024). The proposed Nam Sang Wai WCP lies directly adjacent to the Project Boundary.

8.5 Ecological Baseline Information (Literature Review)

8.5.1 Literature Review

8.5.1.1 Recent EIA studies, along with books, scientific papers and other available information, such as Hong Kong Environmental Database (HKED) (EPD, 2025), on the ecological characteristics of the Assessment Area were collected and reviewed (**Table 8.2**). The information collected was evaluated to identify any gaps related to the assessment of potential ecological impacts.

Table 8.2 Baseline Information of Ecological Resources within the Assessment Area

Relevant Literature	Habitat and Vegetation	Avifauna	Mammal	Herpetofauna	Terrestrial Insects	Freshwater Community
EIA Report for Development at Ngau Tam Mei Area (AEIAR- 271/2025) (CEDD, 2025)	✓	✓	✓	✓	✓	✓
Annual Report of Yuen Long Bypass Floodway Habitat Creation Area, 2024 (AFCD, 2025)		✓				
EIA Report for Ngau Tam Mei Water Treatment Works Extension (AEIAR-262/2024) (WSD, 2024)	✓	✓	✓	✓	✓	✓
EIA Report for First Phase Development of The New Territories North – San Tin / Lok Ma Chau Development Node – Investigation (AEIAR-261/2024) (CEDD & PlanD, 2024)	✓	✓	✓	✓	✓	✓
Project Profile of Ground Investigation Works for Northern Link within Lam Tsuen Country Park and Conservation Area (PP-664/2024) (MTRCL, 2024)	✓	✓	✓	✓	✓	
EIA Report for Northern Link (AEIAR-259/2024) (MTRCL, 2023)	✓	✓	✓	✓	✓	✓
EIA Report of Housing Site in Yuen Long South (AEIAR-215/2017) (CEDD & PlanD, 2017)	✓					
EIA Report for Proposed Low-rise and Low-density Residential Development at Various Lots and their Adjoining Government Land in D.D. 104, East of Kam Pok Road, Mai Po, Yuen Long. New Territories (AEIAR-205/2017) (Glory Queen Limited, 2016)	✓	✓	✓	✓	✓	✓
EIA Report for Comprehensive Development and Wetland Protection near Yau Mei San Tsuen (AEIAR-189/2015) (Asia King Development Limited, 2015)	✓	✓	✓	✓	✓	✓
EIA Report for Proposed Residential Cum Passive Recreation Development within "Recreation" Zone and "Residential (Group C)" Zone at Various Lots in DD 104, Yuen Long, N.T. (AEIAR-182/2014) (Capital Chance Ltd, 2013)	✓	✓	✓	✓	✓	
EIA Report for North East New Territories New Development Areas (AEIAR-175/2013) (CEDD, 2013)	✓	✓	✓	✓	✓	✓
EIA Report of Hong Kong Section of Guangzhou-Shenzhen-Hong Kong Express Rail Link (AEIAR-143/2009) (MTRCL, 2009)	✓	✓	✓	✓	✓	
EIA Report of Construction of Cycle Tracks and the Associated Supporting Facilities from Sha Po Tsuen to Shek Sheung River (AEIAR-133/2009) (CEDD, 2008)	✓	✓	✓	✓	✓	
EIA Report of Sheung Shui to Lok Ma Chau Spur Line (AEIAR-052/2002) (KCRC, 2002)	✓					

8.5.2 Habitat and Vegetation

8.5.2.1 A total of 15 habitat types were previously identified within the proposed 500 m Assessment Area, which included woodland, mixed woodland, shrubland, grassland, plantation, village/orchard, developed area/wasteland, agricultural land, seasonally wet grassland, marsh/reed, pond, natural watercourse, semi-natural watercourse, modified watercourse, mitigation wetland (CEDD, 2025; CEDD & PlanD, 2024; MTRCL, 2023; MTRCL, 2024; WSD, 2024).

General Description of Habitats within Assessment Area

8.5.2.2 The Project Boundary for Northern Metropolis – San Tin section starts from Kam Tin River in the southwest, spanning towards the northeast across Ngau Tam Mei, Ki Lun Tsuen, and ending at Tit Hang near Chau Tau Village. The 500 m Assessment Area from the Project Boundary mainly covers urbanised habitats (village and developed area), wooded habitats (across hills and hillside area), with only a small portion of wetland habitats mainly on the southwest near Kam Tin River and scattered over the Assessment Area.

8.5.2.3 Pond habitat was present within the Assessment Area mainly near Nam Sang Wai and Pok Wai to the west of San Tin Highway, at Ngau Tam Mei (NTM) area, as well as some scattered ponds near Ki Lun Tsuen. Other wetland habitats previously recorded include marsh/reed, natural watercourse, semi-natural watercourse and modified watercourse. These wetland habitats were relatively scattered. While these habitats supported relatively low diversity and abundance of flora and fauna, ecological and hydrological linkage between these habitats and ponds were observed.

8.5.2.4 Woodland habitats (e.g. woodland and mixed woodland) were mainly found at the southern part at lower hillside of Kai Kung Leng and Ngau Tam Shan. This habitat comprised of common tree and herb species. Grassland and shrubland habitats were mainly found at the northern part at Hadden Hill and Ngau Tam Shan. These habitats mainly comprise common herb and shrub species.

8.5.2.5 Urbanised habitats (e.g. developed area/wasteland and village/orchard) were scattered throughout the Assessment Area. These urbanised habitats consisted of villages (such as Ngau Tam Mei Tsuen and Ki Lun Tsuen), roads, open storage areas and garages. These habitats were often subjected to high level of human disturbance and dominated by exotic and ornamental vegetation.

8.5.3 Fauna

Avifauna

8.5.3.1 The avifauna community in the vicinity of the Assessment Area mainly comprise of waterbirds and wetland-dependent species recorded near wetland habitats of the Assessment Area (e.g. pond and watercourse). A total of 62 species of conservation importance were recorded in previous studies (CEDD & PlanD, 2024; CEDD, 2025; CEDD and PlanD, 2022; DSD, 2018; MTR, 2024a; MTR, 2024b; WSD, 2024).

Mammal

8.5.3.2 Mammal species previously recorded within the Assessment Area include bats and non-flying terrestrial mammal species, a total of 20 species of conservation

importance were recorded in previous studies (CEDD, 2025; CEDD & PlanD, 2024; Glory Queen Limited, 2016; MTR, 2024a; MTR, 2024b; WSD, 2024).

Butterfly

8.5.3.3 Butterfly species previously recorded within the Assessment Area mainly near vegetated habitats (e. g. marsh/reed, woodland, plantation, shrubland and grassland), including 22 species of conservation importance (CEDD, 2025; CEDD & PlanD, 2024; DSD, 2021; MTR, 2009; MTR, 2024a).

Odonate

8.5.3.4 Odonate species previously recorded within the Assessment Area mainly near wetland habitats (e.g. marsh/reed, pond and semi-natural watercourse), including ten species of conservation importance (CEDD, 2025; CEDD & PlanD, 2024; MTR, 2009; MTR, 2024a).

Herpetofauna

8.5.3.5 Herpetofauna species previously recorded within the Assessment Area mainly near wetland habitats (e.g. woodland, plantation and village/orchard), including three amphibian and seven reptile species of conservation importance (CEDD, 2010; CEDD, 2013; CEDD, 2025; CEDD & PlanD, 2024; DSD, 2021; Glory Queen Limited, 2016; MTR, 2024a).

Freshwater Fish and Invertebrate

8.5.3.6 Freshwater species previously recorded within the Assessment Area in aquatic habitat (e.g. natural watercourse, semi-natural watercourse and modified watercourse). Three crustacean and two freshwater fish species of conservation importance were recorded within the Assessment Area in previous studies (CEDD, 2010; CEDD, 2025; CEDD & PlanD, 2024; DSD, 2018; DSD, 2021; WSD, 2024).

8.6 Ecological Baseline Information (Survey Findings)

8.6.1 Overview of Ecological Baseline Conditions

8.6.1.1 Consistent with the findings from literature review, a total of 15 habitat types were identified during the ecological survey within the 500 m Assessment Area, which included woodland, mixed woodland, shrubland, grassland, plantation, village/orchard, developed area/wasteland, agricultural land, seasonally wet grassland, marsh/reed, pond, natural watercourse, semi-natural watercourse, modified watercourse, mitigation wetland. Ecological resources within the Assessment Area are presented in the following sections. A summary of all habitats identified within the Assessment Area is presented in **Table 8.3**.

Table 8.3 Summary of Habitats Identified within the Assessment Area

Habitat Types	Assessment Area (ha)	Percentage (%)
Woodland	134.13	9.0%
Mixed Woodland	112.33	7.5%
Shrubland	178.80	12.0%
Grassland	283.36	18.9%
Plantation	94.1	6.3%
Village/Orchard	143.18	9.6%
Developed Area/Wasteland	428.44	28.6%
Agricultural Land	19.88	1.3%
Seasonally Wet Grassland	0.84	0.1%
Marsh/Reed	17.75	1.2%
Pond	33.21	2.2%
Natural Watercourse	5.02 (4.39 km)	0.3%
Semi-natural Watercourse	3.20 (7.94 km)	0.2%
Modified Watercourse	31.2 (13.31 km)	2.1%
Mitigation Wetland	10.42	0.7%
Total	**1495.87**	**100.00**

Note:
(1) The size and percentage are subject to rounding adjustments. Any discrepancies between total and sums of individual numbers listed therein are due to rounding.

8.6.2 Habitat and Vegetation

Woodland

8.6.2.1 Woodland patches were identified near San Tin Barracks, north of Ngau Tam Mei Water Treatment Works (NTMWTW), hillside of Kai Kung Leng, Ki Lun Tsuen, and foothill of Ngau Tam Shan near Cheung Lek.

8.6.2.2 Woodlands of 5 m to 7 m semi-open canopy were composed by native tree species such as Lance-leaved Sterculia (*Sterculia lanceolata*), Yellow Cow Wood (*Cratoxylum cochinchinense*) and Chekiang Machilus (*Machilus chekiangensis*). Native herb species such as Wild Coffee (*Psychotria asiatica*), Wood-fern (*Cyclosorus parasiticus*) and Oriental Blechnum (*Blechnum orientale*) were commonly encountered in the understory.

8.6.2.3 A 6 m to 8 m closed canopy was observed at the hillside woodland at Kai Kung Leng, part of the hillside woodland at Kai Kung Leng falls within CA and LTCP. The canopy composed of native tree species such as Chinese Banyan and Lance-leaved Sterculia, interspersed with ornamentation tree species such as Taiwan Acacia and Brisbane Box. Native herb and climber species such as Elephants-foot (*Elephantopus scaber*), Umbellate Merremia (*Merremia umbellata* subsp. *orientalis*) and Fan-leaved Maidenhair (*Adiantum flabellulatum*) were commonly encountered.

8.6.2.4 Four flora species of conservation importance, namely Incense Tree (*Aquilaria sinensis*), Cycad-fern (*Brainea insignis*), Emarginate-leaved Ormosia (*Ormosia emarginata*) and Lamb of Tartary (*Cibotium barometz*) were found within this habitat in, to the south of Ngau Tam Mei, near NTMWTW and near Kai Kung Leng (**Figures 8.5.3** to **8.5.6** refer).

Mixed Woodland

8.6.2.5 Patches of mixed woodland were identified at Tit Hang, Ngau Tam Mei and at the west of Tsing Long Highway. Mixed woodland habitat was characterized by the tree canopies which consisted of both exotic and native trees and semi-closed canopies. Mixed woodland habitat was connected to adjacent developed area/wasteland and subject to human disturbance.

8.6.2.6 Semi-open canopies ranged from 5 m to 8 m semi-open canopy were recorded and composed by native tree species such as Chinese Hackberry (*Celtis sinensis*) and Camphor Tree (*Cinnamomum camphora*), scattered with cultivated fruit tree species such as Jackfruit. Planted exotic tree species such as Paper-bark Tree, Slash Pine, Brisbane Box and Big-leaved Acacia (*Acacia mangium*) were also encountered. Native herb and shrub species such as Chinese Fevervine (*Paederia scandens*), Snail Seed (*Cocculus orbiculatus*), Desmos (*Desmos chinensis*) and Wild Coffee, and cultivated shrub species such as Orange-jessamine *(Murraya paniculata*) and exotic species such as Lantana were found at the understory.

8.6.2.7 Three flora species of conservation importance, namely *Aralia* chinensis, Incense Tree and Lamb of Tartary (*Cibotium barometz*) were found within this habitat in Ngau Tam Mei area (**Figure 8.5.6 and 8.5.9** refers).

Shrubland

8.6.2.8 Patches of shrubland including mature and tall shrub areas were identified at Tit Hang, Hadden Hill, Ngau Tam Shan, Kai Kung Leng, and along Tsing Long Highway. This habitat was often interspersed with contiguous area of grassland. Part of the hillside shrubland in Hadden Hill, Ngau Tam Mei and Kai Kung Leng was located within CA. Small extent of the hillside shrubland of Kai Kung Leng was also located within the LTCP. Native shrub and herb species such as Rose Myrtle (*Rhodomyrtus tomentosa*), Wax Tree (*Rhus succedanea*) and Hong Kong Hawthorn (*Rhaphiolepis indica*) were commonly encountered in this habitat.

8.6.2.9 Roadside shrubland at Nam Sang Wai area was succeeded from former pond bund which has now been overgrown. Weedy species such as White Popinac, Gairo Morning Glory and Mile-a-minute Weed were commonly encountered. Native tree and herb species such as White Mulberry (*Morus alba*), Opposite-leaved Fig and Many-flowered Silvergrass (*Miscanthus floridulus*) were also encountered.

8.6.2.10 One flora species of conservation importance namely Incense Tree was recorded in this habitat to the northeast of Hadden Hill (**Figure 8.5.11** refers).

Grassland

8.6.2.11 Grassland habitat identified includes hillside grassland and low-lying grassland. Extensive hillside grassland was found in Tit Hang, Hadden Hill, Ngau Tam Shan, Kai Kung Leng, and Ho Hok Shan. In contrast, low-lying grassland was identified at lowland locations scattered by developed area/wasteland, village/orchard, in Ki Lun Tsuen, Ngau Tam Mei, and along San Tin Highway. Part of the hillside grassland in Hadden Hill, Ngau Tam Shan and Ho Hok Shan were located within CA, while part of the hillside grassland in Kai Kung Leng was located within both CA and LTCP.

8.6.2.12 The hillside grasslands are susceptible to periodic hill fires, which arrest ecological succession. Flora composition of these grassland habitats composed of pioneer

light-demanding herb and shrub species such as Dichotomy Forked Fern (*Dicranopteris pedata*), Dwarf Mountain Pine (*Baeckea frutescens*) and Rose Myrtle with interspersed tree species such as Wax Tree, Pond Spice (*Litsea glutinosa*) and India-charcoal Trema (*Trema tomentosa*).

8.6.2.13 Low-lying grassland was largely succeeded from fallow fields or dried up ponds. Ruderal herbs species such as Many-flowered Silvergrass, Lalang Grass (*Imperata cylindrica* var. *major*), *Cuscuta campestris* were commonly recorded.

8.6.2.14 Two flora species of conservation importance namely Chinese Frienge Lily (*Thysanotus chinensis*) and Cycad-fern were recorded in hillside grassland habitat in Kai Kung Leng and Ngau Tam Shan area (**Figure 8.5.4 and 8.5.8** refers).

Plantation

8.6.2.15 Patches of plantation were identified at Tit Hang, Hadden Hill, Ngau Tam Shan, Kai Kung Leng, and along San Tin Highway. Hillside plantation was relatively intact, while roadside plantation is scattered by developed area/wasteland and subject to disturbance from adjacent traffic. For hillside plantation, plantation tree species such as *Acacia* spp. were commonly encountered, scattered with native tree species such as Yellow Cow Wood, forming a 5 m to 7 m semi-open canopy. Roadside plantations consisted of predominately exotic plantation trees such as *Acacia* spp. and Horsetail Tree (*Casuarina equisetifolia*). The understory was rather limited and made of exotic herb species such as *Bidens alba*, Lantana and Redtop (*Melinis repens*).

8.6.2.16 Part of the hillside plantation on Kai Kung Leng falls within CA and LTCP. This area of hillside plantation was dominated by exotic tree species such as *Acacia* spp. and *Eucalyptus* spp., scattered with native tree species such as Chekiang Machilus, forming a 6 m to 8 m tall dense canopy. Native herb and climber species such as Long Stephania (*Stephania longa*), Short-spiked Pepper (*Piper sarmentosum*) and Umbellate Merremia *w*ere occasionally encountered.

8.6.2.17 Two flora species of conservation importance namely Incense Tree and Cycad-fern were found in this habitat to the east of Pok Wai, Ngau Tam Shan and Tit Hang area (**Figures 8.5.3**, **8.5.8** and **8.5.12** refer).

Village/Orchard

8.6.2.18 Village/orchard habitat refers to areas with low-rise village houses interspersed with patches of fruit tree cultivation. Village/orchard was scattered in Tit Hang and along San Tin Highway, while concentrated in Ki Lun Tsuen and Ngau Tam Mei. Cultivated fruit tree species such as Guava (*Psidium guajava*), Longan and Jackfruit were commonly encountered. Ruderal herb species such as Thyme-leaved Spurge (*Euphorbia thymifolia*), Montane Kudzu (*Pueraria lobata* var. *montana*) and *Cleome rutidosperma* were also encountered.

8.6.2.19 Two flora species of conservation importance, namely Incense Tree and Hong Kong Pavetta (*Pavetta hongkongensis*) were found within this habitat (**Figure 8.5.6** refers).

Developed Area/Wasteland

8.6.2.20 Developed area/wasteland habitat comprised various residential areas, brownfield operations, open storage, access roads and wastelands, etc. It is the dominant

habitat in the Assessment Area. Tree species such as Paper-bark Tree, Horsetail Tree and China-berry were commonly encountered. Weedy species such as *Wedelia trilobata*, *Bidens alba* and *Asystasia micrantha* were also widely encountered in this habitat.

8.6.2.21 No flora species of conservation importance was recorded within this habitat.

Agricultural Land

8.6.2.22 Patches of dry agricultural land were identified at Tit Hang, Ki Lun Tsuen, Hang Tau, Ngau Tam Mei, and along San Tin Highway. Small patches of dry agricultural land were scattered in Tit Hang and Ki Lun Tsuen while patches of considerable size were found in Hang Tau and Ngau Tam Mei, with most of them fenced off or surrounded by village houses. Small scale farming with fruit and crop species was observed, such as Common Banana, Purple Granadilla (*Passiflora edulis*) and Sugar-apple (*Annona squamosa*). Inactive fields were interspersed among the active fields. They may be temporarily or permanently abandoned and overgrown with ruderal herbs such as Green Amaranth (*Amaranthus viridis*), Guinea Grass and Redtop.

8.6.2.23 No flora species of conservation importance was recorded within this habitat.

Seasonally Wet Grassland

8.6.2.24 Seasonally wet grassland was identified at the east of Sha Po Marsh. This habitat was succeeded from poorly drained abandoned agricultural land. This habitat was wet most of the time in wet season and dry in dry season, while aquatic plant such as Job's tears (*Coix lacryma-jobi*), Blunt Signal-grass and Giant Alocasia were commonly encountered. Exotic weedy species such as White Popinac, Mile-a-minute Weed and *Wedelia trilobata* were also encountered.

8.6.2.25 No flora species of conservation importance was recorded within this habitat.

Marsh/Reed

8.6.2.26 Several scattered small areas of marsh/reed habitat were identified, mostly on the southwest of the Assessment Area, along Kam Tin River to the west of San Tin Highway. This includes areas within WBA, WCA (such as Nam Sang Wai and Pok Wai), as well as areas outside WBA and WCA such as Sha Po and Ngau Tam Mei.

8.6.2.27 Most of the marsh/reed habitat within the Assessment Area were naturally derived from abandoned ponds. The floral diversity was generally low and dominated by aquatic herbs such as Sessile Alternanthera (*Alternanthera sessilis*) and Interrupted Tri-vein Fern (*Cyclosorus interruptus*), and some ruderal herbs on the edges such as Napier Grass (*Pennisetum purpureum*) and *Bidens alba*. Reed species such as Common Reedgrass (*Phragmites australis*) and Burma-reed (*Neyraudia reynaudiana*) were occasionally encountered.

8.6.2.28 Sha Po Marsh is a restored brackish marsh, native mangrove species such as Leather Fern (*Acrostichum aureum*) and exotic mangrove species such as *Sonneratia* spp. were commonly encountered. Native herb species such as Short-leaved Malacea Galingale (*Cyperus malaccensis* var. *brevifolius*) was also commonly encountered in this restored habitat.

8.6.2.29 No flora species of conservation was recorded within this habitat.

Pond

8.6.2.30 Similar to marsh/reed, pond habitat was mainly concentrated along Kam Tin River, to the west of San Tin Highway, and in Ngau Tam Mei. Scattered ponds were identified in Tit Hang and Ki Lun Tsuen.

8.6.2.31 Majority of the ponds identified to the west of San Tin Highway were located within WBA, WCA. These ponds are mostly inactive, with a few active fishponds observed near Pok Wai. For ponds in Ngau Tam Mei, they were located between village areas, thus relatively fragmented and subject to regular disturbance from adjacent human activities, except for a few larger ponds west of NTMWTW. Most of the ponds in Ngau Tam Mei are active fishponds practicing Koi culturing. Limited vegetation were observed from ponds in both areas (west of San Tin Highway, and in Ngau Tam Mei), mainly comprising Weedy species such as Blunt Signal-grass (*Brachiaria mutica*), Mile-a-minute Weed (*Mikania micrantha*), while pond bunds in Ngau Tam Mei also include cultivated fruit tree species such as Longan (*Dimocarpus logan*) and Common Banana (*Musa* x *paradisiaca*), as well as the presence of some White Popinac.

8.6.2.32 For Tit Hang and Ki Lun Tsuen, these ponds were relatively small and fragmented, located adjacent to village and/or urbanised areas. These small ponds supported limited vegetation, mainly comprising weedy species on the pond bund such as Guinea Grass (*Panicum maximum*), White Popinac and Mile-a-minute Weed.

8.6.2.33 No flora species of conservation was recorded within this habitat.

Natural Watercourse

San Tin

8.6.2.34 ST-S3a was an abandoned meander connecting to the eastern bank of Sheung Yue River and ran through plantation habitat. ST-3a was approximately 5 m to 15 m in width with approximately 1.5 m water depth and slow to moderate water flow. Aquatic plant such as Freshy Lady-fern (*Callipteris esculenta*), Giant Alocasia (*Alocasia macrorrhizos*) and Umbrella Plant (*Cyperus involucratus*) was commonly encountered. Some seasonal waterlines, i.e. ST-S9, ST-S10, ST-S11, ST-S12, ST-S13, ST-S14, ST-S15, ST-S16 and ST-S17, were in approximately 0.5 m to 1.0 m in width and limited water flow were recorded originated from the uphill of Ngau Tam Shan and Hadden Hill which ran through grassland woodland and mixed woodland habitat.

Ngau Tam Mei

8.6.2.35 NTM-S3 originates from woodland within LTCP to the south of the Assessment Area and was a fast-flowing watercourse with clear waterflow. NTM-S3 flows northwards through woodland, mixed woodland, plantation and village/orchard habitats (refer to **Figure 8.5.6**). The stream forks into NTM-S3a before discharging to Ngau Tam Mei Drainage Channel (NTMDC). Native tree species such as Chinese Buttonbush (*Adina pilulifera*), Common Yellow Steg-fig (*Ficus fistulosa*) and Lidded Cleistocalyx (*Cleistocalyx nervosum*) were encountered at the riparian zone.

8.6.2.36 NTM-S6 was an abandoned meander connecting to northern bank of NTMDC which ran through grassland and developed area/wasteland habitat. NTM-S6 flows from east to west and was a slow to moderate-flowing watercourse with fair water

quality. Exotic species such as *Bidens alba*, Diffuse Day-flower (*Commelina diffusa*) and Mile-a-minute Weed were encountered.

Yuen Long

8.6.2.37 YL-S1a was a section of watercourse ran through developed area/wasteland, and grassland habitat before flowed towards KTMDC. Aquatic plant species such as Short-leaved Malacea Galingale and Umbrella Plan were recorded.

8.6.2.38 YL-S2 was an abandoned meander of KTMDC, which is channelized, with concrete paved bank and bottom. YL-S2 was approximately 25 m in width and 3 m water depth with moderate flow rate, subject to tidal influence. Clear water with garbage was observed. Brackish aquatic plants such as Malacea Galingale, Common Reedgrass and *Sonneratia caseolaris* were recorded.

8.6.2.39 YL-S3 is an abandoned meander of KTMDC which was brackish and under tidal influence. This YL-S3 was approximately 25 m to 50 m in width and was connected to the KTMDC. Brackish plant species such as Common Reedgrass and *Sonneratia caseolaris* were recorded.

8.6.2.40 No flora species of conservation importance was recorded within this habitat.

Semi-natural Watercourse

San Tin

8.6.2.41 ST-S1 and ST-S1a were partially modified watercourse, some sections were found with concrete bank and bottom, while the remaining sections remain natural with sand and gravels as substrate. ST-S1 was approximately 2 m to 6 m in width, with 0.3 m water depth and slow water flow. ST-S1a was approximately ranged from 2 m to 4 m in width, with 0.3 m water depth and rapid speed of flow. Turbid water and garbage were observed in ST-S1 while clear water was observed in ST-S1a. Aquatic plant species such as Umbrella Plant (*Cyperus involucratus*) and Giant Alocasia, also weedy species such as Mile-a-minute Weed were commonly encountered.

8.6.2.42 ST-S2 and ST-S2a were partially modified watercourse with concrete bottom with some section observed to be sandy with boulders. Both watercourses flowed from the east towards the northwest which ran along developed area/wasteland, village/orchard, grassland and shrubland habtiat. Both ST-S2 and ST-S2a had water depth of about 0.1 m and slow water flow. Riparian vegetation was dominated by Elephant's Ear (*Macaranga tanarius* var. *tomentosa*) and Kudzu (*Pueraria* spp.).

8.6.2.43 ST-S7 was an upstream section of the San Tin Eastern Main Drainage Channel flowing towards the north. This watercourse was semi-natural with some modification from surrounding village/orchard habitat. The water depth was approximately 0.1m to 0.2 m with slow water flow. Dominant vegetation included Water Lettuce (*Pistia stratiotes*) and *Ludwigia perennis.*

8.6.2.44 ST-S8 was a semi-natural watercourse which ran along developed area, mixed woodland and village/orchard habitat. The water depth was approximately 0.2 m with moderate water flow. The bottom of this watercourse was sandy with boulders. Dominant riparian vegetation included Giant Alocasia and Oriental Blechnum (*Blechnum orientale*).

Ngau Tam Mei

8.6.2.45 NTM-S2 was partially modified watercourse, the bank was concrete paved while gravels and sand were found as substrate. NTM-S2 was approximately 2 m in width, with 0.3 m water depth and moderate speed of flow. Clear water with garbage was observed. Aquatic species such as Chinese Knotweed (*Persicaria chinensis*) and Freshy Lady-fern, also weedy species such as Mile-a-minute Weed were commonly encountered.

8.6.2.46 NTM-S3a was partially modified watercourse, the bank was concrete paved while gravels and sand were found as substrate. NTM-S3a was approximately 5 m in width, with 0.3 m water depth and rapid speed of flow. Aquatic plant species such as Diffuse Day-flower and Giant Alocasia and weedy species such as Mile-a-minute Weed were commonly encountered.

8.6.2.47 The upper section of NTM-S4 and NTM-S4a were partially modified watercourse, the bank was concrete-paved and some sections were modified as culvert, while sand and gravels were found as substrate. Upper section of NTM-S4 was approximately 6 m in width, with 0.3 m water depth and rapid speed of flow. Clear water with garbage was observed. Aquatic plant species such as Diffuse Day-flower, Sessile Alternanthera, and Chinese Knotweed were commonly encountered.

8.6.2.48 NTM-S5 was a partially modified watercourse, part of the watercourse was channelized and modified as culvert. NTM-S5 was approximately 3 m in width, with 0.3 m water depth and rapid speed of flow. Clear water with garbage was observed. Native herb species such as Virid-leaved Boehmeria (*Boehmeria nivea* var. *tenacissima*), also aquatic plant species such as Freshy Lady-fern (*Callipteris esculenta*) Giant Alocasia and were commonly encountered.

8.6.2.49 NTM-S6a and NTM-S7 were partially modified watercourse with altered banks and relatively natural substrates such as sand and mud. These two semi-natural watercourses were approximately 0.3 m to 2 m in width with moderate flow and turbid to fair water observed. Exotic species such as *Bidens alba* and White Popinac were commonly encountered.

Yuen Long

8.6.2.50 YL-S1b was section of drainage channels ran through developed area/wasteland, habitat before discharging to main drainage channel and flowed towards KTMDC. Aquatic plant species such as Short-leaved Malacea Galingale, Sessile Alternanthera and Umbrella Plant. Mangrove species such as Spiny Bears Breech (*Acanthus ilicifolius*) and *Sonneratia caseolaris* were also encountered.

8.6.2.51 YL-S6 was a partially modified watercourse, with concrete paved bank, and the bottom was generally muddy, with rocky and boulder substrate upstream. YL-S6 was approximately 4 m to 11 m in width, with 0.5 m water depth and moderate flow. Clear water with garbage was observed. Native tree species such as Elephant's Ear (*Macaranga tanarius* var. *tomentosa*) and Opposite-leaved Fig (*Ficus hispida*), and aquatic plant species such as Giant Alocasia were recorded.

8.6.2.52 YL-S8 was partially modified watercourse, part of the watercourse was modified as culvert underneath Sha Po Sewage Pumping Station, while the remaining natural section flows around the Sha Po Marsh towards and eventually to Kam Tin River towards the west. Sand and mud were found as substrate. YL-S8 was approximately 3m in width, with 0.5 m water depth and slow flow, subject to tidal

influence. Turbid water with garbage was observed. Weedy species such as Mile-a-minute Weed and Gairo Morning Glory, also exotic mangrove species such as *Sonneratia caseolaris* were commonly encountered.

8.6.2.53 No flora species of conservation importance was recorded within this habitat.

Modified Watercourse

San Tin

8.6.2.54 ST-S3 was a section of Sheung Yue River, with concrete paved bank and bottom. ST-S3 was approximately in 15 m width, with 1.5 m water depth and rapid speed of flow. Turbid water and the presence of garbage were observed during the survey period. Aquatic plant species such as Umbrella Plant, Short-leaved Malacea Galingale and Diffuse Day-flower were commonly encountered.

8.6.2.55 ST-S4 was a drainage channel downstream of Kwu Tung Reservoir. ST-S4 was approximately range from 4 m to 16 m in width, however seasonal flow was observed only during wet season. Both the bank and bottom of the watercourse was concrete paved and channelized. Weedy species such as Lantana (*Lantana camara*), Mile-a-minute Weed and Guinea Grass were encountered along the bank.

8.6.2.56 ST-S5, ST-S5a and ST-S5b were drainage channels ran along developed area, plantation, village/orchard, agricultural land and along Fanling Highway. ST-S5 was approximately 5 m in width and 0.3 m water depth with slow water flow. Both the bank and bottom of the watercourse were concrete-paved and channelized. Limited space was allowed for vegetation growth. Turbid water and the presence of garbage were observed during the survey period. Seedling of exotic tree species such as China-berry (*Melia azedarach*), Paper-bark Tree (*Melaleuca cajuputi* subsp. *cumingiana*) and Chittagong Chickrassy (*Chukrasia tabularis*) were occasionally encountered along the bank.

8.6.2.57 ST-S6 is a drainage channel ran through developed area and plantation habitat which flow towards ST-S5. ST-S6 was approximately 1m to 4m in width with muddy and silty bottom and slow water flow. Common flora species included Giant Alocasia, *Wedelia trilobata* and Diffuse Day-flower were encountered.

Ngau Tam Mei

8.6.2.58 NTM-S1 (i.e. NTMDC) is a modified drainage channel in Ngau Tam Mei area with channel width between 6 m and 8 m. NTMDC was channelized, with concrete paved bank and bottom. NTMDC was trapezoidal in shape with concrete and grasscrete lining and subject to tidal influence at its downstream section. During low tide and dry season, water flow was restricted to the dry weather flow channel of about 0.5m wide in the middle. Aquatic plant such as *Ludwigia perennis*, Sessile Alternanthera (*Alternanthera sessilis*) and Umbrella Plant (*Cyperus involucratus*) were commonly encountered. Weedy species such as Blunt Signal-grass (*Brachiaria mutica*), White Popinac (*Leucaena leucocephala*) and Mile-a-minute Weed (*Mikania micrantha*) were also encountered. NTM-S1b was an upper section of the NTMDC which ran beside the NTMWTW and connected to NTM-S1a before flowing to the NTMDC.

8.6.2.59 NTM-S2a and NTM-S4 were channelized, with concrete paved bank and bottom. NTM-S2a was approximately 6 m in width and 0.3 m water depth with moderate flow rate. While NTM-S4 was approximately 10 m in width, with 0.3 m water depth and rapid speed of flow. Clear water with large amount of garbage was observed.

Aquatic plant species such as Diffuse Day-flower and weedy species such as Blunt Signal-grass and *Bidens alba* were commonly encountered.

8.6.2.60 NTM-S7a was a modified watercourse lined by gabion baskets filled with rocks and gravels with approximately 5 m in width. It ran through developed area/wasteland habitat with moderate flow rate. Species such as Mile-a-minute Weed and Chinese Knotweed were encountered.

Yuen Long

8.6.2.61 YL-S1 (i.e. KTMDC) is a modified section of Kam Tin River, with channel width of between 65 m and 130 m. KTMDC was channelized, with concrete paved bank and bottom. It is highly subjected to tidal influence, in which benthic riverbed near the embankments would be exposed in low tide. Part of the KTMDC occurs within WBA, WCA, CA and Priority Site for Enhanced Conservation (Deep Bay Wetland outside Ramsar Site). YL-S1c and YL-S1d were sections of drainage channels ran through developed area/wasteland, plantation and marsh/reed habitat along the main drainage channel and flowed towards KTMDC. Aquatic plant species such as Short-leaved Malacea Galingale, Sessile Alternanthera and Umbrella Plant. Mangrove species such as Spiny Bears Breech (*Acanthus ilicifolius*) and *Sonneratia caseolaris* were also encountered.

8.6.2.62 YL-S2a and YL-S2b were channelized watercourses with concrete paved bank and muddy and sandy bottom. YL-S2a and YL-S2b were approximately 4 m to 10 m in width, with 0.3 m water depth and moderate flow. Turbid water with garbage was observed. Weedy species such as Mile-a-minute Weed, Gairo Morning Glory and White Popinac were recorded.

8.6.2.63 YL-S3a was a modified watercourse which ran through developed area/wasteland habitat with approximately 10 m to 20 m width. Turbid water and slow water flow was observed. Rural plant species such as White Popinac and *Wedelia trilobata* were recorded.

8.6.2.64 YL-S4, YL-S4a and YL-S5 were channelized watercourses with concrete paved bank and muddy and sandy bottom. They were both approximately 8 m width, with 0.5 m water depth and moderate flow. Turbid water with garbage was observed. No vegetation was recorded due to concrete paved bank limiting the space for vegetation growth.

8.6.2.65 YL-S6a, YL-S6b, YL-S7 and YL-S7a were minor drainage channels ran along the developed area/wasteland habitat on both sides of San Tin Highway with about 5m width and slow water flow. Rural plant species such as White Popinac and *Wedelia trilobata* were recorded.

8.6.2.66 No flora species of conservation importance was recorded within this habitat.

Mitigation Wetland

8.6.2.67 Mitigation wetland is a constructed wetland habitat along KTMDC, which consisted of ponds and roadside plantation vegetation, this West Rail Compensatory Wetland is implemented under the Environmental Permit No. FEP-24/004/1998/J. Part of the mitigation wetland was located within CA. Aquatic plant species such as Malacea Galingale, Blunt Signal-grass and Sessile Alternanthera were commonly encountered. Exotic plantation tree species such as Tree Cotton (*Bombax ceiba*),

Tall Bottle-brush (*Callistemon viminalis*) and Paper-bark Tree were found at margin of the habitat.

8.6.2.68 Another mitigation wetland (i.e. YLBFEW) occurs on the west of Kam Tin River, as described in **Section** Error! Reference source not found.. This wetland mainly c omprises pond habitats with small areas of marsh/reed. Generally low to moderate floral diversity was identified within these wetland habitats within YLBFEW, with aquatic plant species identified. Native aquatic plant species such as Polygonum barbatum (*Persicaria barbata*), *Commelina diffusa* and *Alocasia macrorrhizos* were commonly encountered. Native shrub species such as *Psychotria asiatica*, Microcos (*Microcos nervosa*), and *Cinnamomum camphora* were also found surrounding the wetland.

8.6.2.69 No flora species of conservation importance was recorded within this habitat.

8.6.3 Fauna

Avifauna

8.6.3.1 A total of 99 avifauna species were recorded within the Assessment Area, including 46 avifauna species of conservation importance. Over half of the avifauna species of conservation importance were wetland-dependent species recorded in modified watercourse habitat. A total of 14 avifauna species of conservation importance were recorded in mitigation wetland habitat within the Assessment Area including YLBFEW. Among the species of conservation importance recorded within mitigation wetland, except Oriental Magpie (*Pica serica*), all of them were wetland-dependent species. Summary of avifauna species of conservation importance recorded within the Assessment Area is provided in **Table 8.13** below.

8.6.3.2 Majority of the general avifauna species within the Assessment Area were recorded in modified watercourse, plantation and grassland habitats. Waterbirds and wetland-dependent species (e.g. various waders and ardeids species) were observed foraging on the exposed riverbed of KTMDC, i.e. one of the modified watercourses, during low tide.

Night Roost and Flight Line

8.6.3.3 Two night roosts (i.e. Ko Po Road Night Roost and Sha Po Night Roost) were surveyed at the southern boundary of the Assessment Area (**Figure 8.6** refers). Chinese Pond Heron, Eastern Cattle Egret, Great Egret and Little Egret were recorded in Ko Po Road Night Roost. Over half of the ardeids were recorded flying through Ko Po Flight Line 4 from the east to Ko Po Road Night Roost, along the Kam Tin River. On the other hand, ardeid species including Chinese Pond Heron, Great Egret, Grey Heron and Little Egret were recorded during the flight line survey in Sha Po Night Roost. Similar with Ko Po Road Night Roost, majority of the ardeids utilised flight lines along the Kam Tin River (i.e. Sha Po Flight Line 2 and Flight Line 3).

8.6.3.4 Location of avifauna species of conservation importance are presented in **Figure 8.5**. The full list of avifauna species recorded in the Assessment Area is given in **Appendix 8.3** and the result of flight line survey is given in **Appendix 8.4**.

Mammal

8.6.3.5 A total of 24 mammal species were recorded within the Assessment Area, including 20 mammal species of conservation importance. A total of 11 bat species were recorded within the Assessment Area, all bat species are considered as species of conservation importance. These bat species were all recorded in flight during night surveys, above developed area/wasteland and modified watercourse habitat. Nine non-flying mammal species of conservation importance namely East Asian Porcupine (*Hystrix brachyura*), Leopard Cat (*Prionailurus bengalensis*), Masked Palm Civet (*Paguma larvata*), Pallas's Squirrel (*Callosciurus erythraeus*), Red Muntjac (*Muntiacus vaginalis*), Rhesus Macaque (*Macaca mulatta*)), Small Asian Mongoose (*Herpestes javanicus*), Small Indian Civet (*Viverricula indica*) and Small-toothed Ferret Badger (*Melogale moschata*) were mainly found in woodland, mixed woodland, plantation and village/orchard in scarce abundance. Summary of mammal species of conservation importance recorded within the Assessment Area is provided in **Table 8.13** below.

8.6.3.6 A day-roosting site utilized by Himalayan Leaf-nosed Bat was recorded at one of the classrooms of the deserted Pok Wai Public School (**Figure 8.1** refers). A thin layer of guano (bat excrements) was present on the floor, suggesting the roost had been utilized for a relatively short period of time. The highest number of roosting bats observed in the day-roost was 78 individuals recorded in July 2025 during the survey period.

8.6.3.7 Location of mammal species of conservation importance are presented in **Figure 8.5**. The full list of mammal species recorded in the Assessment Area is given in **Appendix 8.3**.

Butterfly

8.6.3.8 A total of 71 butterfly species were recorded within the Assessment Area, including five butterfly species of conservation importance, namely Blackvein Sergeant (*Athyma ranga*), Danaid Eggfly (*Hypolimnas misippus*), Little Branded Swift (*Pelopidas agna*), Silver Royal (*Ancema blanka*) and Small Three-ring (*Ypthima norma*). Most of the butterfly species were recorded in plantation and grassland habitats, while butterfly species of conservation importance were recorded in pond, mixed woodland, plantation, grassland and village/orchard habitats. Summary of butterfly species of conservation importance recorded within the Assessment Area is provided in **Table 8.13** below.

8.6.3.9 Location of butterfly species of conservation importance are presented in **Figure 8.5**. The full list of butterfly species recorded in the Assessment Area is given in **Appendix 8.3**.

Odonate

8.6.3.10 A total of 30 odonate species were recorded within the Assessment Area, including three odonate species of conservation importance namely Dusk-hawker (*Gynacantha sp*.), Emerald Cascader (*Zygonyx iris insignis*) and Ruby Darter (*Rhodothemis rufa*). Most of the recorded odonate species were widely distributed in different habitats throughout Hong Kong. Odonate species were recorded in various habitats within the Assessment Area, mostly near water sources at pond, village/orchard and developed area/wasteland habitat, while odonate species of conservation importance were recorded in mitigation wetland, plantation and

developed area/wasteland habitat. Summary of odonate species of conservation importance recorded within the Assessment Area is provided in **Table 8.13** below.

8.6.3.11 Location of odonate species of conservation importance are presented in **Figure 8.5,** with full list of odonate species in the Assessment Area in **Appendix 8.3**.

Herpetofauna

8.6.3.12 A total of 11 amphibian species were recorded within the Assessment Area. Generally low abundance of herpetofauna was recorded, which are mostly common species that are widely distributed in Hong Kong. Amphibian species were mostly recorded in pond and village / orchard habitat. One amphibian species of conservation importance, namely Chinese Bullfrog (*Hoplobatrachus rugulosus*), was recorded at pond and developed area/wasteland habitat in Ngau Tam Mei (**Figure 8.5.6**).

8.6.3.13 A total of 19 reptile species were recorded within the Assessment Area. Reptile species were mostly recorded in plantation, village/orchard and developed area/wasteland. Six reptile species of conservation importance were recorded, namely Banded Krait (*Bungarus fasciatus*), Burmese Python (*Python bivittatus*), Chinese Cobra (*Naja atra*), Chinese Soft-shelled Turtle (*Pelodiscus sinensis*), Four-clawed Gecko (*Gehyra mutilata*) and Reeves' Turtle (*Mauremys reevesii*). The three snake species of conservation importance were recorded in grassland and developed area/wasteland habitats; Chinese Soft-shelled Turtle recorded in semi-natural watercourse; while Reeves' Turtle was recorded in developed area/wasteland. Four-clawed Gecko was recorded in plantation, village/orchard and developed area/wasteland habitat in various locations within the Assessment Area.

8.6.3.14 Summary of herpetofauna species of conservation importance recorded within the Assessment Area is provided in **Table 8.13** below. Location of herpetofauna species of conservation importance are presented in **Figure 8.5**. The full list of herpetofauna species recorded in the Assessment Area is given in **Appendix 8.3**.

Freshwater Fish and Invertebrate

8.6.3.15 A total of 61 freshwater community species were recorded within the Assessment Area, mainly in habitats such as semi-natural watercourses and modified watercourse. Five species of conservation importance were recorded, including two crab species namely *Cryptopotamon anacoluthon* and *Nanhaipotamon hongkongense*, and three fish species namely Blotched Snakehead (*Channa maculata*), Great Blue Spotted Mudskipper (*Boleophthalmus pectinirostris*) and Small Snakehead (*Channa asiatica*). The two crab species of conservation importance were recorded in natural watercourse, mitigation wetland and developed area/wasteland habitat. Great Blue Spotted Mudskipper was recorded in modified watercourse (i.e. KTMDC), while Blotched Snakehead was recorded in semi-natural watercourse (ST-S1) outside the Project Boundary and Small Snakehead was recorded in semi-natural watercourse (NTM-S4) within the Project Boundary. Aside from species of conservation importance, an invasive snail species, Apple Snail (*Pomacea canaliculata*), was also recorded in pond and modified watercourse habitat.

8.6.3.16 Summary of freshwater fish and invertebrate species of conservation importance recorded within the Assessment Area is provided in **Table 8.13** below. Location of

freshwater fish and invertebrate species of conservation importance are presented in **Figure 8.5**. The full list of freshwater fish and invertebrate species recorded in the Assessment Area is given in **Appendix 8.3**.

Firefly

8.6.3.17 A total of four firefly species were recorded within the Assessment Area in generally low abundance, mainly recorded in marsh/reed and developed area/wasteland habitat. An endemic firefly species of conservation importance, Motschulsky's Starworm (*Rhagophthalmus motschulskyi*), was recorded in woodland, village/orchard and developed area/wasteland habitat. A total of 23 adult flightless female were recorded occurring in woodland habitat in Ki Lun Tsuen during survey in April 2025. Summary of firefly species of conservation importance recorded within the Assessment Area is provided in **Table 8.13** below.

8.6.3.18 Location of firefly species of conservation importance are presented in **Figure 8.5**. The full list of firefly species recorded in the Assessment Area is given in **Appendix 8.3**.

8.7 Evaluation of Ecological Values

8.7.1 Habitats within the Assessment Area

8.7.1.1 The ecological importance of recorded habitats was evaluated in accordance with the EIAO-TM Annex 8 criteria and presented in **Table 8.4** to **Table 8.12** below. Species of conservation importance identified from literature review and surveys are summarised in **Table 8.13**.

Table 8.4 Ecological Evaluation for Woodland and Mixed Woodland within the Assessment Area

Criterion	Woodland	Mixed Woodland
Naturalness	Moderate	Low to moderate
Size	134.13 ha	112.33 ha
Diversity	Moderate diversity of flora; Low to moderate diversity of fauna	Moderate to high diversity of flora (consisting of native ruderal and exotic species); Moderate diversity of fauna
Rarity	Common habitat in Hong Kong Current Survey Four flora species of conservation importance (i.e. Incense Tree, Cycad-fern, Lamb of Tartary and Emarginate-leaved Ormosia) and 13 fauna species of conservation importance (i.e. Chinese Francolin, Collared Scops Owl, Chinese Noctule, Himalayan Leaf-nosed Bat, Japanese Pipistrelle, Lesser Bamboo Bat, Short-nosed Fruit Bat, East Asian Porcupine, Leopard Cat, Pallas's Squirrel, Red Muntjac, Small Indian Civet and Motschulsky's Starworm)	Common habitat in Hong Kong Current Survey Three flora species of conservation importance (i.e. Incense Tree, *Aralia chinensis* and Lamb of Tartary) and 20 fauna species of conservation importance (i.e. Chinese Pond Heron, Collared Scops Owl, Common Emerald Dove, Greater Coucal, Grey Treepie, Indian Cuckoo, Lesser Coucal, Oriental Magpie, Rufous-capped Babbler, Chinese Noctule, Japanese Pipistrelle, Lesser Yellow Bat, Unknown Vespertilionidae species 1, East

Criterion	Woodland	Mixed Woodland
	Literature Review Five flora species of conservation importance (i.e. *Aralia chinensis*, Emarginate-leaved Ormosia, Eurya-leaved Camellia, Incense Tree and Lamb of Tartary) and 35 fauna species of conservation importance (i.e. Ashy Drongo, Asian Barred Owlet, Besra, Black Kite, Black-throated Laughingthrush, Chinese Hwamei, Chinese Sparrowhawk, Collared Crow, Collared Scops Owl, Common Emerald Dove, Crested Serpent Eagle, Greater Coucal, Grey Heron, Grey-chinned Minivet, Grey-headed Canary-flycatcher, Lesser Shortwing, Pygmy Wren-babbler, Rufous-capped Babbler, Silver-eared Leiothrix, Slaty-legged Crake, Speckled Piculet, White-bellied Erpornis, White-shouldered Starling, Wood Sandpiper, East Asian Porcupine, Japanese Pipistrelle, Leopard Cat, Masked Palm Civet, Pallas's Squirrel, Red Muntjac, Short-nosed Fruit Bat, Small Asian Mongoose, Small Indian Civet, Unknown Vespertilionidae species 1 and Spotted Royal)	Asian Porcupine, Leopard Cat, Pallas's Squirrel, Red Muntjac, Rhesus Macaque, Small Indian Civet and Blackvein Sergeant) Literature Review Two flora species of conservation importance (i.e. *Aralia chinensis* and Incense Tree) and 25 fauna species of conservation importance (i.e. Asian Barred Owlet, Black-throated Laughingthrush, Chinese Hwamei, Collared Crow, Collared Scops Owl, Common Kestrel, Crested Goshawk, Crested Serpent Eagle, Greater Coucal, Grey-chinned Minivet, Rufous-capped Babbler, Speckled Piculet, White-bellied Erpornis, Chinese Noctule, Himalayan Leaf-nosed Bat, Japanese Pipistrelle, Leopard Cat, Lesser Bamboo Bat, Pallas's Squirrel, Red Muntjac, Unknown Vespertilionidae species 1, Forget-me-not, Dingy Dusk-hawker and Chinese Cobra)
Re-creatability	Low to moderate. Re-creatable if time is given for natural succession	Moderate. Re-creatable if time is given for natural succession
Fragmentation	Low to moderate	Moderate
Ecological linkage	Part of the woodland near NTMWTW and Kai Kung Leng occur within CA and LTCP Structurally and functionally linked to adjacent shrubland and grassland	Part of the mixed woodland on Ho Hok Shan occurs within CA Structurally and functionally linked to adjacent woodland, plantation, shrubland and grassland
Potential value	Moderate	Moderate
Nursery / breeding ground	No notable nursery and breeding behaviour observed	No notable nursery and breeding behaviour observed
Age	N/A	N/A
Abundance / richness of wildlife	Moderate	Moderate

Criterion	Woodland	Mixed Woodland
Ecological value	**Moderate**	**Low to moderate**

Table 8.5 Ecological Evaluation for Shrubland and Grassland within the Assessment Area

Criterion	Shrubland	Grassland
Naturalness	Moderate	Moderate for hillside grassland Low for low-lying grassland
Size	178.80 ha	283.36 ha
Diversity	Moderate diversity of flora; Low to moderate diversity of fauna	Moderate diversity of flora and fauna
Rarity	Common habitat in Hong Kong <u>Current Survey</u> One flora species of conservation importance (i.e. Incense Tree) and 11 fauna species of conservation importance (i.e. Black-throated Laughingthrush, Collared Scops Owl, Greater Coucal, Grey Treepie, Long-tailed Shrike, Rufous-capped Babbler, East Asian Porcupine, Pallas's Squirrel, Red Muntjac, Small Indian Civet and Small Three-ring) <u>Literature Review</u> Two flora species of conservation importance (i.e. Cycad-fern and Incense Tree) and 24 fauna species of conservation importance (i.e. Black-throated Laughingthrush, Chinese Hwamei, Crested Serpent Eagle, Greater Coucal, Grey Heron, Rufous-capped Babbler, Siberian Rubythroat, Speckled Piculet, East Asian Porcupine, Himalayan Leaf-nosed Bat, Japanese Pipistrelle, Leopard Cat, Lesser Bamboo Bat, Pallas's Squirrel, Red Muntjac, Small Indian Civet, Broad Spark, Dark Brown Ace, Malayan, Rounded 6-line Blue, Blue-spotted Dusk-hawker,	Common habitat in Hong Kong <u>Current Survey</u> Two flora species of conservation importance (i.e. Cycad-fern and Chinese Frienge Lily) and 19 fauna species of conservation importance (i.e. Asian Barred Owlet, Black-faced Bunting, Black-throated Laughingthrush, Chinese Francolin, Chinese Hwamei, Crested Serpent Eagle, Greater Coucal, Sooty-headed Bulbul, White-throated Kingfisher, Zitting Cisticola, Japanese Pipistrelle, East Asian Porcupine, Red Muntjac, Small Indian Civet, Danaid Eggfly, Silver Royal, Small Three-ring, Burmese Python and Chinese Cobra) <u>Literature Review</u> Three flora species of conservation importance (i.e. Chinese Frienge Lily, Cycad-fern and Incense Tree) and 22 fauna species of conservation importance (i.e. Amur Falcon, Black Kite, Chinese Francolin, Chinese Hwamei, Collared Crow, Collared Scops Owl, Crested Serpent Eagle, Greater Coucal, Little Egret, Chinese Noctule, Himalayan Leaf-nosed Bat, Japanese Pipistrelle, Pallas's Squirrel, Short-nosed Fruit Bat, Unknown Vespertilionidae species 2, Danaid Eggfly, Forget-me-not, Small Three-ring,

Criterion	Shrubland	Grassland
	Orange-backed Threadtail, Ruby Darter and King Cobra)	Spotted Angle, Tiny Grass Blue, Scarlet Basker and King Cobra)
Re-creatability	Moderate	Moderate
Fragmentation	Low to moderate	Low to moderate for hillside grassland High for low-lying grassland
Ecological linkage	Part of the hillside shrubland in Ngau Tam Shan and Ho Hok Shan falls within CA, while the hillside shrubland in Kai Kung Leng occurs within CA and LTCP Hillside shrubland was structurally and functionally linked to adjacent woodland, mixed woodland, plantation and grassland	Part of the hillside grassland in Hadden Hill, Ngau Tam Shan and Ho Hok Shan falls within CA, while part of the hillside grassland in Kai Kung Leng occurs within CA and LTCP Hillside grassland was structurally and functionally linked to adjacent woodland, plantation and shrubland
Potential value	Moderate	Low
Nursery / breeding ground	No notable nursery and breeding behaviour observed	No notable nursery and breeding behaviour observed
Age	N/A	N/A
Abundance / richness of wildlife	Low	Moderate
Ecological value	**Low to moderate**	**Low to moderate** for hillside grassland **Low** for low-lying grassland

Table 8.6 Ecological Evaluation for Plantation within the Assessment Area

Criterion	Plantation
Naturalness	Low
Size	94.10 ha
Diversity	Moderate to high diversity of flora (consisting of native ruderal and exotic species) and fauna
Rarity	Common habitat in Hong Kong Presence of Ko Po Road Night Roost <u>Current Survey</u> Two flora species of conservation importance (i.e. Incense Tree and Cycad-fern) and 32 fauna species of conservation importance (i.e. Asian Barred Owlet, Black-throated Laughingthrush, Chinese Hwamei, Collared Scops Owl, Common Emerald Dove, Common Rosefinch, Crested Serpent Eagle, Greater Coucal, Grey Nightjar, Lesser Coucal, Rufous-capped Babbler, Chinese Noctule, Himalayan Leaf-nosed Bat, Japanese Pipistrelle, Lesser Bamboo Bat, Lesser Bent-winged Bat, Short-nosed Fruit Bat, Unknown Vespertilionidae species 1, East Asian Porcupine, Leopard Cat, Masked Palm Civet, Pallas's Squirrel, Red Muntjac, Rhesus Macaque, Small Asian Mongoose, Small Indian Civet, Small-toothed Ferret Badger, Danaid Eggfly, Little Branded Swift, Small Three-ring, Dusk-hawker and Four-clawed Gecko)

Criterion	Plantation
	Literature Review Three flora species of conservation importance (i.e. *Aralia chinensis*, Cycad-fern and Incense Tree) and 35 fauna species of conservation importance (i.e. Besra, Black Kite, Black-throated Laughingthrush, Collared Scops Owl, Common Emerald Dove, Crested Goshawk, Crested Serpent Eagle, Greater Coucal, Rufous-capped Babbler, Slaty-legged Crake, Speckled Piculet, White-bellied Erpornis, White-throated Kingfisher, Chinese Noctule, East Asian Porcupine, Himalayan Leaf-nosed Bat, Intermediate Horseshoe Bat, Japanese Pipistrelle, Leopard Cat, Lesser Bamboo Bat, Lesser Yellow Bat, Masked Palm Civet, Red Muntjac, Small Asian Mongoose, Small Indian Civet, Unknown Vespertilionidae species 1, Unknown Vespertilionidae species 2, Banded Demon, Baron, Common Dart, Danaid Eggfly, Forget-me-not, Green Skirt Baron, Tiny Grass Blue and Chinese Water Dragon)
Re-creatability	Moderate to high for hillside plantation High for roadside plantation
Fragmentation	Low to moderate
Ecological linkage	Part of the hillside plantation in Ngau Tam Shan falls within CA, while part of the hillside plantation in Kai Kung Leng falls within CA and LTCP Hillside plantation was structurally and functionally linked to adjacent shrubland and grassland No notable ecological linkage for roadside plantation
Potential value	Low to moderate for hillside plantation Low for roadside plantation
Nursery / breeding ground	No notable nursery and breeding behaviour observed
Age	N/A
Abundance / richness of wildlife	Moderate
Ecological value	**Low to moderate** for hillside plantation and plantation of Ko Po Road Night Roost **Low** for roadside plantation

Table 8.7 Ecological Evaluation for Village/Orchard and Developed Area/Wasteland within the Assessment Area

Criterion	Village/Orchard	Developed Area/Wasteland
Naturalness	Low	Low
Size	143.18 ha	428.44 ha
Diversity	Moderate to high diversity of flora (consisting of native ruderal and exotic species); Moderate diversity of fauna	Moderate to high diversity of flora (consisting of native ruderal and exotic species); Moderate diversity of fauna
Rarity	Common habitat in Hong Kong	Common habitat in Hong Kong Presence of Sha Po Night Roost and Day-Roosting of Himalayan Leaf-nosed Bat

Criterion	Village/Orchard	Developed Area/Wasteland
	Current Survey Two flora species of conservation importance (i.e. Incense Tree and Hong Kong Pavetta) and 18 fauna species of conservation importance (i.e. Brown Fish Owl, Collared Scops Owl, Great Egret, Greater Coucal, Little Egret, Chinese Horseshoe Bat, Chinese Noctule, Himalayan Leaf-nosed Bat, Japanese Pipistrelle, Lesser Bamboo Bat, Lesser Yellow Bat, Short-nosed Fruit Bat, Unknown Vespertilionidae species 1, Unknown Vespertilionidae species 2, Pallas's Squirrel, Red Muntjac, Four-clawed Gecko, Motschulsky's Starworm) Literature Review Three flora species of conservation importance (i.e. *Aralia chinensis*, Hong Kong Pavetta and Incense Tree) and 21 fauna species of conservation importance (i.e. Asian Barred Owlet, Besra, Chinese Grosbeak, Collared Scops Owl, Common Emerald Dove, Eastern Buzzard, Greater Coucal, Chinese Noctule, Himalayan Leaf-nosed Bat, Japanese Pipistrelle, Lesser Yellow Bat, Pallas's Squirrel, Unknown Vespertilionidae species 1, Unknown Vespertilionidae species 2, Common Birdwing, Common Rose, Pale Palm Dart, Peacock Royal, Chinese Bullfrog, Common Rat Snake and Copperhead Racer)	Current Survey 22 fauna species of conservation importance (i.e. Chinese Pond Heron, Greater Coucal, Little Egret, Oriental Magpie, Rufous-capped Babbler, White-throated Kingfisher, Chinese Noctule, Himalayan Leaf-nosed Bat, Japanese Pipistrelle, Lesser Bamboo Bat, Lesser Bent-winged Bat, Lesser Yellow Bat, Unknown Vespertilionidae species 1, Unknown Vespertilionidae species 2, Pallas's Squirrel, Emerald Cascader, Chinese Bullfrog, Banded Krait, Four-clawed Gecko, Reeves' Turtle, *Nanhaipotamon hongkongense* and Motschulsky's Starworm) Literature Review Two flora species of conservation importance (i.e. *Aralia chinensis* and Incense Tree) and 25 fauna species of conservation importance (i.e. Black Kite, Chinese Pond Heron, Collared Crow, Crested Serpent Eagle, Great Egret, Greater Coucal, Grey Heron, Little Egret, Red-billed Starling, Chinese Horseshoe Bat, Chinese Noctule, Himalayan Leaf-nosed Bat, Japanese Pipistrelle, Lesser Bamboo Bat, Pallas's Squirrel, Unknown Myotis sp., Unknown Vespertilionidae species 1, Unknown Vespertilionidae species 2, Danaid Eggfly, Grass Demon, Tiny Grass Blue, Chinese Bullfrog, Buff-striped Keelback, Chinese Cobra and Indo-chinese Rat Snake)
Re-creatability	High	High
Fragmentation	Low to moderate for Ki Lun Tsuen and Ngau Tam Mei Moderate to high for the remaining scattered village/orchard	Low to moderate
Ecological linkage	No notable ecological linkage	No notable ecological linkage
Potential value	Low	Low
Nursery / breeding ground	No notable nursery and breeding behaviour observed	No notable nursery and breeding behaviour observed

Criterion	Village/Orchard	Developed Area/Wasteland
Age	N/A	N/A
Abundance / richness of wildlife	Moderate	Moderate
Ecological value	**Low to moderate**	**Low to moderate** for Sha Po Night Roost and Day-Roosting of Himalayan Leaf-nosed Bat **Low** for remaining developed area/wasteland

Table 8.8 Ecological Evaluation for Agricultural Land within the Assessment Area

Criterion	Agricultural Land
Naturalness	Low. A man-made habitat for the purpose of crop production.
Size	19.88 ha
Diversity	Low diversity of flora and fauna
Rarity	Common habitat in Hong Kong <u>Current Survey</u> Five fauna species of conservation importance (i.e. Black-throated Laughingthrush, Chinese Pond Heron, Common Snipe, Green Sandpiper and Japanese Pipistrelle) <u>Literature Review</u> Six fauna species of conservation importance (i.e. Black Kite, Chinese Pond Heron, Greater Coucal, Lesser Bamboo Bat, Unknown Vespertilionidae species 2 and Chinese Bullfrog)
Re-creatability	High
Fragmentation	Low to moderate for agricultural land in Ngau Tam Mei High for the remaining agricultural land
Ecological linkage	Structurally and functionally linked to nearby ponds, semi natural watercourse and grassland
Potential value	Low to moderate
Nursery / breeding ground	No notable nursery and breeding behaviour observed
Age	N/A
Abundance / richness of wildlife	Low
Ecological value	**Low**

Table 8.9 Ecological Evaluation for Seasonally Wet Grassland and Marsh/Reed within the Assessment Area

Criterion	Seasonally Wet Grassland	Marsh/Reed
Naturalness	Low to moderate	Moderate. Succeeded from man-made habitats (e.g. ponds and agricultural lands) through natural processes
Size	0.84 ha	17.75 ha

Criterion	Seasonally Wet Grassland	Marsh/Reed
Diversity	Low diversity of flora and fauna	Low to moderate diversity of flora and fauna
Rarity	Uncommon habitat in Hong Kong <u>Current Survey</u> One fauna species of conservation importance (i.e. Greater Coucal) <u>Literature Review</u> One fauna species of conservation importance (i.e. Chinese Pond Heron)	An uncommon habitat in Hong Kong <u>Current Survey</u> Eight fauna species of conservation importance recorded (i.e. Chinese Pond Heron, Greater Coucal, Grey Heron, Little Egret, Japanese Pipistrelle, Lesser Bamboo Bat, Lesser Yellow Bat and Unknown Vespertilionidae Species 1) <u>Literature Review</u> Two flora species of conservation importance (i.e. Coastal Bulrush and Prince's Feather) and 20 fauna species of conservation importance recorded (i.e. Chinese Pond Heron, Great Egret, Greater Coucal, Grey Heron, Little Egret, Purple Heron, Siberian Rubythroat, Von Schrenck's Bittern, Zitting Cisticola, Chinese Horseshoe Bat, Intermediate Horseshoe Bat, Japanese Pipistrelle, Lesser Bamboo Bat, Small Asian Mongoose, Unknown Vespertilionidae species 1, Unknown Vespertilionidae species 2, Danaid Eggfly, Tiny Grass Blue, Four-spot Midget and Mangrove Skimmer)
Re-creatability	Moderate	Low to moderate
Fragmentation	High	Low to moderate for Sha Po marsh High for the remaining marsh scattered across the Assessment Area
Ecological linkage	Structurally and functionally linked to adjacent marsh/reed and low-lying grassland	Sha Po Marsh is hydrologically connected to KTMDC Some of the marsh/reed falls within WCA, WBA, CA and Priority Site for Enhanced Conservation (Deep Bay Wetland outside Ramsar Site) Structurally and functionally linked to adjacent wetland such as ponds, modified watercourse, and mitigation wetland

Criterion	Seasonally Wet Grassland	Marsh/Reed
Potential value	Low to moderate	Moderate to high for Sha Po Marsh, which are relatively larger in size and currently maintained and managed actively Moderate for remaining marsh/reed which could be enhanced if managed as wildlife habitat
Nursery / breeding ground	No notable nursery and breeding behaviour observed	Sha Po Marsh is one of the few known nursery/breeding grounds of Four-spot Midget No notable nursery and breeding behaviour in other marsh/reed
Age	N/A	N/A
Abundance / richness of wildlife	Low	Moderate for Sha Po Marsh Low to moderate in other marsh/reed
Ecological value	**Low**	**Moderate to high** for Sha Po Marsh **Low to moderate** for other marsh/reed

Table 8.10 Ecological Evaluation for Pond within the Assessment Area

Criterion	Pond within WBA and WCA	Pond near NTMWTW	Other Pond
Naturalness	Low. A man-made habitat for the purpose of aquaculture and landscaping		
Size	20.81 ha	2.67 ha	9.73 ha
Diversity	Low to moderate diversity of flora and fauna	Low to moderate diversity of flora and fauna	Low to moderate diversity of flora and fauna
Rarity	Uncommon. Mainly located at northwestern New Territories <u>Current Survey</u> Eight fauna species of conservation importance recorded (i.e. Black Kite, Chinese Pond Heron, Great Cormorant, Great Egret, Grey Heron, Little Egret, Little Grebe and Unknown Vespertilionidae species 1) <u>Literature Review</u>	Uncommon. Mainly located at northwestern New Territories <u>Current Survey</u> Eight fauna species of conservation importance recorded (i.e. Asian Barred Owlet, Chinese Pond Heron, Great Egret, Greater Coucal, Grey Heron, Little Egret, Small Indian Civet, and Blackvein Sergeant) <u>Literature Review</u>	Uncommon. Mainly located at northwestern New Territories <u>Current Survey</u> Seven fauna species of conservation importance recorded (i.e. Great Cormorant, Grey Heron, Green Sandpiper, Chinese Pond Heron, Great Egret, White-throated Kingfisher and Chinese Bullfrog) <u>Literature Review</u>

Criterion	Pond within WBA and WCA	Pond near NTMWTW	Other Pond
	15 fauna species of conservation importance recorded (i.e. Black Kite, Chinese Pond Heron, Collared Crow, Great Cormorant, Great Egret, Greater Coucal, Grey Heron, Kentish Plover, Little Egret, Little Grebe, Purple Heron, Red-billed Starling, White-throated Kingfisher, Japanese Pipistrelle and Unknown Vespertilionidae species 1)	11 fauna species of conservation importance recorded (i.e. Black-winged Stilt, Chinese Pond Heron, Great Egret, Greater Coucal, Grey Heron, Little Egret, White-throated Kingfisher, Japanese Pipistrelle, Unknown Vespertilionidae species 1, Dingy Dusk-hawker and Chinese Bullfrog)	Nine fauna species of conservation importance recorded (i.e. Chinese Pond Heron, Little Egret, Purple Heron, Japanese Pipistrelle, Lesser Bamboo Bat, Lesser Yellow Bat, Unknown Vespertilionidae species 1, Coastal Glider and Chinese Bullfrog)
Re-creatability	High	High	High
Fragmentation	Low to moderate	Moderate	Moderate for Kwu Tung Reservoir Moderate to high for remaining ponds
Ecological linkage	Ponds within WCA and WBA are structurally and functionally connected to the nearby agricultural land, grassland and marsh/reed	Structurally and functionally connected with the tributaries of NTMDC and marsh/reed habitat for ponds to the east of NTM-S4 and S4a	Some functional linkage with ponds within WBA Structurally and functionally connected with the adjacent mixed woodland near Kwu Tung Reservoir
Potential value	Moderate	Moderate for pond of larger size Ponds could be enhanced with ecologically friendly aquacultural measures	Moderate for pond of larger size Ponds could be enhanced with ecologically friendly aquacultural measures Low for remaining small and scattered pond
Nursery / breeding ground	No notable nursery and breeding behaviour observed	No notable nursery and breeding behaviour observed	No notable nursery and breeding behaviour observed
Age	N/A	N/A	N/A

Criterion	Pond within WBA and WCA	Pond near NTMWTW	Other Pond
Abundance / richness of wildlife	Moderate	Low to moderate	Low to moderate
Ecological value	**Moderate**	**Low to moderate**	**Low to moderate** for Kwu Tung Reservoir **Low** for remaining ponds

Table 8.11 Ecological Evaluation for Natural Watercourse, Semi-Natural Watercourse and Modified Watercourse within the Assessment Area

Criterion	Natural Watercourse	Semi-Natural Watercourse	Modified Watercourse
Naturalness	Moderate to high	Low to moderate	Low
Size	5.02 ha (4.39 km)	3.20 ha (7.94 km)	31.20 ha (13.31 km)
Diversity	Low diversity of flora and fauna	Low to moderate diversity of flora and moderate diversity of fauna	Low to moderate diversity of flora; Moderate diversity of fauna
Rarity	Common habitat in Hong Kong <u>Current Survey</u> Five fauna species of conservation importance (i.e. Eurasian Coot, Grey Heron, Little Egret, Lesser Yellow Bat and *Cryptopotamon anacoluthon)*	Common habitat in Hong Kong <u>Current Survey</u> Nine fauna species of conservation importance (i.e. Chinese Pond Heron, Common Emerald Dove, Greater Coucal, Little Egret, Japanese Pipistrelle, Leopard Cat, Chinese Soft-shelled Turtle, Blotched Snakehead and Small Snakehead)	Common habitat in Hong Kong <u>Current Survey</u> 32 fauna species of conservation importance (i.e. Black Kite, Black-faced Spoonbill, Black-winged Stilt, Chinese Pond Heron, Collared Crow, Common Greenshank, Common Redshank, Eurasian Curlew, Eurasian Teal, Eurasian Wigeon, Great Cormorant, Great Egret, Greater Coucal, Green Sandpiper, Grey Heron, Indian Cuckoo, Little Egret, Northern Pintail, Northern Shoveler, Oriental Magpie, Pied Avocet, Spotted Redshank, White-throated Kingfisher, Wood Sandpiper, Chinese Noctule, Japanese Pipistrelle, Lesser Bamboo Bat, Unknown Myotis

Criterion	Natural Watercourse	Semi-Natural Watercourse	Modified Watercourse
	Literature Review One flora species of conservation importance (i.e. Lamb of Tartary) and 12 fauna species of conservation importance (i.e. Chinese Pond Heron, Greater Coucal, Grey Heron, Little Egret, Japanese Pipistrelle, White-banded Shadowdamsel, Brown Wood Frog, Chinese Bullfrog, Lesser Spiny Frog, *Cryptopotamon anacoluthon, Nanhaipotamon hongkongense* and *Somanniathelphusa zanklon*)	Literature Review Nine fauna species of conservation importance (i.e. Chinese Pond Heron, Greater Coucal, Grey Heron, Little Egret, Chinese Noctule, Japanese Pipistrelle, Mangrove Skimmer, Chinese Bullfrog and Small Snakehead)	Species, Unknown Vespertilionidae species 1, Unknown Vespertilionidae species 2, Leopard Cat and Great Blue Spotted Mudskipper) Literature Review 23 fauna species of conservation importance (i.e. Black-winged Stilt, Chinese Pond Heron, Collared Crow, Common Greenshank, Common Redshank, Eurasian Teal, Eurasian Wigeon, Great Cormorant, Great Egret, Greater Coucal, Grey Heron, Grey-headed Lapwing, Little Egret, Little Grebe, Marsh Sandpiper, Northern Shoveler, Pied Avocet, Wood Sandpiper, Japanese Pipistrelle, Unknown Vespertilionidae species 2, Chinese Bullfrog, *Gobiopterus macrolepis* and *Somanniathelphusa zanklon*)
Re-creatability	Low	Moderate to high	High
Fragmentation	Moderate to high	Moderate to high	No major fragmentation for KTMDC Low for the remaining modified watercourse
Ecological linkage	NTM-S3 was structurally and hydrologically linked to NTMDC downstream	YL-S1 is structurally and hydrologically linked to Sha Po Marsh and KTMDC. Remaining semi-natural watercourses were structurally and functionally linked to adjacent wetland such as pond, and watercourses	Part of the KTMDC occurs within WBA, WCA, CA and Priority Site for Enhanced Conservation (Deep Bay Wetland outside Ramsar Site) Structurally and functionally linked to adjacent ponds, natural watercourse, semi natural

Criterion	Natural Watercourse	Semi-Natural Watercourse	Modified Watercourse
			watercourse, mitigation wetland
Potential value	Low to moderate	Low for ST-S1 to ST-S5 (except ST-S1a) and YL-S6 Moderate for YL-S1 Low to moderate for the remaining semi-natural watercourse	Low to moderate for ST-S3, ST-S5, KTMDC, modified watercourse in Ngau Tam Mei Low for the remaining modified watercourse
Nursery / breeding ground	No notable nursery and breeding behaviour observed	No notable nursery and breeding behaviour observed	No notable nursery and breeding behaviour observed
Age	N/A	N/A	N/A
Abundance / richness of wildlife	Low	Low to moderate	Moderate
Ecological value	**Low to moderate**	**Low for** ST-S1 to ST-S5 (except ST-S1a) and YL-S6 **Moderate** for YL-S1 **Low to moderate** for the remaining semi natural watercourse	**Moderate** for KTMDC **Low to moderate** for ST-S3, ST-S5, modified watercourse in Ngau Tam Mei **Low** for the remaining modified watercourse

Table 8.12 Ecological Evaluation for Mitigation Wetland within the Assessment Area

Criterion	Mitigation Wetland
Naturalness	Low to moderate. A man-made habitat near Yuen Long Bypass Floodway and along Kam Tin River created to compensate the loss of wetlands and are under active management
Size	10.42 ha
Diversity	Low to moderate diversity of flora and fauna for YLBFEW; Low to moderate diversity of flora and low diversity of fauna for West Rail Compensatory Wetland
Rarity	An uncommon habitat to Hong Kong <u>Current Survey</u> 17 fauna species of conservation species (i.e. Black Kite, Chinese Pond Heron, Common Greenshank, Collared Crow, Eurasian Curlew, Eurasian Teal, Great Cormorant, Great Egret, Grey Heron, Little Egret, Little Grebe, Northern Shoveler, Oriental Magpie, White-throated Kingfisher, Japanese Pipistrelle, Ruby Darter and *Nanhaipotamon hongkongense*) <u>Literature Review</u> 24 fauna species of conservation importance (i.e. Black Kite, Black-winged Stilt, Chinese Pond Heron, Cinnamon Bittern, Collared Crow, Crested Serpent Eagle, Eastern Marsh Harrier, Eurasian Coot, Eurasian Curlew, Eurasian Teal, Great Cormorant, Great Egret, Grey Heron, Intermediate Egret, Little Egret, Little Grebe, Northern Shoveler, Pied Kingfisher, Tufted Duck, Western Osprey, White-

Criterion	Mitigation Wetland
	shouldered Starling, White-throated Kingfisher, Unknown Vespertilionidae species 1 and Common Rat Snake)
Re-creatability	High
Fragmentation	Low to moderate
Ecological linkage	Part of the mitigation wetland occurs within CA Structurally and functionally linked to adjacent marsh/reed, and modified watercourse
Potential value	Moderate. West Rail Compensatory Wetland is currently maintained and managed actively but are mostly surrounded by developed area YLBFEW is connected to the nearby wetland habitat such as marsh/reed and pond
Nursery / breeding ground	No notable nursery and breeding behaviour observed
Age	N/A
Abundance / richness of wildlife	Low to moderate for YLBFEW Low for West Rail Compensatory Wetland
Ecological value	**Moderate** for YLBFEW **Low to moderate** for West Rail Compensatory Wetland

Note:
The higher number of species of conservation importance recorded in Assessment Area in some habitats was due to the accumulation of findings with the recorded species of conservation importance consolidated from different studies with different survey transects coverage in the past ten years. The species of conservation importance from the current study focuses on the project-specific Assessment Area within a 12-month period, as such recorded with lower abundance and diversity in comparison to the accumulated findings from literature review. Nonetheless, findings from desktop studies are also taken into account in the impact assessment.

8.7.2 Species of Conservation Importance

8.7.2.1 A summary of flora and fauna species of conservation importance recorded within the Assessment Area were gathered from previous studies and from recent surveys, and presented in **Table 8.13** below.

Table 8.13 Species of Conservation Importance Recorded within the Assessment Area from Previous Studies and Recent Survey

Common Name (*Scientific Name*)	Distribution in Hong Kong[1]	Protection and Conservation Status	Habitat Recorded	
			Previous Studies	Current Study
Flora				
Aralia chinensis	Common	Vulnerable(6)	Woodland, Mixed Woodland, Plantation, Village/Orchard, Developed area/Wasteland (25,30)	Mixed Woodland*
Chinese Frienge Lily (*Thysanotus chinensis*)	Rare	-	Grassland (25)	Grassland
Coastal Bulrush (*Schoenoplectus subulatus*)	Restricted	Rare(20)	Marsh/Reed (25)	-
Cycad-fern (*Brainea insignis*)	Restricted	Category II(8); (Vulnerable)(12) Category II (Vulnerable)(13)	Plantation, Shrubland, Grassland (23,25,26,30)	Woodland, Plantation*, Grassland
Emarginate-leaved Ormosia (*Ormosia emarginata*)	Common	Category II(8)	Woodland (26)	Woodland
Eurya-leaved Camellia (*Camellia euryoides*)	Restricted	Cap.96(4)	Woodland (27)	-
Hong Kong Pavetta (*Pavetta hongkongensis*)	Common	Cap.96(4)	Village/Orchard (30)	Village/Orchard*
Incense Tree (*Aquilaria sinensis*)	Common	Cap. 586(5); Vulnerable(6);Category II(8); Vulnerable(9); Near Threatened(11); Vulnerable(12); Category 2 & 3 (Near Threatened)(13)	Woodland, Mixed Woodland, Plantation, Shrubland, Grassland, Village/Orchard, Developed Area (23,25,26,27,28,30)	Woodland*, Mixed Woodland*, Plantation*, Shrubland*, Village/Orchard
Lamb of Tartary (*Cibotium barometz*)	Common	Cap. 586(5); Category II(8); Category 2 (Vulnerable)(13)	Natural Watercourse, Woodland (25,26,27,30)	Woodland, Mixed Woodland
Prince's Feather (*Persicaria orientalis*)	Very rare	-	Marsh/Reed (25)	-

Common Name (*Scientific Name*)	Distribution in Hong Kong[1]	Protection and Conservation Status	Habitat Recorded	
			Previous Studies	Current Study
Avifauna[3]				
Amur Falcon (*Falco amurensis*)	Uncommon autumn passage migrant	Cap.586[5]; Class II[14]; Near threatened[16]	Grassland [25]	-
Ashy Drongo (*Dicrurus leucophaeus*)	Uncommon winter visitor	LC[2]	Woodland [27]	-
Asian Barred Owlet (*Glaucidium cuculoides*)	Locally common resident	Cap.586[5]; Class II[14]	Woodland, Mixed Woodland, Village/Orchard [25]	Grassland, Plantation, Pond*
Besra (*Accipiter virgatus*)	Common resident and migrant	Cap.586[5]; Class II[14]	Woodland, Plantation, Village/Orchard [25]	-
Black-faced Bunting (*Emberiza spodocephala*)	Common winter visitor and passage migrant	MR[33]	-	Grassland*
Black-faced Spoonbill[19] (*Platalea minor*)	Common winter visitor.	PGC[2] Class 1[14] Endangered[6][15][16][17] MR[33]	-	Modified Watercourse*
Brown Fish Owl (*Ketupa zeylonensis*)	Widely distributed in Hong Kong.	RC[2] Cap. 586[5] Class II[14] Endangered [16][17] MR*[33]	-	Village/Orchard
Black Kite[19] (*Milvus migrans*)	Common resident and winter visitor	Class II[14]	Pond, Mitigation Wetland, Woodland, Plantation, Dry Agricultural Land, Grassland, Developed Area/Wasteland [23,25,27,30,34]	Pond, Modified Watercourse, Mitigation Wetland, In Flight*
Black-throated Laughingthrush (*Pterorhinus chinensis*)	Common resident	Class II[14]; Near threatened[16]	Woodland, Mixed Woodland, Plantation, Shrubland [30]	Shrubland*, Grassland*, Plantation*, Agricultural Land
Black-winged Stilt[19]	Common migrant and winter visitor	RC[2]	Pond, Mitigation Wetland, Watercourse [25,30,34]	Modified Watercourse*

Common Name (*Scientific Name*)	Distribution in Hong Kong[1]	Protection and Conservation Status	Habitat Recorded	
			Previous Studies	Current Study
(*Himantopus himantopus*)				
Chinese Francolin (*Francolinus pintadeanus*)	Common resident	MR(33)	Grassland (25,30)	Woodland, Grassland*
Chinese Grosbeak (*Eophona migratoria*)	Common winter visitor	LC(2)	Village/Orchard (30)	-
Chinese Hwamei (*Garrulax canorus*)	Common resident	Cap.586(5); Class II(14)	Woodland, Mixed Woodland, Shrubland, Grassland (23,25,30)	Grassland*, Plantation
Chinese Pond Heron(19) (*Ardeola bacchus*)	Common resident	PRC(2)	Marsh/Reed, Pond, Mitigation Wetland, Natural Watercourse, Semi-natural Watercourse, Modified Watercourse, Watercourse, Dry Agricultural Land, Seasonally Wet Grassland, Developed Area/Wasteland (22,23,24,25,30, 34)	Mixed Woodland*, Marsh/Reed*, Pond*, Developed Area/Wasteland, Semi-natural Watercourse*, Modified Watercourse*, Agricultural Land, Mitigation Wetland
Chinese Sparrowhawk (*Accipiter soloensis*)	Common passage migrant	Cap.586(5); Class II(14)	Woodland (30)	-
Cinnamon Bittern(19) (*Ixobrychus cinnamomeus*)	Uncommon passage migrant and scarce summer visitor	HR(33)	Mitigation Wetland (34)	-
Collared Crow(19) (*Corvus torquatus*)	Locally common resident	LC(2); Vulnerable(6); MR(33)	Pond, Watercourse, Mitigation Wetland, Woodland, Mixed Woodland, Grassland, Developed Area/Wasteland (25,27,30, 34)	Modified Watercourse, Mitigation Wetland
Collared Scops Owl (*Otus lettia*)	Common resident	Cap.586(5); Class II(14)	Woodland, Mixed Woodland, Plantation, Grassland, Village/Orchard (23,30)	Woodland, Mixed Woodland*, Shrubland, Plantation*, Village/Orchard*
Common Emerald Dove (*Chalcophaps indica*)	Uncommon but widespread resident	Vulnerable(15)	Woodland, Plantation, Village/Orchard (30)	Mixed Woodland, Plantation, Semi-natural Watercourse*

Common Name (*Scientific Name*)	Distribution in Hong Kong[1]	Protection and Conservation Status	Habitat Recorded	
			Previous Studies	Current Study
Common Greenshank[19] (*Tringa nebularia*)	Abundant winter visitor and migrant	RC[2]; HR[33]	Watercourse, Modified Watercourse [25,30]	Modified Watercourse, Mitigation Wetland
Common Kestrel (*Falco tinnunculus*)	Common autumn migrant and winter visitor	Cap.586[5]; Class II[14]; HR[33]	Mixed Woodland [30]	-
Common Redshank[19] (*Tringa totanus*)	Abundant passage migrant and winter visitor	RC[2]	Watercourse [24,25]	Modified Watercourse
Common Rosefinch (*Carpodacus erythrinus*)	Rare winter visitor and migrant	LC[2]; HR[33]	-	Plantation*
Common Snipe[19] (*Gallinago gallinago*)	Common passage migrant and winter visitor.	HR[33]	-	Agricultural Land
Crested Goshawk (*Accipiter trivirgatus*)	Common resident	Cap.586[5]; Class II[14]	Mixed Woodland, Plantation [25,30]	-
Crested Serpent Eagle (*Spilornis cheela*)	Common resident	Cap.586[5]; Class II[14]; Vulnerable[15]	Mitigation Wetland, Woodland, Mixed Woodland, Plantation, Shrubland, Grassland, Developed Area/Wasteland [25,30,34]	Grassland, Plantation, In Flight*
Eastern Buzzard[19] (*Buteo japonicus*)	Common winter visitor	Cap.586[5]; Class II[14]	Village/Orchard [25]	In Flight*
Eastern Marsh Harrier[19] (*Circus spilonotus*)	Common winter visitor and passage migrant	LC[2]; Class II[14] ; HR[33]	Mitigation Wetland [34]	-
Eurasian Coot[19] (*Fulica atra*)	Uncommon winter visitor.	RC[2] HR[33]	Mitigation Wetland [34]	Natural Watercourse
Eurasian Curlew[19] (*Numenius arquata*)	Abundant winter visitor.	RC[2] Class II[14]	Mitigation Wetland [34]	Modified Watercourse, Mitigation Wetland
Eurasian Hobby (*Falco subbuteo*)	Uncommon autumn passage migrant	Cap. 586[5]; Class II[14]; HR[33]	-	In Flight
Eurasian Teal[19] (*Anas crecca*)	Common winter visitor	RC[2]; HR*[33]	Watercourse, Mitigation Wetland [25]	Modified Watercourse, Mitigation Wetland
Eurasian Wigeon[19] (*Mareca penelope*)	Winter visitor	RC[2]	Watercourse [25]	Modified Watercourse*

Common Name (*Scientific Name*)	Distribution in Hong Kong[1]	Protection and Conservation Status	Habitat Recorded	
			Previous Studies	Current Study
Great Cormorant[19] (*Phalacrocorax carbo*)	Common winter visitor	PRC[2]	Pond, Mitigation Wetland, Watercourse [25,30,34]	Pond, Modified Watercourse*, Mitigation Wetland, In Flight
Great Egret[19] (*Ardea alba*)	Common resident, migrant and winter visitor	PRC[2]	Marsh/Reed, Pond, Watercourse, Mitigation Wetland, Developed Area/Wasteland [25,30,34]	Village/Orchard, Pond, Modified Watercourse* , Mitigation Wetland
Greater Coucal (*Centropus sinensis*)	Common resident	Class II[14]; Vulnerable[15]	Marsh/Reed, Pond, Natural Watercourse, Semi-natural Watercourse, Watercourse, Woodland, Mixed Woodland, Plantation, Dry Agricultural Land, Shrubland, Grassland, Village/Orchard, Developed Area/Wasteland [23,25,30]	Mixed Woodland, Shrubland, Grassland*, Plantation*, Village/Orchard*, Developed Area/Wasteland, Seasonally Wet Grassland, Pond*, Marsh/Reed, Semi-natural Watercourse*, Modified Watercourse
Green Sandpiper[10] (*Tringa ochropus*)	Common migrant and winter visitor	HR[33]	-	Agricultural Land, Pond, Modified Watercourse,
Grey Heron[19] (*Ardea cinerea*)	Common winter visitor	PRC[2]; MR	Marsh/Reed, Pond, Watercourse, Modified Watercourse, Mitigation Wetland, Woodland, Shrubland, Developed Area/Wasteland [23,25,30,34]	Marsh/Reed, Village/Orchard*, Pond*, Natural Watercourse, Modified Watercourse*, Mitigation Wetland, In Flight
Grey-chinned Minivet (*Pericrocotus solaris*)	Locally common resident	LC[2]	Woodland, Mixed Woodland [30]	-
Grey-headed Canary-flycatcher (*Culicicapa ceylonensis*)	Uncommon winter visitor	LC[2]	Woodland [30]	-
Grey-headed Lapwing[19] (*Vanellus cinereus*)	Locally common winter visitor and migrant	LC[2]; HR[33]	Watercourse [25]	-
Grey Nightjar (*Caprimulgus jotaka*)	Scarce passage migrant	LC[2]	-	Plantation

Common Name (*Scientific Name*)	Distribution in Hong Kong[1]	Protection and Conservation Status	Habitat Recorded	
			Previous Studies	Current Study
Grey Treepie (*Dendrocitta formosae*)	Locally common resident	LC(2)	-	Mixed Woodland, Shrubland
Indian Cuckoo (*Cuculus micropterus*)	Locally common spring and summer visitor	HR(33)	-	Mixed Woodland*, Modified Watercourse
Intermediate Egret(19) (*Ardea intermedia*)	Resident and passage migrant	RC(2)	Mitigation Wetland (25,34)	-
Kentish Plover(19) (*Charadrius alexandrinus*)	Abundant winter visitor and scarce migrant	RC(2)	Pond (25)	-
Lesser Coucal (*Centropus bengalensis*)	Uncommon resident	Class II(14); Vulnerable(15); MR(33)	-	Mixed Woodland, Plantation
Lesser Shortwing (*Brachypteryx leucophris*)	Locally common resident and winter visitor	LC(2)	Woodland (30)	-
Little Egret(19) (*Egretta garzetta*)	Common resident, migrant and winter visitor	PRC(2)	Marsh/Reed, Pond, Mitigation Wetland, Natural Watercourse, Semi-natural Watercourse, Modified Watercourse, Watercourse, Grassland, Developed Area/Wasteland (23,25,30,34)	Marsh/Reed, Village/Orchard*, Developed Area/Wasteland, Pond*, Natural Watercourse, Semi-natural Watercourse*, Modified Watercourse*, Mitigation Wetland
Little Grebe(19) (*Tachybaptus ruficollis*)	Common resident	LC(2)	Pond, Watercourse, Mitigation Wetland (25,34)	Pond, Mitigation Wetland
Long-tailed Shrike (*Lanius schach*)	Common resident	MR(33)	-	Shrubland
Marsh Sandpiper(19) (*Tringa stagnatilis*)	Abundant winter visitor and migrant	RC(2); HR(33)	Watercourse (25)	-
Northern Pintail(19) (*Anas acuta*)	Abundant winter visitor.	RC(2); HR(33)	-	Modified Watercourse*
Northern Shoveler(19) (*Spatula clypeata*)	Abundant winter visitor	RC(2)	Mitigation Wetland, Watercourse (25,34)	Modified Watercourse*, Mitigation Wetland
Oriental Magpie (*Pica serica*)	Common resident	HR(33)	-	Mixed Woodland, Village/Orchard, Modified

Common Name (*Scientific Name*)	Distribution in Hong Kong[1]	Protection and Conservation Status	Habitat Recorded	
			Previous Studies	Current Study
				Watercourse*, Mitigation Wetland
Pied Avocet[19] (*Recurvirostra avosetta*)	Abundant winter visitor	RC[2]	Watercourse [25]	Modified Watercourse*
Pied Kingfisher[19] (*Ceryle rudis*)	Common resident	HR[33]	Mitigation Wetland [34]	-
Purple Heron[19] (*Ardea purpurea*)	Uncommon passage migrant	RC[2]; HR[33]	Marsh/Reed, Pond [24,25]	-
Pygmy Wren-babbler (*Pnoepyga pusilla*)	Locally common resident	LC[2]	Woodland [30]	-
Red-billed Starling[19] (*Spodiopsar sericeus*)	Abundant winter visitor	GC[2]; HR[33]	Pond, Developed Area/Wasteland [25,30]	-
Rufous-capped Babbler (*Cyanoderma ruficeps*)	Common resident	LC[2]	Woodland, Mixed Woodland, Plantation, Shrubland [23,30]	Mixed Woodland, Shrubland*, Plantation*, Developed Area/Wasteland*
Siberian Rubythroat (*Calliope calliope*)	Common winter visitor and passage migrant	Class II[14]; MR[33]	Marsh/Reed, Shrubland [30]	-
Silver-eared Leiothrix (*Leiothrix argentauris*)	Common resident	Cap.586[5]; Class II[14]	Woodland [30]	-
Slaty-legged Crake (*Rallina eurizonoides*)	Locally common breeding season visitor, migrant and scarce winter visitor	Vulnerable[16][17]	Woodland, Plantation [30]	-
Sooty-headed Bulbul (*Pycnonotus aurigaster*)	Common resident	HR*[33]	-	Grassland
Speckled Piculet (*Picumnus innominatus*)	Rare resident	LC[2]	Woodland, Mixed Woodland, Plantation, Shrubland [30]	-
Spotted Redshank[19] (*Tringa erythropus*)	Common spring passage migrant	RC[2] ; HR[33]	-	Modified Watercourse
Tufted Duck[19] (*Aythya fuligula*)	Abundant winter visitor	LC[2]	Mitigation Wetland [34]	-

Common Name (*Scientific Name*)	Distribution in Hong Kong[1]	Protection and Conservation Status	Habitat Recorded	
			Previous Studies	Current Study
Von Schrenck's Bittern[19] (*Ixobrychus eurhythmus*)	Scarce passage migrant	RC[2]; HR[33]	Marsh/Reed [25]	-
Western Osprey (*Pandion haliaetus*)	Common winter visitor	RC[2]; Class II[14] ; MR[33]	Mitigation Wetland [34]	-
White-bellied Erpornis (*Erpornis zantholeuca*)	Uncommon resident	LC[2]	Woodland, Mixed Woodland, Plantation [30]	-
White-shouldered Starling (*Sturnia sinensis*)	Locally common passage migrant and uncommon winter visitor	(LC)[2]	Woodland, Mitigation Wetland [25,27]	-
White-throated Kingfisher[19] (*Halcyon smyrnensis*)	Common resident	Class II[14]	Pond, Mitigation Wetland, Plantation [30,34]	Grassland*, Developed Area/Wasteland*, Pond, Modified Watercourse*, Mitigation Wetland
Wood Sandpiper [19] (*Tringa glareola*)	Common migrant and winter visitor	LC[2]; HR[33]	Woodland, Modified Watercourse [25,30]	Modified Watercourse
Zitting Cisticola (*Cisticola juncidis*)	Common passage migrant and winter visitor	LC[2]; HR[33]	Marsh/Reed [30]	Grassland
Mammal[3]				
Chinese Horseshoe Bat (*Rhinolophus sinicus*)	Widely distributed in countryside areas throughout Hong Kong	-	Marsh/Reed, Developed Area [23,25]	Village/Orchard*
Chinese Noctule (*Nyctalus plancyi*)	Fairly widely distributed in countryside areas throughout Hong Kong	PRC[2]	Semi-natural Watercourse, Mixed Woodland, Plantation, Grassland, Village/Orchard, Developed Area/Wasteland [23,30]	Woodland, Mixed Woodland, Plantation*, Village/Orchard*, Developed Area/Wasteland*, Modified Watercourse
East Asian Porcupine (*Hystrix brachyura*)	Very widely distributed in countryside areas throughout Hong Kong, except for Lantau Island	PGC[2]	Woodland, Plantation, Shrubland [23,25,30]	Woodland*, Mixed Woodland*, Shrubland*, Plantation, Grassland

Common Name (*Scientific Name*)	Distribution in Hong Kong[1]	Protection and Conservation Status	Habitat Recorded	
			Previous Studies	Current Study
Himalayan Leaf-nosed Bat (*Hipposideros armiger*)	Widely distributed in countryside areas throughout Hong Kong	(LC)[(2)]	Mixed Woodland, Plantation, Shrubland, Grassland, Village/Orchard, Developed Area/Wasteland [(23,25,30)]	Woodland*, Plantation*, Village/Orchard*, Developed Area/Wasteland
Intermediate Horseshoe Bat (*Rhinolophus affinis*)	Widely distributed in countryside areas throughout Hong Kong	(LC)[(2)]	Marsh/Reed, Plantation [(24,25)]	-
Japanese Pipistrelle (*Pipistrellus abramus*)	Widely distributed throughout Hong Kong	-	Marsh/Reed, Pond, Natural Watercourse, Semi-natural Watercourse, Modified Watercourse, Woodland, Mixed Woodland, Plantation, Shrubland, Grassland, Village/Orchard, Developed Area/Wasteland [(25,27,30)]	Woodland*, Mixed Woodland*, Grassland*, Plantation*, Village/Orchard*, Developed Area/Wasteland*, Agricultural Land, Marsh/Reed, Semi-natural Watercourse, Modified Watercourse, Mitigation Wetland
Leopard Cat (*Prionailurus bengalensis*)	Widely distributed in countryside areas throughout Hong Kong, except for Lantau Island	Cap.586[(5)]; Class II[(14)]; Vulnerable[(15)(16)(17)]	Woodland, Mixed Woodland, Plantation, Shrubland [(23,25,30)]	Woodland*, Mixed Woodland, Plantation*, Semi-natural Watercourse, Modified Watercourse*
Lesser Bamboo Bat (*Tylonycteris fulvida*)	Fairly widely distributed in countryside areas throughout Hong Kong	(LC)[(2)]; Rare[(15)]	Marsh/Reed, Pond, Mixed Woodland, Plantation, Shrubland, Dry Agricultural Land, Village/Orchard, Developed Area/Wasteland [(23,25,30)]	Woodland*, Plantation, Village/Orchard*, Developed Area/Wasteland*, Marsh/Reed, Modified Watercourse
Lesser Bent-winged Bat (*Miniopterus pusillus*)	Fairly widely distributed in countryside areas throughout Hong Kong	-	-	Plantation, Developed Area/Wasteland*
Lesser Yellow Bat (*Scotophilus kuhlii*)	Fairly widely distributed in countryside areas throughout Hong Kong	(LC)[(2)]	Marsh/Reed, Pond, Plantation, Village/Orchard [(30)]	Marsh/Reed, Mixed Woodland, Village/Orchard*, Developed Area/Wasteland*, Natural Watercourse

Common Name (*Scientific Name*)	Distribution in Hong Kong[1]	Protection and Conservation Status	Habitat Recorded	
			Previous Studies	Current Study
Masked Palm Civet (*Paguma larvata*)	Widely distributed in countryside areas throughout Hong Kong, except for Lantau Island and northwestern New Territories	PRC(2); Near threatened(15)	Woodland, Plantation (23,25,30)	Plantation*
Pallas's Squirrel (*Callosciurus erythraeus*)	Fairly widely distributed	-	Woodland, Mixed Woodland, Shrubland, Grassland, Village/Orchard, Developed Area/Wasteland (23,25,27,30)	Woodland, Mixed Woodland*, Shrubland, Plantation*, Village/Orchard*, Developed Area/Wasteland*
Red Muntjac (*Muntiacus vaginalis*)	Very widely distributed in countryside areas throughout Hong Kong	PRC(2); Near threatened(15)	Woodland, Mixed Woodland, Plantation, Shrubland (23,25,26,27,30)	Woodland*, Mixed Woodland*, Shrubland*, Grassland*, Plantation*, Village/Orchard*
Rhesus Macaque (*Macaca mulatta*)	Mainly distributed in Kam Shan, Shing Mun and Tai Po Kau; also found in Ma On Shan, Sai Kung, Tai Lam Country Parks and the North District	Cap. 586(5); Class II(14); Vulnerable(15)	-	Mixed Woodland, Plantation*
Short-nosed Fruit Bat (*Cynopterus sphinx*)	Very widely distributed in urban and countryside areas throughout Hong Kong	-	Woodland, Grassland(23,27)	Woodland*, Plantation, Village/Orchard
Small Asian Mongoose (*Herpestes javanicus*)	Fairly widely distributed in countryside areas in the New Territories	Vulnerable(16)(17)	Marsh/Reed, Woodland, Plantation (25,30)	Plantation
Small Indian Civet (*Viverricula indica*)	Very widely distributed in countryside areas throughout Hong Kong, except for Lantau Island	Class I(14); Vulnerable(16)	Woodland, Plantation, Shrubland (23,25,26,30)	Woodland*, Mixed Woodland*, Shrubland*, Grassland* Plantation*, Pond*
Small-toothed Ferret Badger (*Melogale moschata*)	Widely distributed	-	-	Plantation
Unknown Myotis sp.	-	-	Developed Area(23)	Modified Watercourse
Unknown Vespertilionidae species 1	-	-	Marsh/Reed, Pond, Mitigation Wetland, Mixed Woodland, Mitigation Woodland, Plantation,	Marsh/Reed, Mixed Woodland, Plantation*, Village/Orchard*, Developed Area/Wasteland*, Pond, Modified Watercourse

Common Name (*Scientific Name*)	Distribution in Hong Kong[1]	Protection and Conservation Status	Habitat Recorded	
			Previous Studies	Current Study
			Village/Orchard, Developed Area/Wasteland (23,25,30)	
Unknown Vespertilionidae species 2	-	-	Marsh/Reed, Modified Watercourse, Plantation, Agricultural Land, Grassland, Village/Orchard, Developed Area (23,25,30)	Village/Orchard*, Developed Area*, Modified Watercourse
Butterfly				
Banded Demon (*Notocrypta paralysos*)	Widely distributed throughout Hong Kong	-	Plantation (30)	-
Baron (*Euthalia aconthea*)	Widely distributed throughout Hong Kong	LC(2)	Plantation (30)	-
Blackvein Sergeant (*Athyma ranga*)	Shing Mun, Ngau Ngak Shan, Tai Mong Tsai, Tai Mo Shan, Tai Po Kau, Cloudy Hill	LC(2)	-	Mixed Woodland*, Pond*
Broad Spark (*Sinthusa chandrana*)	Widely distributed throughout Hong Kong	-	Shrubland (23)	-
Common Birdwing (*Troides helena*)	Uncommon	Cap.170(3) Cap.586(5)	Village/Orchard (23)	-
Common Dart (*Potanthus pseudomaesa*)	Hok Tau	LC(2)	Plantation (31)	-
Common Rose (*Pachliopta aristolochiae*)	Widely distributed throughout Hong Kong	-	Village/Orchard (23)	-
Danaid Eggfly (*Hypolimnas misippus*)	Ngau Ngak Shan, Lung Kwu Tan, Hong Kong Wetland Park, Mount Parker, Cloudy Hill, Lin Ma Hang	LC(2)	Marsh/Reed, Plantation, Grassland, Developed Area/Wasteland (25,30)	Grassland*, Plantation
Dark Brown Ace (*Halpe porus*)	Lam Tsuen, Tsiu Hang, Lung Kwu Tan, Ho Sheung Heung	LC(2)	Shrubland (23)	-
Forget-me-not (*Catochrysops strabo*)	Pui O, Tai Po Kau, Fung Yuen, Shing Mun, Sha Lo Wan	-	Mixed Woodland, Plantation, Grassland (25,30)	-

Common Name (*Scientific Name*)	Distribution in Hong Kong[1]	Protection and Conservation Status	Habitat Recorded	
			Previous Studies	Current Study
Grass Demon (*Udaspes folus*)	Widely distributed throughout Hong Kong	Rare(18)	Developed Area/ Wasteland (30)	-
Green Skirt Baron (*Cynitia whiteheadi*)	North New Territories	-	Plantation (30)	-
Little Branded Swift (*Pelopidas agna*)	Widely distributed throughout Hong Kong.	Rare(18)	-	Plantation*
Malayan (*Megisba malaya*)	North Lantau Island	LC(2)	Shrubland (23)	-
Pale Palm Dart (*Telicota colon*)	Widely distributed throughout Hong Kong	LC(2)	Village/Orchard (30)	-
Peacock Royal (*Tajuria cippus*)	Ma On Shan, Wu Kau Tang, Tai Po Kau, Mount Nicholson, Victoria Peak, Cloudy Hill	LC(2)	Village/Orchard (30)	-
Rounded 6-line Blue (*Nacaduba berenice*)	Tai Po, Sha Lo Wan	Rare(18)	Shrubland (23)	-
Silver Royal (*Ancema blanka*)	-	Rare(18)	-	Grassland
Small Three-ring (*Ypthima norma*)	Pak Tam Chung, Ngau Kwo Lo, Kai Shan	LC(2); HR*(33),VR(34)	Grassland (25)	Grassland, Shrubland*, Plantation
Spotted Angle (*Caprona alida*)	Lantau South Country Park, Nam Chung, Lung Kwu Tan, Pat Sin Lang	LC(2); HR*(33)	Grassland (25)	-
Spotted Royal (*Tajuria maculata*)	Tai Po Kau, Shing Mun, Tai Mo Shan	LC(2)	Woodland (30)	-
Tiny Grass Blue (*Zizula hylax*)	Lung Kwu Tan, Fung Yuen, Sha Lo Wan	-	Marsh/Reed, Plantation, Grassland, Developed Area/Wasteland (25,30)	-
Odonate				
Blue Chaser (*Potamarcha congener*)	Widely distributed in the New Territories; Scattered	LC(2)	Modified Watercourse, Developed Area/Wasteland (23,25)	-
Blue-spotted Dusk-hawker (*Gynacantha japonica*)	Widespread in well-shaded woodlands; Sparse	LC(2); MR(33)	Shrubland (23)	-

Common Name (*Scientific Name*)	Distribution in Hong Kong[1]	Protection and Conservation Status	Habitat Recorded	
			Previous Studies	Current Study
Coastal Glider (*Macrodiplax cora*)	Frequents marshes and ponds with dense vegetation, especially adjacent to coastal areas; Sparse	LC(2)	Pond (25)	-
Dingy Dusk-hawker (*Gynacantha subinterrupta*)	Found in well-shaded woodlands. Sparse	LC(2)	Pond, Mixed Woodland (30)	-
Dusk-hawker (*Gynacantha sp.*)	-	LC(2)	-	Plantation
Emerald Cascader (*Zygonyx iris insignis*)	Widely distributed in moderately clean, rapidly flowing forested streams throughout Hong Kong; Widespread	PGC(2)	-	Developed Area/Wasteland
Four-spot Midget (*Mortonagrion hirosei*)	Found mainly in the Northeast and Northwest New Territories and west Lantau; Sparse	GC(2)	Marsh/Reed (25)	-
Mangrove Skimmer (*Orthetrum poecilops poecilops*)	Occurs where small freshwater streams cascade into tidal mangroves. Scattered	GC(2); VU(6)	Marsh/Reed, Watercourse (25)	-
Orange-backed Threadtail (*Prodasineura croconota*)	Widely distributed in well wooded streams throughout Hong Kong; Scattered	LC(2)	Shrubland (31)	-
Ruby Darter (*Rhodothemis rufa*)	Widely distributed in ponds and marshes with dense floating plants; Scattered	LC(2)	Shrubland (31)	Mitigation Wetland
Scarlet Basker (*Urothemis signata signata*)	Common in areas with abandoned fish ponds throughout Hong Kong; Scattered	LC(2)	Grassland (23)	-
White-banded Shadowdamsel (*Protosticta taipokauensis*)	Widely distributed in mature forests with permanent streams throughout Hong Kong; Sparse	GC(2)	Natural Watercourse (30)	-
Herpetofauna				
Amphibian				

Common Name (*Scientific Name*)	Distribution in Hong Kong[1]	Protection and Conservation Status	Habitat Recorded	
			Previous Studies	Current Study
Brown Wood Frog (*Hylarana latouchii*)	Distributed in woodlands in western and central New Territories	LC(2)	Natural Watercourse (30)	-
Chinese Bullfrog (*Hoplobatrachus rugulosus*)	Widely distributed in Hong Kong	PRC(2); Class II(14); Endangered(16); MR(33)	Pond, Natural Watercourse, Semi-natural Watercourse, Modified Watercourse, Plantation, Wet Agricultural Land, Village/Orchard, Developed Area (21,23,30,32)	Developed Area/Wasteland, Pond
Lesser Spiny Frog (*Quasipaa exilispinosa*)	Widely distributed in upland forest streams throughout Hong Kong	PGC(2); Vulnerable(16)(17)	Natural Watercourse (30)	-
Reptile				
Banded Krait *(Bungarus fasciatus*)	Locally restricted in Hong Kong	RC(2); Endangered(15)(16)(17)	-	Developed Area/Wasteland
Buff-striped Keelback (*Amphiesma stolatum*)	Distributed in lowland in central and northern New Territories and Lantau Island	LC(2); MR(33)	Developed Area(29)	-
Burmese Python (*Python bivittatus*)	Widely distributed throughout Hong Kong	PRC(2); Vulnerable(6); Critically Endangered(15)(16)(17)	-	Grassland
Chinese Cobra (*Naja atra*)	Common and widely distributed in Hong Kong	PRC(2); Cap.586(5); Vulnerable(6)(15)(16)(17)	Mixed Woodland, Developed Area (23)	Grassland
Chinese Water Dragon (*Physignathus cocincinus*)	Distributed in the New Territories, Kowloon and Hong Kong Island	Vulnerable(6); Endangered(16)(17)	Plantation (30)	-
Chinese Soft-shelled Turtle (*Pelodiscus sinensis*)	Locally found in reservoirs and fishponds in Deep Bay area	GC(2); Vulnerable(6)(15); Endangered(16)(17); MR(33)	-	Semi-natural Watercourse
Common Rat Snake (*Ptyas mucosus*)	Widely distributed throughout Hong Kong	PRC(2); Endangered(15)(16)(17)	Mitigation Wetland, Village/Orchard (29,30)	-
Copperhead Racer (*Coelognathus radiatus*)	Widely distributed throughout Hong Kong	PRC(2); Class II(14); Endangered(15)(16)(17)	Village/Orchard (30)	-

Common Name (*Scientific Name*)	Distribution in Hong Kong[1]	Protection and Conservation Status	Habitat Recorded	
			Previous Studies	Current Study
Four-clawed Gecko (*Gehyra mutilata*)	Widely distributed throughout Hong Kong	Vulnerable(16)(17)	-	Plantation, Village/Orchard, Developed Area/Wasteland
Indo-chinese Rat Snake (*Ptyas korros*)	Widely distributed throughout Hong Kong	PRC(2); Endangered(15); Vulnerable(16)	Developed Area (25)	-
King Cobra (*Ophiophagus hannah*)	Common and widespread in Hong Kong	PRC(2);Cap. 586(5); Vulnerable(6); Class II(14); Critically Endangered(15) Endangered(16)(17); MR(33)	Grassland/Shrubland (24)	-
Reeves' Turtle (*Mauremys reevesii*)	Widespread in Hong Kong	GC(2); Class II(14); Endangered(6)(16)(17); MR(33)	-	Developed Area/Wasteland
Freshwater Fish and Invertebrate				
Blotched Snakehead (*Channa maculata*)	Uncommon in the wild	HR(33)	-	Semi-natural Watercourse
Cryptopotamon anacoluthon	Common ; Endemic	PGC(2)	Natural Watercourse (27,30)	Natural Watercourse
Gobiopterus macrolepis	Recorded in Deep Bay	Vulnerable(16)	Channelized Watercourse (29)	-
Great Blue Spotted Mudskipper (*Boleophthalmus pectinirostris*)	Common in Deep Bay	Vulnerable(6)	-	Modified Watercourse
Nanhaipotamon hongkongense	Presumably endemic to Hong Kong, but been recently discovered in Shenzhen; Common	PGC(2)	Natural Watercourse (30)	Developed Area/Wasteland, Mitigation Wetland
Small Snakehead (*Channa asiatica*)	Uncommon in the wild.	LC(2); MR(33)	Semi-natural Watercourse (30)	Semi-natural Watercourse*
Somanniathelphusa zanklon	Endemic	GC(2)	Natural Watercourse, Modified Watercourse (21,23,28)	-
Firefly				
Motschulsky's Starworm	Endemic	-	-	Woodland*, Village/Orchard, Developed Area/Wasteland

Common Name (*Scientific Name*)	Distribution in Hong Kong[1]	Protection and Conservation Status	Habitat Recorded	
			Previous Studies	Current Study
(*Rhagophthalmus motschulskyi*)				

Notes:
(1) Distribution in Hong Kong and Rarity follows:
Flora: Wu and Lee (2000); Xing and Chau (2000); Siu (2000).
Fauna: AFCD (2021); Karsen et al. (1998); Shek (2006a); Reels (2019).
(2) Fellowes et al. (2002): GC=Global Concern; LC=Local Concern; RC=Regional Concern; PRC=Potential Regional Concern; PGC: Potential Global Concern.
Letters in parentheses indicate that the assessment is on the basis of restrictedness in nesting and/or roosting sites rather than in general occurrence.
(3) Cap. 170: Protected under Wild Animals Protection Ordinance (Cap. 170).
(4) Cap. 96: Forests and Countryside Ordinance (Cap. 96).
(5) Cap. 586: Protection of Endangered Species of Animals and Plants Ordinance (Cap.586).
(6) IUCN (2025).
(7) Hong Kong Herbarium (2021).
(8) List of Wild Plants Under State Protection (promulgated by the National Forestry and Grassland Administration in 2021)
(9) Fu and Chin (1992). (China Red Data Book, Plant)
(10) Feng et al. (2002). Study on Rare and Endangered Plants and National Key Protected Plants in Guangdong. Journal of South China Agricultural University 3:24-27.
(11) Wu and Hu (1988). (Illustration of Rare and Endangered Plant in Guangdong)
(12) Qin et al. (2017). (Threatened Species List of China's Higher Plant)
(13) Hu et al. (2003) (Rare and Precious Plant in HK)
(14) The List of Endangered and Protected Species of China (promulgated by the National Forestry and Grassland Administration in 2021)
(15) Zheng and Wang (1998). (China Red Data Book)
(16) Jiang et al. (2016). (Red List of Chinese Vertebrate)
(17) Ministry of Ecology and Environment & Chinese Academy of Sciences (2023). Redlist of China's Biodiversity. (In Chinese only).
(18) Lee, H.Y., Chan, C.H., Chan, C.M., Chow, W.S., Mak, H.W., Shue, P.H., So, C.F. and Wong, K.Y. (2026). An Updated Checklist of Butterflies of Hong Kong. Hong Kong Biodiversity. Issue No. 28: 11-19.
(19) Wetland-dependent species (including wetland-dependent species and waterbirds).
(20) AFCD (2010) (Hong Kong Biodiversity Issue No. 19)
(21) CEDD (2010)
(22) CEDD and PlanD (2022)
(23) CEDD and PlanD (2024)
(24) Glory Queen Limited (2016)
(25) MTR (2024a)
(26) MTR (2024b)
(27) WSD (2024
(28) DSD (2018)
(29) DSD (2021)
(30) CEDD (2025)

(31) MTR (2009)
(32) CEDD (2013)
(33) Or and Chan (2025). The State of Hong Kong Biodiversity 2025: MR = Moderate Risk of Extinction in Hong Kong species status assessment; HR = High Risk of Extinction in Hong Kong species status assessment; with a single asterisk (*) signifying a one-step change and two asterisks (**) indicating a two-step change.
(34) AFCD (2025)
(35) Habitat with * = recorded within Project Boundary, or recorded within both Project Boundary and Assessment Area

8.8 Identification of Potential Ecological Impacts

8.8.1 Construction Phase – Direct Impact

Direct Loss of Habitat and Vegetation

8.8.1.1 Direct impacts on terrestrial and aquatic habitats would arise from the proposed alignment, including at-grade works (e.g. roads, slope works, tunnel portals, ancillary facilities, etc.), viaduct supports (e.g. pile and pier-related works), and associated temporary works (e.g. works area, temporary access, and storage areas) with the Project boundary.

8.8.1.2 Around two-thirds of the areas that are subject to direct habitat loss would result from temporary works such as works areas, temporary access (e.g. haul roads), and storage area. These temporary works areas would be released and reinstated upon completion of the construction works as far as practicable. Only a relatively small areas of habitats would be lost due to the proposed alignment (i.e. at-grade works and viaduct supports). The extent of direct habitat loss associated with the proposed alignment and temporary works within the Project Boundary is presented in **Table 8.14** below.

8.8.1.3 Due consideration was given to the land requirements for both temporary at-grade construction activities and permanent at-grade structures during the development of the layout design for the Project. Sensitive ecological resources were identified early in the ecological baseline study, leading to corresponding adjustments to the alignment and layout of associated works. These adjustments included opting for tunnel and/or viaduct sections to minimise at-grade works and fine-tuning the alignment to minimise encroachment on sensitive areas. This approach aimed to avoid direct impacts on the concerned ecological resources or keep them to a minimum where possible.

8.8.1.4 A substantial portion of the direct loss area would consist of man-made habitats of low ecological value, such as developed area/wasteland, grassland, and village/orchard habitats. In contrast, wetland habitats including marsh/reed, pond, semi-natural watercourse, and modified watercourse would account for only a small proportion of the total direct loss area during the construction phase of the Project.

Table 8.14 Direct Loss of Terrestrial and Aquatic Habitats within Project Boundary during Construction Phase

Habitat	Direct Loss (ha) from proposed alignment (including at-grade works (i.e. roads, slope works, tunnel portals, ancillary facilities, etc) and viaduct supports (i.e. pile and pier related works))	Direct Loss (ha) from Temporary Works (i.e. works area, access and storage area) (1) (2)	Total Direct Loss (ha)
Woodland	10.35	6.28	16.63
Mixed Woodland	3.74	9.02	12.76

Habitat	Direct Loss (ha) from proposed alignment (including at-grade works (i.e. roads, slope works, tunnel portals, ancillary facilities, etc) and viaduct supports (i.e. pile and pier related works))	Direct Loss (ha) from Temporary Works (i.e. works area, access and storage area) (1) (2)	Total Direct Loss (ha)
Shrubland	8.68	8.26	16.94
Grassland	14.81	10.17	24.98
Plantation	3.16	12.89	16.05
Village/Orchard	7.86	18.60	26.46
Developed Area/Wasteland	10.93	49.03	59.96
Agricultural Land	0.04	0.72	0.76
Seasonally Wet Grassland	-	-	-
Marsh/Reed	0.06	1.07	1.13
Pond	Ponds near NTMWTW: 0.05 Other Ponds: 0.02	Ponds near NTMWTW(3): 0.75 Other Ponds: 0.36	Ponds near NTMWTW: 0.80 Other Ponds: 0.38
Natural Watercourse	0.03 (0.30 km)	Negligible	0.03 (0.30 km)
Semi-natural Watercourse	0.11 (0.41 km)	0.59 (1.33 km)	0.70 (1.74 km)
Modified Watercourse	KTMDC: 0.03 (0.05 km) Other Modified Watercourses: Negligible	KTMDC(3): 0.85 (1.00 km) Other Modified Watercourses: 0.66 (0.05 km)	KTMDC: 0.88 (1.05 km) Other Modified Watercourses: 0.66 (0.05 km)
Mitigation Wetland	-	-	-
Total (ha)	**59.87**	**119.25**	**179.12**

Notes:

(1) Within the Project Boundary, approximately 1 ha of wetland habitats (including marsh/reed, pond, and natural watercourse habitats) and 0.05 ha of woodland habitat are overlapping with the concurrent Development at Ngau Tam Mei project. The loss of these habitats (i.e. marsh/reed, pond, natural watercourse and woodland habitats) has already been assessed and addressed under the approved EIA for the concurrent project. Thus, they are excluded from the present calculation of habitat loss for this Project.

(2) As "no work zone" is incorporated into the Project design to delineate areas where no construction activities will be undertaken, direct impact to habitats within this area is not expected, hence, the area is excluded from the calculation.

(3) To minimise potential impact to the ponds near NTMWTW and the KTMDC, phasing of construction works would be adopted in the area (i.e. pond habitat near NTMWTWT and KTMDC) with actual construction activities would not occupying more than half of the area designated for temporary works, at any given time, and the remaining half of the area would be maintained as pond habitat or watercourse. The impacted area by temporary works will be reinstated to their pre-existing condition as far as practicable prior to the commencement of construction works on

the remaining half of the area. On this basis, only half of these areas are counted as direct loss in this context

Direct Impact on Recognised Sites of Conservation Importance

8.8.1.5 Within the Project Boundary, four recognised sites of conservation importance have been identified, namely Conservation Area (CA) under OZP No. S/YL-KTN/11, S/YL-NTM/15, and S/STT/2; the Wetland Conservation Area (WCA); the Wetland Buffer Area (WBA); and the Deep Bay Wetland outside Ramsar Site Priority Site for Enhanced Conservation. While Lam Tsuen Country Park (LTCP) lies directly adjacent to the Project Boundary, careful design measures have been adopted to avoid direct encroachment into this site of conservation importance.

8.8.1.6 Part of the Project Boundary encroaches on CA to the west of Kai Kung Leng (OZP No. S/YL-KTN/11), on Ngau Tam Shan (OZP No. S/YL-NTM/15), and on Hadden Hill (S/STT/2) (**Figures 8.1** and **8.5** refer). For the CA west of Kai Kung Leng, the proposed works include viaduct and at-grade roads, along with associated slope works. This area of CA is in close proximity to developed area / wasteland habitat, which is currently subject to disturbance from existing human activities and highway traffic. In the case of Ngau Tam Shan and Hadden Hill, there are minor areas of habitat loss, specifically at the tunnel portal near NTMWTW, the slip road near Ki Lun Tsuen, and the associated temporary works area. Nonetheless, the direct loss resulting from the proposed alignment (including at-grade works and viaduct supports) is limited. The temporary works areas are anticipated to be of limited extent for storage areas and temporary access, and will be reinstated upon completion of construction as far as practicable. Furthermore, the alignment at Ngau Tam Shan and Hadden Hill primarily comprise tunnel sections, which significantly minimises the loss of aboveground habitats and redconuce disturbance. Consequently, the anticipated direct impact to CA is expected to be low in the absence of mitigation measures.

8.8.1.7 Part of the Project Boundary also encroaches upon the eastern edge of the WBA (to the east of Pok Wai and Nam Sang Wai) and a small area and the southern corner of the WCA and Priority Site for Enhanced Conservation (**Figures 8.1** and **8.5.2** to **8.5.3** refer). The habitats within these encroached areas are mainly limited to developed area/wasteland and roadside plantation along the existing San Tin Highway and Yuen Long Highway, and small sections of modified watercourse, including Kam Tin River and culverts. The proposed works in this encroached area mainly involve temporary works adjacent to the proposed alignment. The direct loss resulting from the proposed alignment will be confined to pile caps and viaduct structure on man-made habitats, namely developed area/wasteland, roadside plantation, and modified watercourse habitat only. These habitats are immediately adjacent to existing highways and are currently subject to regular disturbance from traffic and human activities. In addition, “no work zone” has been incorporated into the Project design along the Project Boundary and the wetland habitats (e.g. modified watercourse) in the WCA and Deep Bay Wetland outside Ramsar Site Priority Site for Enhanced Conservation. No construction works would be carried out in these natural habitats. Considering the encroachment is limited to developed area/wasteland, roadside plantation, and modified watercourse, along with the small scale of direct loss from the proposed alignment (such as pile caps), the anticipated direct impact on the recognised sites of conservation importance – the

WCA, WBA, and Priority Sites - is anticipated to be low in the absence of mitigation measures.

Direct Impact on Other Ecologically Sensitive Resources

8.8.1.8 Direct impact to the other ecologically sensitive resources (e.g. IBA, Sha Po Night Roost, Ko Po Night Roost, planned bat shelter, Sha Po Marsh, YLBFEW, proposed Nam Sang Wai Wetland Conservation Park, planned compensation wetland for NTM Development Area and planned wetland compensation site in Kam Tin for the Northern Link) are avoided. Nonetheless, direct impact on KTMDC is unavoidable, the direct impact on KTMDC is detailed in **Section 8.8.1.18** below. The direct impact on other ecological sensitive resources is anticipated to be low.

Direct Impact on Terrestrial Habitats and Associated Wildlife

Loss of Wooded and Vegetated Habitats

8.8.1.9 **Woodland**: A total of 10.35 ha of woodland habitat in Ngau Tam Shan and Kai Kung Leng would be subjected to direct loss due to the proposed alignment, including at-grade works and pile and pier-related works. Additionally, 6.28 ha of woodland would be impacted by temporary construction works, and are expected to be of limited extent for storage areas and temporary access. The impacted area from temporary works would also be reinstated by planting of native tree species as far as practicable. The woodland habitat is structurally connected to adjacent wooded habitats and is partially located within the CAs under OZP No. S/YL-NTM/15 and S/STT/2. The directly affected woodland at Kai Kung Leng is also situated adjacent to the LTCP and CA under OZP No. S/YL-KTN/11. The woodland habitat is relatively intact and mature, supporting a moderate diversity and abundance of flora and fauna, including species of conservation importance, and is considered to have moderate ecological value. Considering the maturity and ecological value of this habitat, as well as the size of the affected area, the anticipated direct impact on the woodland is expected to be low to moderate, if unmitigated.

8.8.1.10 **Mixed Woodland**: A total of 3.74 ha of mixed woodland habitat on Hadden Hill, Ki Lun Tsuen, and Ngau Tam Mei area, would be subjected to direct loss due to the proposed alignment, while about 9.02 ha of habitat would be impacted by temporary works. Similar to the woodland habitat, the mixed woodland within the temporary works area is also considered to be lost. The areas affected by the Project primarily consist of the fringes of the mixed woodland, where the existing disturbance level is high due to edge effects and proximity to heavily trafficked roads, such as the adjacent Fanling Highway and village areas. The ecological value of mixed woodland is considered as low to moderate. While mixed woodland would be subjected to direct loss, considering its existing disturbance and fragmented nature, and with the impacted area from temporary works expected to be reinstated by planting of native tree species as far as practicable, the direct impact to mixed woodland is considered to be low.

8.8.1.11 **Shrubland**: A total of 8.68 ha shrubland habitat in Tit Hang, Hadden Hill, Ki Lun Tseun, and next to NTMWTW would be subjected to direct loss due to the proposed alignment; while about 8.26 ha of the habitat would be subjected to impact from temporary works. The affected hillside shrubland is structurally and functionally linked to adjacent wooded habitats. Considering the generally low to moderate diversity and low abundance of wildlife, the relatively young age of the habitat, and

its low to moderate ecological value, with the impacted area from temporary works is expected to be reinstated by planting or hydroseeding as far as practicable, the direct impact to shrubland is considered to be low.

8.8.1.12 **Grassland:** A total of 14.81 ha of grassland in Tit Hang, Hadden Hill, Ki Lun Tsuen, NTMWTW, and Kai Kung Leng would be subjected to direct loss due to the proposed alignment. Additionally, about 10.17 ha of this habitat would also be impacted by temporary construction activities. Nonetheless, the impacted area from temporary works is expected to be reinstated by planting or hydroseeding as far as practicable. Given that the floral structure of the grassland is relatively simple and mainly comprised common herbaceous species, along with its low potential value and low and low-to-moderate ecological values, the direct impact to grassland is considered to be low.

8.8.1.13 **Plantation:** A total of 3.16 ha of plantation habitat in Tit Hang, Ngau Tam Mei, Kai Kung Leng, and along Tsing Long Highway would be subjected to direct loss due to the proposed alignment. Additionally, about 12.89 ha of the habitat would be subjected to impact from temporary construction activities. While some flora species of conservation importance have been recorded within the hillside plantation in Kai Kung Leng, the remaining roadside plantation areas are of relatively low ecological values, characterised by high existing disturbance level due to proximity to roads of heavy traffic. Considering that plantation habitat is common and has low naturalness in Hong Kong, the direct impact on both hillside and roadside plantations is deemed to be low. Nonetheless, efforts will be made to reinstate the impacted areas by planting of native tree species as far as practicable following temporary works.

Loss of Wetland Habitats and Watercourses

8.8.1.14 **Marsh/Reed:** A total of 0.06 ha of marsh/reed habitat would be subjected to direct loss due to the proposed alignment, including at-grade works and the proposed pile and pier-related works in Ngau Tam Mei area. Additionally, about 1.07 ha of this habitat may be subjected to impact from temporary construction activities, for example by being decked over to facilitate the pile and pier installation. The ecological value of the marsh/reed habitat is assessed as low to moderate, with generally low to moderate diversity and abundance of wildlife recorded. The species are predominantly common and widespread across Hong Kong. Furthermore, the habitat is fragmented and subject to disturbance from adjacent developed area/wasteland and village/orchard environments. Considering the small size of the affected area, the direct loss of marsh/reed habitat is therefore considered to be of low to moderate significance, if unmitigated.

8.8.1.15 **Pond:** A total of 0.07 ha of pond habitat, including associated riparian vegetation, would be subjected to direct loss due to the proposed alignment (i.e. 0.05 ha near NTMWTW and 0.02 ha at Ki Lun Tsuen). In addition, ponds near NTMWTW (0.75 ha), north of Kwu Tung Reservoir, and at Ki Lun Tsuen (about 0.36 ha) would be subjected to impact from temporary works such as temporary access, storage areas, and decking to facilitate pile and pier works. These temporarily impacts are expected to be of minor significance. The ponds near NTMWTW are considered to have low to moderate ecological value, owing their relative intactness and connectivity with other wetlands, including adjacent ponds, marsh/reed habitats, and tributaries of NTMDC (i.e. NTM-S4 and NTM-S4a). In the absence of mitigation, the direct impact on ponds near NTMWTW is considered to be low to moderate. Mitigation measures are outlined in **Section 8.11**, including phasing of works to

minimise the area affected at any one time, as well as wetland compensation and enhancement measures for unavoidable losses. On the other hand, the other affected pond habitats (e.g. those at Ki Lun Tsuen and North of Kwu Tung Reservoir) are relatively small, fragmented, and of low ecological value. Nonetheless, these other ponds would be reinstated after the completion of construction works as far as practicable. Considering the limited size of these areas and would be reinstated after the completion of construction works, the direct impact is anticipated to be low.

8.8.1.16 **Natural Watercourse:** The natural watercourse NTM-S3, located south of NTM, was identified along the proposed NMH alignment and within the Project Boundary. The proposed pile cap locations have been refined and adjusted to avoid direct encroachment. In addition, an exclusion zone will be established along the natural watercourse, including the riparian vegetation, to avoid encroachment by construction activities during the construction phase. A total of 0.03 ha of natural watercourse, comprising seasonal waterlines, would be subjected to direct loss due to proposed alignment at the foothill of Ngau Tam Shan (i.e. ST-S15, ST-S16 and ST-S17). Considering the small size of affected area, the direct impact on these seasonal waterlines is anticipated to be low.

8.8.1.17 **Semi-natural Watercourse:** Several semi-natural watercourses, primarily located south of NTM, as well as Ki Lun Tsuen and Tit Hang, fall within the Project Boundary. Although the proposed pile cap locations have been refined to minimise direct impacts, some unavoidable direct loss, including riparian vegetation, due to the proposed alignment would arise from the pile and pier related works, at-grade constructions (including the proposed interchange and site office south of Ngau Tam Mei Tsuen, i.e. NTM-S5), and associated temporary works. A total of 0.11 ha of semi-natural watercourse would be subjected to direct loss due to the proposed alignment, while 0.59 ha would be subjected to impact from temporary works. To mitigate these impacts, water diversion will be implemented in at-grade works areas to the west of NTMWTW and would be reinstated as far as practicable, specifically in the upstream section of NTM-S4a, to maintain its water flow. The riparian habitat would be reinstated by planting or hydroseeding to its pre-existing condition as far as practicable. Overall, the affected semi-natural watercourses are small in size, already highly disturbed by anthropogenic activities, and considered to have low ecological value. Consequently, the impact from the loss of the semi-natural watercourse habitat is assessed as low.

8.8.1.18 **Modified Watercourse:** Modified watercourse identified within the Project Boundary include KTMDC, as well as minor culverts and channels near Pok Wai and south of NTMWTW. Direct impacts are expected along the alignment across KTMDC, including direct habitat loss due to the proposed alignment arising from the proposed pile and pier structures, while temporary impacts may arise from associated construction activities, such as temporary decking during the construction phase. A total of 0.03 ha of modified watercourse would be subjected to direct loss due to proposed alignment, while 0.85 ha of KTMDC and 0.66 ha of other modified watercourses would be subjected to impact from temporary works. Any temporary decking will be removed upon the completion of construction. KTMDC is considered to have moderate ecological value, supported by the moderate diversity and abundance of avifauna, particularly waterbirds recorded during dry season. Accordingly, the impact from direct loss due to the proposed alignment and temporary impact across the KTMDC is considered to be low to moderate, if unmitigated. Minimisation measures are further described in **Section 8.11**, including phasing of works to reduce the area affected at any given time. The

minor culverts near Pok Wai and south of NTMWTW are of low ecological value and would only experience temporary impact during the construction. The resulting low ecological impact is therefore expected to be low.

Loss of Man-made Habitats

8.8.1.19 **Agricultural Land:** A total of 0.04 ha of agricultural land in Ngau Tam Mei area would be subjected to direct loss due to the proposed alignment such as proposed at-grade works and pile and pier related works. In addition, about 0.72 ha of this habitat would be subjected to impact from temporary construction activities. This habitat was generally small and scattered, and under active management. While it supports low floral diversity, most of the vegetation is artificially introduced, consisting primarily of dry crops and fruit tree species. The land is utilised by low diversity and abundance of common fauna species. Considering the limited size of agricultural land directly impacted by permanent structures and temporary works the ecological impact of its loss is assessed as low. The impacted area by temporary works would be reinstated by planting or hydroseeding as far as practicable.

8.8.1.20 **Village/Orchard:** About 7.86 ha of village/orchard habitat in Ki Lun Tsuen and Ngau Tam Mei area would be subjected to direct loss due to the proposed alignment, whereas about 18.60 ha of the habitat would be subjected to impact from temporary construction activities. This is a man-made habitat and most of the vegetation recorded were common, artificially introduced, or horticultural species. Considering its low ecological value and the man-made nature of the habitat, which is subject to constant anthropogenic disturbance, the ecological impact from the direct loss of this village/orchard habitat is considered as low. The impacted area by temporary works would be reinstated by planting or hydroseeding as far as practicable.

8.8.1.21 **Developed Area/Wasteland:** A total of 10.93 ha of developed area/wasteland would be subjected to direct loss due to the proposed alignment, whereas about 49.03 ha of the habitat would be affected by temporary construction activities. This man-made habitat represent the largest proportion of habitat loss under the Project. While the affected area is relatively extensive, most of the vegetation recorded consists of common, artificially introduced, or horticultural species, primarily native ruderal and exotic species. Considering the high degree of anthropogenic disturbance and the low ecological value of this habitat, the ecological impact from direct loss of developed area/wasteland is considered low. The impacted area by temporary works would be reinstated by planting or hydroseeding as far as practicable.

Direct Impact on Associated Wildlife

8.8.1.22 Given the generally high mobility of avifauna, mammal, butterfly, odonate and reptile species, which are mostly widespread and common species in Hong Kong, and the presence of similar habitats adjacent to the Project Boundary, construction activities are not expected to result in major direct injury or mortality to these faunal groups. The natural watercourses to be impacted by the Project comprise small scale seasonal waterlines supporting limited freshwater community species, while only parts of the semi-natural watercourses and modified watercourses would be directly impacted by the Project. Given the natural watercourses support only limited freshwater community species and the availability of remaining sections of the impacted watercourses, major direct impacts on freshwater community species in watercourses are not anticipated. The majority of the recorded amphibian

species, except the species of conservation importance, are widely distributed species in Hong Kong and were recorded with scarce or occasional abundance within the Project Boundary. Accordingly, the direct impact to the amphibian species is anticipated to be low. Two firefly species were recorded within the Project Boundary, namely Rimmed Window Firefly which was recorded in scarce abundance within the Project Boundary and a firefly species of conservation importance, i.e. Motschulsky's Starworm which was recorded occasionally within the Project Boundary. The Rimmed Window Firefly is a widespread species in Hong Kong and commonly occurs in wide range of habitats. Given the low abundance of this species recorded within the Project Boundary, potential direct impact to this species is anticipated to be low. Potential direct impacts to the species of conservation importance are further assessed in **Section 8.8.1.26** to **8.8.1.36** below.

Fragmentation of Terrestrial Habitats

8.8.1.23 Fragmentation refers to the discontinuity of habitat, reducing the carrying capacity and hindering the movement of wildlife species, resulting in less favourable habitat for flora and fauna species. Fragmentation of habitat may also result in isolation of some population of species in the area.

Fragmentation and Isolation of Vegetated Areas

8.8.1.24 Under the Project, the at-grade works at Kai Kung Leng, Ngau Tam Shan and Hadden Hill may potentially lead to fragmentation of existing vegetated habitats by obstructing the movement of associated wildlife, such as non-flying mammals and herpetofauna species, and limiting access to their foraging grounds. Nonetheless, the proposed works are largely situated along the margins of the wooded habitats. Moreover, a significant portion of the alignment in Ngau Tam Shan, as well as a section in the southeast of Hadden Hill, has been designed as underground tunnels, effectively minimising and avoiding direct impact and fragmentation to the wooded habitats. As such, the potential fragmentation and isolation impacts arising from the Project during the construction phase is expected to be low.

Direct Impact on Species of Conservation Importance

Flora

8.8.1.25 As described in **Section 8.6.2**, ten flora species of conservation importance were recorded within the Assessment Area, based on both previous studies and recent surveys. Most of these recorded flora species of conservation importance were located outside the Project Boundary; however, four flora species of conservation importance (i.e. *Aralia chinensis*, Cycad-fern, Incense Tree and Hong Kong Pavetta) were recorded within the Project Boundary. In the absence of mitigation measures, the Project would result in the direct loss of these flora species of conservation importance, which is assessed as a low to moderate impact. Appropriate mitigation measure, such as transplantation, are further discussed in **Section 8.11.** In addition, a “no work zone’ will be implemented in woodland habitat to the east of Pok Wai, thereby avoiding direct impacts on flora species of conservation importance located in this area.

Avifauna

8.8.1.26 A total of 33 avifauna species of conservation importance were recorded within the Project Boundary, based on both literature review and recent surveys. These include waterbirds, forest birds, and common resident generalist in different habitats. No notable breeding sites for these avifauna species were identified within the Project Boundary. Given the high mobility of avifauna species and the availability of similar habitats adjacent to the Project Boundary (e.g. LTCP), construction activities are not expected to result in major direct injury or mortality. Hence, direct impacts on the avifauna species are expected to be low.

8.8.1.27 One avifauna roosting site, namely Sha Po Night Roost, was identified within the Project Boundary along Kam Tin River, approximately 25 m from the proposed alignment. No direct loss is anticipated at Sha Po Night Roost due to the proposed alignment. Temporary works will be carefully designed and planned to avoid direct encroachment on the roost. As a result, direct impacts on this night roost are not anticipated.

Mammal

8.8.1.28 A total of 12 flying mammal species of conservation were recorded from both literature review and recent surveys. These are all bat species, recorded from various habitats within the Project Boundary, frequently in village/orchard, developed area/wasteland, and woodland. These bat species were mostly recorded foraging and commuting above the Assessment Area. One particular roosting site, namely Pok Wai bat roost, was recorded within the Project Boundary. This roost is subjected to direct loss under the Northern Link, with a bat shelter proposed to mitigate the loss (MTRCL, 2024a), and is located within the Project Boundary. The alignment and works area have been carefully designed and avoid direct impact to the planned bat shelter. No direct impact is anticipated on the flying mammal species as well as the planned bat shelter.

8.8.1.29 Seven non-flying mammal species of conservation importance were recorded within the Project Boundary, mostly on hillside habitats such as woodland and mixed woodland. Two species, namely East Asian Porcupine and Small Indian Civet were recorded within the Project Boundary, but outside the footprint of at-grade works and pile and pier works. Considering the high mobility of the species, the proposed temporary works are not expected to result in direct injury or mortality, while direct impact on other mammal species outside the Project Boundary is also not anticipated. The loss of associated habitats and foraging ground is considered to be low to moderate for woodland habitat while low for mixed woodland habitat which has been discussed in **Section 8.8.1.9** and **Section 8.8.1.10**.

8.8.1.30 On the other hand, during the ecological surveys, no field signs or individuals of Eurasian Otter (*Lutra lutra)* were recorded within the Project Boundary or Assessment Area. Thus, direct impact to Eurasian Otter is not anticipated.

Butterfly

8.8.1.31 A total of 11 butterfly species of conservation importance were recorded within the Project Boundary from both literature review and recent survey. These rare or very rare species were recorded mostly on grassland within the Project Boundary. These species were not recorded with particular breeding ground or host plants within the Project Boundary. Given the high mobility and availability of similar habitats

adjacent to the Project Boundary, potential direct impact due to direct injury or mortality of butterfly species of conservation importance and loss of associated habitats is considered to be low.

Odonate

8.8.1.32 Two odonate species of conservation importance were recorded within the Project Boundary from literature review. These odonate species of conservation importance were recorded with low abundance in mixed woodland, plantation and village/orchard within the Project Boundary. Given the low abundance and high mobility of these species, and availability of similar habitats adjacent to the Project Boundary, potential direct impact due to direct injury or mortality of odonate species of conservation importance and loss of associated habitats is considered to be low.

Herpetofauna

8.8.1.33 One amphibian species of conservation importance (Chinese Bullfrog) was recorded in village/orchard from literature review within the Project Boundary, which is also within the proposed NTM Development Area. The recorded individuals would be subject to translocation under the aforementioned project (CEDD, 2025). Individuals of Chinese Bullfrog was also recorded from current survey, but outside the Project Boundary. No other breeding or nursery ground of these amphibian species of conservation importance was observed. As such, no direct injury or mortality is anticipated on this species of conservation importance.

8.8.1.34 Two reptile species of conservation importance (Chinese Cobra and Chinese Water Dragon) were recorded from literature review in mixed woodland and plantation within the Project Boundary, but outside the at-grade works and pile and pier works. Temporary works during the construction stage may potentially affect the recorded species. Nonetheless, both species were recorded in low abundance and of high mobility, which are expected to move away from the source of disturbance. No notable breeding or nursery ground of reptile species of conservation importance was observed. Potential direct impact due to direct injury or mortality of reptile species of conservation importance is considered low.

Firefly

8.8.1.35 A general low abundance of fireflies was recorded, with only one species of conservation importance, Motschulsky's Starworm, recorded in woodland at Ki Lun Tsuen. Although individuals were recorded within the Project Boundary, they were located outside the footprint of any at-grade works or pile and pier structures. Due to the low mobility of the flightless female Motschulsky's Starworm, potential temporary impacts and disturbances may occur during the construction stage. Direct injury or mortality on this species could result in low to moderate ecological impact. To avoid unnecessary habitat loss, a “no work zone” will be designated within part of the woodland habitat at Ki Lun Tsuen where Motschulsky's Starworm was recorded. Appropriate mitigation measures, such as pre-construction surveys and translocation, are further detailed in **Section 8.11.**

Aquatic Communities

8.8.1.36 One freshwater fish species of conservation importance, Small Snakehead, along with another freshwater crab species of conservation importance (*Somanniathelphusa zanklon*), was recorded in the semi-natural watercourse and

modified watercourse within the Project Boundary. *Somanniathephusa zanklon* was identified only through literature review in the at-grade road works area near Fanling Highway along Kwu Tung Road. However, based on the current survey, this location has been heavily concretised and subjected to heavy traffic, with no individuals of *Somanniathelphusa zanklon* recorded. Therefore, ecological impacts on this freshwater crab species are no longer expected. The other Small Snakehead was recorded outside the proposed at-grade works and pile/pier structures but within the temporary works area (NTM-S4). In the absence of mitigation, low to moderate ecological impacts may arise from potential direct impacts, such as direct injury or mortality. Appropriate mitigation measures, including pre-construction surveys and translocation, are further detailed in **Section 8.11.**

Bird Collision

8.8.1.37 Construction activities involving heavy machineries (e.g. cranes), installations of noise barriers, and building facades with highly transparent or reflective materials may pose risks of bird collisions during the construction phase. Of particular concern are the viaduct and the associated pile and pier works along Tsing Long Highway crossing KTMDC, as flight line surveys recorded ardeids and other waterbirds frequently using KTMDC as major flight corridor. Nonetheless, the proposed viaduct sections crossing KTMDC under this Project would be of similar height to the existing viaduct structures of Tsing Long Highway and Yuen Long Highway. Survey observations indicate that avifauna species are generally able to avoid direct collision with these existing structures, flying either above or below them. Thus, in the absence of mitigation, the potential bird collision impact is assessed as low to moderate. To further reduce risks, appropriate mitigation measures will be implemented, including the use of non-transparent or non-glaring materials. These measures are descried in detail in **Section 8.11**.

8.8.2 Construction Phase – Indirect Impact

Disturbance Impact to Recognised Sites of Conservation Importance

8.8.2.1 Construction activities will be confined within the Project Boundary. However, temporary disturbance arising from construction, such as noise, glare, dust, vibration, and other human activities, may generate indirect impacts on the surrounding areas. These effects are of particular concern for nearby natural habitats, especially those located within recognized sites of conservation importance and other ecologically sensitive areas.

Deep Bay Wetland outside Ramsar Site Priority Site for Enhanced Conservation

8.8.2.2 The Priority Site is located within Nam Sang Wai area and extends partially into the Project Boundary. Within the Project Boundary, however, the habitats present are limited to developed area/wasteland and modified watercourse. All natural habitats within the Priority Site lie outside the Project Boundary. Furthermore, existing disturbances can be observed, including traffic along multiple viaducts across KTMDC (i.e., Yuen Long Highway and Tsing Long Highway). Consequently, construction-related disturbance impacts on the Priority Site are assessed as low. To further minimise potential impacts, mitigation measures such as the installation of noise barriers and proper site control will be implemented.

LTCP and CA

8.8.2.3 The LTCP lies outside the Project Boundary, south of the proposed alignment. Certain at-grade works area and pile and pier works are directly adjacent to woodland habitats within the LTCP at Kai Kung Leng. In the absence of mitigation, disturbance of low to moderate impact may occur, particularly affecting woodland habitats of moderate ecological value. The Project alignment has been refined and carefully designed, with strict control over site boundary and access (**Section 8.11** refers). Without mitigation, disturbance impact to the LTCP is considered low to moderate. With the implementation of additional minimisation measures, such as noise barriers and stringent site control, adverse ecological impacts from construction disturbance are not anticipated.

8.8.2.4 Several CAs in Hadden Hill, Ngau Tam Shan, and Kai Kung Leng may be subject to construction disturbance. As discussed in **Section 8.8.1.6**, the alignment primarily comprises underground tunnel across Ngau Tam Shan, thereby minimising direct loss and disturbance to aboveground habitats (e.g. grassland and shrubland). The remaining CA areas within the Project Boundary are mostly temporary works areas, expected to involve construction activities of low intensity such as use as storage areas and temporary access. Furthermore, the CA zones are already subject to existing disturbance from the nearby developed area/wasteland habitats and tomb-sweeping activities. As such, additional disturbance impacts to these sites of conservation importance are assessed to be low.

WCA and WBA

8.8.2.5 Within the Project Boundary, a small corner of the WCA is situated on the southeastern edge of the Nam Sang Wai area, while a small portion of the WBA is situated along the west side of the existing San Tin Highway at Wing Kei Tsuen. The WCA and WBA within the Project Boundary are mainly composed of developed area/wasteland, roadside plantations, and modified watercourses, all of which are already subject to regular disturbances such as heavy traffic and road maintenance. Outside the Project Boundary, the WCA and WBA extend westward within the Assessment Area. The WCA in this area includes wetland habitats such as marsh/reed, ponds, and modified watercourses (e.g. as part of Nam Sang Wai), alongside disturbed habitats such as developed areas. The WBA within the Assessment Area is mainly composed of developed areas and wasteland. The proposed works footprint within the WCA and WBA is small in scale and limited to temporary works areas (e.g. material storage or road access) located within already disturbed habitats such as developed area/wasteland and plantation habitats. Notably, “no work zone” has been incorporated into the Project design along the Project Boundary and applied to the wetland habitats (e.g. modified watercourses) within the WCA, thereby minimising disturbance impacts to nearby wetland habitats (**Section 8.11** refers). As a result, disturbance impacts to the WCA and WBA are considered low.

Disturbance Impact to other Ecologically Sensitive Resources

Inner Deep Bay and Shenzhen River Catchment Important Bird Area (IBA)

8.8.2.6 The IBA is located largely outside the Project Boundary, with the nearest point approximately 400 meters away at Nam Sang Wai. While some pond habitats within the Nam Sang Wai area form part of the IBA, the majority of the IBA lies at a considerable distance. Additionally, developed area/wasteland habitats, such as highways and residential areas, form a buffer zone between the IBA and the Project

Boundary. As such, the potential disturbance impact from the Project on the IBA is considered as very low.

Roosting Sites

8.8.2.7 The Sha Po Night Roost was identified within the Project Boundary along KTMDC, approximately 25 m east of the proposed viaduct. The Ko Po Road Night Roost is located about 360 m southeast, outside the Project Boundary. Both night roosts are located within urbanised area and are directly adjacent to human activities and disturbance, including heavy traffic along existing highway and residential area with nighttime lighting. Nonetheless, the presence of roosting ardeids means that the construction activities nearby could reduce usage of the Sha Po Night Roost, particularly during the overwintering season, potentially resulting in low to moderate ecological impacts in the absence of mitigation measures. Specific mitigation measures for the Sha Po Night Roost, such as establishing buffer zones and applying timing controls to construction activities, are further described in **Section 8.11**. Considering the distance of the Ko Po Road Night Roost from the Project Boundary and the existing level of disturbance, the ecological impact on this roost is anticipated to be low.

Sha Po Marsh

8.8.2.8 Sha Po Marsh is located adjacent to both the Project Boundary and the existing San Tin Highway, where it is already subject to disturbance from heavy traffic. The portion of the Project Boundary directly adjoining Sha Po Marsh is limited to a temporary works area, which currently consists of developed area / wasteland and plantation habitats. Furthermore, a designated “no work zone” is proposed at the planned wetland compensation site under Ngau Tam Mei Development (further detailed in **Section 8.11.3.5**). This zone lies between the proposed alignment and Sha Po Marsh, providing an additional buffer. As such, the additional disturbance associated with the Project is expected to result in a low ecological impact on Sha Po Marsh.

Yuen Long Bypass Floodway Engineered Wetland (YLBFEW)

8.8.2.9 The YLBFEW is situated about 40 m from the Project Boundary within the Nam Sang Wai area. Direct impacts on the wetland have been avoided, and activities within the adjacent Project Boundary are limited to temporary works supporting the viaduct construction. Some existing disturbance from the adjacent highway is observed. Taking into account the current disturbances from Nam Sang Wai Road, Yuen Long Highway, and Tsing Long Highway, the potential disturbance impact on the YLBFEW is expected to be low to moderate in the absence of mitigation measures. To further minimise impacts, a designated “no work zone” has been incorporated into the Project design along the Project Boundary across the KTMDC. The inclusion of “no work zone” will help reduce disturbance from temporary works to the surrounding environment. With these measures in place, adverse ecological impacts on YLBFEW are not expected.

Kam Tin Main Drainage Channel (KTMDC)

8.8.2.10 The KTMDC is situated in the southern part of the Project Boundary. Construction activities associated with the viaduct, pile and pier works may potentially deter and obstruct fauna species from utilizing the channel. However, existing disturbances from traffic on the adjacent highway above the channel have already been observed.

Given the moderate abundance of waterbird species recorded in KTMDC, the disturbance impacts are anticipated to be low to moderate if unmitigated, particularly during the overwintering period. To address these potential impacts, several measures are proposed (see **Section 8.11**), including the designation of "no work zone" along the Project Boundary over the channel to limit disturbance, as well as phased construction to minimise the affected area at any given time. For phased construction in KTMDC, steel casing would be implemented to isolate the surrounding riverbed when the bored pile construction method is adopted. As noted in **Section 4** of this report, the construction programme does not require percussive pilling, instead, silent pilling using large-diameter bored piles would be adopted. Pre-casting of bridge decking would also be implemented wherever practicable to reduce on-site activities and associated disturbance. Additional mitigation measures such as good site practices to limit noise emissions at the source, as well as the use of temporary noise barriers, enclosures, insulating fabric, silencer, and other noise-control measures to screen construction works and equipment. Furthermore, in view of the higher usage by waterbirds during low tide level in KTMDC during dry season, piling works would be scheduled to conduct during high tidal level (i.e. above 1.0 mCD) during dry season when the riverbed of KTMDC would be mostly submerged. Excavated material within the steel casing will be confined during pile reinforcement and concreting, further ensuring that the watercourse area outside the casing remains unaffected. With the implementation of these minimisation measures, adverse ecological impact is not expected.

West Rail Compensatory Wetland

8.8.2.11 Part of the West Rail Compensatory Wetland lies within the 500 m Assessment Area of the Project and is located in approximately 325 m to the east of the Project Boundary. Given the considerable distance of the West Rail Compensatory Wetland with the Project Boundary, disturbance impact to the planned wetland compensation site is anticipated to be low.

Pok Wai Bat Roost/Planned Bat Shelter

8.8.2.12 The Pok Wai Bat Roost serves as a day-roost for Himalayan Leaf-nosed Bats, likely used only during the summer months. It is located within the Project Boundary and is expected to be directly impacted by the Northern Link construction (MTRCL, 2024a). As such, disturbance impacts on this roost are not further assessed in this study in this study. A planned bat shelter, intended as an alternative roosting site for the Himalayan Leaf-nosed Bats, is situated near the Pok Wai Bat Roost and lies in close proximity to the viaduct and associates pile and pier works. The Project alignment has been refined to ensure no direct loss of the planned bat shelter. In addition, part of the woodland habitat to the south and east of the planned bat shelter has been designed as "no work zone". As such, potential disturbance impact such as barrier effect, impact on microclimate is anticipated to be low. On the other hand, given bored pile construction method would be adopted, which is a low-vibration method, potential vibration impact to the bat shelter is anticipated to be low.

8.8.2.13 In addition, the surrounding area already experiences significant disturbance from existing sources such as San Tin Highway and human activities in nearby villages. Additional disturbance such as air, noise, glare, dust and other human activities arising from construction activities is expected to result in low to moderate ecological impact in the absence of mitigation measures. To address this,

minimisation measures and good site practices will be implemented to ensure that construction activities would not lead to significant disturbance.

Planned Compensation Wetland for Ngau Tam Mei (NTM) Development Area

8.8.2.14 Under the approved EIA for the NTM Development Area (Register No.: AEIAR-271/2025, CEDD, 2025), a wetland compensation site of about 3.5 ha has been designated adjacent to the Kam Tin River, between Tsing Long Highway and San Tam Road at Sha Po area. Although part of this site overlaps with the proposed NMH alignment, recent project coordination during the preparation of this EIA Report has incorporated "no work zone" into the Project design to clearly delineate areas where construction activities will not occur. As a result, disturbance impacts on the planned compensation wetland during its establishment period are anticipated to be low to moderate, if unmitigated.

Planned Wetland Compensation Site in Kam Tin

8.8.2.15 Under the approved EIA for the Northern Link (Register No.: AEIAR-259/2024), a wetland compensation site of about 1.9 ha is identified in Kam Tin, adjacent to the future Ecological Enhancement Area (EEA) of the Park Yoho Phases 2 and 3 Development and the West Rail Compensatory Wetlands. This site was proposed to compensate for the loss of marsh/reed habitat due to the project. Given the considerable distance of the planned wetland compensation in Kam Tin with the Project Boundary, i.e. approximately 250 m, disturbance impact to the planned wetland compensation site is anticipated to be low.

Proposed Nam Sang Wai Wetland Conservation Park

8.8.2.16 The proposed Nam Sang Wai Wetland Conservation Park is located adjacent to the Project Boundary within the Nam Sang Wai area. Activities within the adjoining Project Boundary are limited to temporary works supporting viaduct construction. Some existing disturbance from the nearby highway has already been observed. Taking into account current disturbances from Nam Sang Wai Road, Yuen Long Highway, and Tsing Long Highway, the potential disturbance impact on the proposed Wetland Conservation Park is expected to be low to moderate in the absence of mitigation measures. To further reduce impacts, a designated "no work zone" has been incorporated into the Project design along the Project Boundary across KTMDC. This measure will help reduce disturbance caused by temporary works to the surrounding environment. As such, adverse ecological impact on the proposed Nam Sang Wai Wetland Conservation Park is not expected.

<u>Disturbance Impact to Habitats and Associated Wildlife</u>

Disturbance to Watercourses

8.8.2.17 The majority of the avifauna species recorded in the modified watercourse habitat within the Project Boundary were ardeids and waterbird species, particularly concentrated in the KTMDC near Nam Sang Wai. As stated in **Section 8.8.2.10**, construction activities associated with pile and pier works may temporarily disturb this habitat by obstructing foraging grounds. However, the Project Boundary encompasses only a small portion of the KTMDC, and similar habitats are available outside the Project Boundary, including downstream section of the KTMDC, NTMDC, and wetland habitats to the west of KTMDC. Affected species are anticipated to utilise these nearby wetland habitats. Disturbance impacts on

species of conservation importance are therefore anticipated to be low to moderate if unmitigated. Project design refinements to further minimise disturbance at the KTMDC are detailed in **Section 8.11.**

8.8.2.18 Watercourses adjacent to the Project Boundary (e.g. NTM-S2, NTM-S4, NTM-S4a, NTM-S5, ST-S3 and ST-S3a etc.) may also experience temporary disturbance from pile and pier construction and associated temporary works. Considering the limited extent of these activities and the generally low abundance and diversity of fauna species recorded in these watercourses, disturbance impact are considered low. Project design will be further refined to minimise encroachment and disturbance to these watercourses, as described in **Section 8.11**.

Disturbance to Other Wetland Habitats

8.8.2.19 Indirect impacts to other wetland habitats adjacent to the Project Boundary, such as those near NTMWTW and Nam Sang Wai, are anticipated due to construction disturbances. They may include air and dust pollution, noise, glare, increased human activities, and potential deterioration of water quality. Runoffs and other pollutants from construction works could also affect wetland habitats, such as marsh/reed and pond.

8.8.2.20 However, these adjacent wetland habitats already experience regular disturbances from surrounding developed area and villages under the existing baseline conditions. Consequently, the indirect disturbance impact on these habitats and their associated wildlife is considered to be low to moderate in the absence of mitigation measures. Implementing appropriate mitigation measures, such as general good site practices, would further help prevent potential environmental degradation to the surrounding habitats. Disturbance to seasonally wet grassland and mitigation wetland near Ko Po Road is also expected to be low, as they are situated away from the Project Boundary. Nonetheless, mitigation measures, such as noise barriers and hoardings, would be implemented to further minimise the potential disturbance impact to the surrounding habitats.

Disturbance to Other Non-Wetland Habitats

8.8.2.21 In addition to wetland habitats adjacent to the Project Boundary, other vegetated habitats, such as mixed woodland, plantation, shrubland, grassland, were recorded in the vicinity. Dust generated from the construction activities may degrade habitat quality, and deposition of dust and pollutants could affect vegetation growth. However, most vegetated habitats near the Project Boundary are already exposed to significant anthropogenic disturbance from surrounding village/orchard and developed area/wasteland habitats. As a result, disturbance impacts on non-wetland habitats are anticipated to be low. Moreover, alignment in Ngau Tam Shan and Hadden Hill is proposed to be underground, which would further reduce the disturbances to surrounding habitats. A designated “no work zone’ has also been incorporated into the Project design within woodland habitat at Pok Wai to minimise potential impact. Disturbance impacts on vegetated habitats within the LTCP and CA are detailed in **Section 8.8.2.4**.

8.8.2.22 Disturbance to urbanized habitats outside the Project Boundary, such as roadside plantation, developed area/wasteland, and village/orchard habitats, is considered low, as these habitats are already subject to constant disturbance under existing baseline conditions.

Potential Obstruction and Disturbance to Flight Corridor

8.8.2.23 The proposed construction works areas across the KTMDC may potentially obstruct the flight corridor of avifauna and disturb ardeids or waterbirds transverse along the KTMDC and the nearby Sha Po Night Roost. Given the moderate abundance of waterbird species recorded in KTMDC, the disturbance impacts are anticipated to be low to moderate if unmitigated. To minimise the potential obstruction and disturbance to flight corridor, several measures are proposed (see **Section 8.11**), including the designation of "no work zone" along the Project Boundary over the channel to limit disturbance, as well as phased construction to minimise the affected area in KTMDC at any given time, allowing passage of avifauna along the watercourse. As mentioned in **Section 8.8.2.10**, steel casing would be implemented to isolate the surrounding riverbed when the bored pile construction method is adopted for the phased construction in KTMDC. As noted in **Section 4** of this report, the construction programme also does not involve percussive pilling, while silent pilling using large-diameter bored piles would be adopted. Pre-casting of bridge decking would also be implemented wherever practicable to reduce on-site activities and associated disturbance to the flight corridor in KTMDC.

Potential Water Quality, Water Holding/Water Flow and Groundwater Drawdown Impact

8.8.2.24 Site formation and construction activities may cause potential water quality impacts to adjacent watercourses and pond habitats. These include deterioration of water quality, elevated suspended solids, and the introduction of potential contaminants. Possible sources are site runoff from exposed soil, construction discharges, and accidental spills of oil, fuel, or other harmful chemicals. Additionally, construction works may alter water flow in watercourses outside the Project Boundary.

8.8.2.25 During pile and pier construction in the KTMDC and Ngau Tam Mei areas, water quality in the KTMDC, adjacent watercourses (e.g. YL-S2 to YL-S5, YL-S7, YL-S8, NTM-S1 to NTM-S6, ST-S1 to ST-S6, ST-S9 to ST-S11, ST-S14 to ST-S17), and ponds may be affected by runoff, discharges, and accidental spillage. Such indirect impacts could degrade habitat quality and affect associated foraging wildlife, extending downstream to the KTMDC and NTMDC, which connect to the boarder wetland habitats including the WBA and WCA. Mitigation measures, such as steel casing during bored pile construction, would isolate the surrounding riverbed and confine the affected areas, thereby reducing water quality impact. Without mitigation, potential ecological impacts from water quality degradation in the KTMDC, pond water-holding regimes, and watercourse flow regimes are considered low to moderate. For outside the Project Boundary, given the separation distance with the Project Boundary, the potential water quality impact is considered to be low.

8.8.2.26 Direct encroachment on watercourses will be avoided as far as practicable. Where unavoidable at-grade works occur, water diversion will be implemented to maintain flow. Although approximately 0.03 ha of natural waterlines would be lost, these are seasonal with limited flow, so that the overall impact is considered low. Ponds within the Project Boundary near NTMWTW are primarily rain-fed, therefore, construction works are not expected to adversely affect their hydrological properties. In view of this, impacts on the water flow regime impact of these ponds are considered low.

8.8.2.27 Underground tunnel construction through Ngau Tam Shan and Hadden Hill may result in groundwater drawdown. However, standard engineering practices, such as waterproofing system and hydraulic cut-offs during deep excavation, would prevent groundwater ingress and minimise drawdown risks. Additional measures, including pre-grouting and recharge wells, would be deployed if necessary to further control potential drawdown.

Night-time Disturbance

8.8.2.28 Uncontrolled lighting and glare from construction sites at night may affect the foraging, predation, and breeding behaviours of nocturnal fauna, including avifauna, bat, herpetofauna, and firefly species in the vicinity, especially for species associated with nearby recognised sites of conservation importance (e.g. LTCP), natural habitats (e.g. woodland in Ki Lun Tsuen area) habitats are relatively close to existing urbanised area, such as adjacent residential neighbourhoods, villages, and main roads, and are already subject to night-time glare impacts. Without appropriate mitigation measures, the night-time glare may result in low to moderate ecological impacts on adjacent habitats and associated faunal groups. Mitigation measures to minimize glare disturbance are discussed in **Section 8.11**.

Potential Ground-borne Noise and Vibration Impact

8.8.2.29 Tunnel construction passing through Ngau Tam Shan and Hadden Hill will generate ground-borne noise and vibration. The affected habitats primarily consisted of hillside plantations and grasslands. Given the absence of roosting or breeding sites and the low abundance and diversity of fauna, particularly vibration-sensitive species such as bats, the potential indirect disturbance impact from ground-borne noise and vibration on associated wildlife is considered to be low.

8.8.3 Operation Phase – Direct Impact

Direct Impacts on Recognised Sites of Conservation Importance

8.8.3.1 Four recognized sites of conservation importance are situated within the Project Boundary: Deep Bay Wetland outside Ramsar Site Priority Site for Enhanced Conservation, CA, WCA, and WBA. Based on the proposed alignment layout for the Project, no further direct impacts on these sites are anticipated in the operation phase.

Direct Impacts on Other Ecologically Sensitive Resources

8.8.3.2 Based on the proposed alignment layout for the Project, no further direct impacts on the other ecologically sensitive resources are anticipated in the operation phase.

Direct Impacts on Habitats and Associated Wildlife

8.8.3.3 According to the proposed alignment for San Tin section of the Project, no further loss of vegetated habitats and riparian vegetation is expected during the operation phase. Hence, additional direct impacts, including fragmentation and obstruction on habitats and associate wildlife as well as alternation to riparian habitats are not anticipated.

Impacts from Direct Injury/Mortality

8.8.3.4 During the operation phase, a significant portion of the alignment will consist of viaduct and tunnel sections, where direct injury or mortality of wildlife is not anticipated. The majority of recorded wildlife species are highly mobile (e.g. birds, butterflies, and flying mammals) and are already experiencing some existing disturbances. These species are anticipated to migrate to similar nearby habitats progressively during the construction phase. Nonetheless, potential direct injury or mortality due to roadkill may occur in at-grade sections situated near vegetated habitats, such as the at-grade section south of NTM, near or within CA and LTCP, where non-flying mammal species such as the Small Indian Civet, Red Muntjac, East Asian Porcupine, and Masked Palm Civet have been recorded in previous studies (**Figure 8.4.5** refers). As such, the impact from direct injury or mortality is anticipated to be low to moderate, if unmitigated. Appropriate mitigation measures, such as animal barriers, would be proposed to minimise and prevent roadkill in these at-grade sections, as detailed in **Section 8.11**.

Impacts on Bird Collision

8.8.3.5 During the operation phase, heavy traffic is expected along the proposed viaduct alignment in the Nam Sang Wai area. Transparent or reflective materials (e.g. glass noise barriers) may pose a risk of bird collision. However, by Implementing mitigation measures such as using non-transparent or non-glaring materials on noise barriers, collision risks during operation can be effectively avoided. Existing structures in the vicinity, including bridges and highways with heavy traffic, already demonstrated that commuting ardeids from the Sha Po Night Roost and Ko Po Road Night Roost are capable of maneuvering within urbanized areas. Observations show these birds flying underneath or over viaducts such as those on Yuen Long Highway and Tsing Long Highway across the KTMDC. As such, bird-vehicle collisions during operation are considered unlikely, as local bird populations are accustomed to traffic conditions. Moreover, previous studies, including the *Shenzhen Western Corridor EIA Study* and *Night-roosting Egrets' Response to Traffic and Bridge Structures*, reported no collisions with bridge structures, piers and vehicles, as avifauna species are able to swiftly adjust their flight paths (HyD, 2002; Stanton and Klick, 2018). Considering the wide range of commuting routes currently used by ardeids and their adaptability, the potential bird collision resulting from the Project's operation is considered low.

Fragmentation and Isolation of Wooded Areas

8.8.3.6 Under the Project, the at-grade permanent structures (e.g. road roads, slope works, ancillary facilities and piles and piers) at Kai Kung Leng, located southwest of the Ngau Tam Mei area, Ngau Tam Shan, and Hadden Hill, may potentially fragment existing vegetated habitats. This fragmentation could obstruct the movement of associated wildlife, including non-flying mammals and herpetofauna species, along with their foraging grounds. Nonetheless, these at-grade permanent structures would primarily be located along the edges of wooded habitats, where significant wildlife crossings between disturbed area (e.g. village/orchard and developed area/wasteland habitats) and wooded habitats are not anticipated. In addition, part of the at-grade section within the vegetated habitats at Kai Kung Leng would take the form of a viaduct, while a substantial portion of the alignment in Ngau Tam Shan and the southeast of Hadden Hill is designed as underground tunnels. This design effectively minimises fragmentation and facilitates animal passage between the vegetated and wooded habitats. In particular, the vegetated and wooded habitats within the CAs in Ngau Tam Shan and Hadden Hill would be preserved through the design of underground tunnel sections, with the proposed alignment positioned

along the margins of wooded habitats to the east of the CA in Hadden Hill. As such, the potential impacts of fragmentation and isolation resulting from the Project during the operation phase is expected to be low.

8.8.4 Operation Phase – Indirect Impact

Disturbance on Recognised Sites of Conservation Importance and Other Ecologically Sensitive Resources

8.8.4.1 During the operation phase, disturbances such as noise, light and glare, dust, traffic, and other human disturbances may arise due to the Project. Four recognized sites of conservation importance (i.e. Deep Bay Wetland outside Ramsar Site Priority Site for Enhanced Conservation, CA, WCA, and WBA) and four other ecologically sensitive resources (i.e. proposed Nam Sang Wai Wetland Conservation Park, the planned wetland compensation site under the Ngau Tam Mei Development, the planned bat shelter in Pok Wai, and Sha Po Night Roost) are situated within the Project Boundary. Additionally, other ecologically sensitive resources (i.e. YLBFEW, Sha Po Marsh, Ko Po Road Night Roost, Important Bird Area, West Rail Compensatory Wetland, and planned wetland compensation site in Kam Tin for the Northern Link) are located directly adjacent or outside the Project Boundary. Given that the proposed alignment would be located along the existing highways in Kam Tin area, the Project is not expected to result in disturbance levels significantly different from current conditions. The recognised sites of conservation importance and ecologically sensitive resources mentioned above are already subject to substantial and ongoing human activities, including existing roads, high traffic volumes, nearby residential and industrial developments, and other infrastructure. As a result, the potential additional disturbance impact on these sites of conservation importance and ecologically sensitive resources during the operation phase is anticipated to be low.

8.8.4.2 In terms of the planned bat shelter in Pok Wai, given the road section adjacent to the planned bat shelter would be located in existing developed area/wasteland habitat in form of viaduct with height of approximately 10 m or above, along with the retained woodland habitat to the south and east of the planned bat shelter, potential disturbance impacts such as barrier effect, impact on microclimate and potential light spill is anticipated to be low. Nevertheless, directional lighting would be adopted during operation phase to further minimise potential light spill.

8.8.4.3 In addition, apart from the KTMDC, the proposed viaduct section would span a small area of marsh/reed and pond habitats (e.g. marsh/reed and pond near the NTMWTW and the planned wetland compensation site under the Ngau Tam Mei Development). The viaduct section above these habitats would be approximately 13 m wide and 10 m high, ranging from +27 mPD to +31 mPD above ground level. Given the high clearance, lateral light will penetrate beneath the deck, resulting in only partial shading of the habitats below. Therefore, the impact from changes in natural light is considered as low.

Disturbance on Species of Conservation Importance, Wildlife and Habitat

8.8.4.4 Noise, glare, dust, traffic, and other human disturbances may also affect habitats and species within or near the Project Boundary during its operation. These habitats and species are already exposed to regular disturbances under the existing baseline conditions, including those arising from anthropogenic activities in surrounding village/orchard habitats, residential areas, and traffic from existing

roads. Therefore, the disturbance impact caused by the operation of the Project is anticipated to be low.

Obstruction and Disturbance to Flight Corridor

8.8.4.5 The proposed viaduct section across the KTMDC may potentially disrupt the avifauna flight corridor and disturb ardeids or waterbirds foraging in the KTMDC and the nearby Sha Po Night Roost. However, since the proposed viaduct section would be of similar height and close to the existing viaduct of the Yuen Long Highway and Tsing Long Highway, avifauna species are expected to continue transversing beneath or above the viaduct sections, similar to current conditions. In terms of disturbance, the anticipated traffic flow on the proposed viaduct section would also be comparable to that of the existing viaducts in the KTMDC, where the surrounding environment is already subjected to constant human disturbance and noise from heavy traffic. In view of these analogous conditions, the potential obstruction and disturbance to the flight corridor and foraging ardeids and waterbirds in the KTMDC and the Sha Po Night Roost are anticipated to be low.

Impact on Water Quality and Water Holding/Water Flow

8.8.4.6 An adequate drainage system would be implemented for the proposed at-grade road, viaduct, and tunnel, as mentioned in **Section 5.6**. The potential impact on water quality due to surface runoff is considered low, and no adverse impacts from non-point source surface runoff is expected. The drainage system would effectively prevented surface runoff from entering ecologically sensitive watercourses, as mentioned in **Section 5.6**. Consequently, the potential water quality impact on the surrounding environment, including ponds, semi-natural watercourses, and natural watercourses, is anticipated to be low, with no adverse effects from non-point source surface runoff expected.

8.8.4.7 The proposed piles and piers located within the KTMDC would be of similar size and orientation to the existing bridge piers of the Tsing Long Highway and Yuen Long Highway. As a result, the impact on the water holding and water flow regime in the KTMDC could be reduced due to the overall cross-sectional area of the KTMDC. Furthermore, there would be no additional works or structures that would affect the water holding capacity of the nearby ponds, semi-natural watercourses, and natural watercourses during the operation phase of the Project. Therefore, the potential impact on the water holding and water flow regime of nearby ponds, semi-natural/natural watercourse, and the KTMDC is considered low.

8.9 Evaluation of Potential Ecological Impacts

8.9.1.1 The potential ecological impacts on habitats within the assessment area arising from both the construction and operation of the Project have been assessed in accordance with Annex 8 of the EIAO-TM. The detailed evaluations are provided in **Table 8.15** to **Table 8.23** and summarised in **Table 8.24**.

Table 8.15 Evaluation of Potential Ecological Impacts to Woodland and Mixed Woodland

Criterion	Woodland	Mixed Woodland
Habitat Quality	Moderate	Low to moderate

Criterion	Woodland	Mixed Woodland
Species/Ecological Resources	Moderate diversity of flora Low to moderate diversity of fauna Moderate abundance of wildlife <u>Current Survey</u> Four flora species of conservation importance (i.e. Incense Tree, Cycad-fern, Lamb of Tartary and Emarginate-leaved Ormosia) and 13 fauna species of conservation importance (i.e. Chinese Francolin, Collared Scops Owl, Chinese Noctule, Himalayan Leaf-nosed Bat, Japanese Pipistrelle, Lesser Bamboo Bat, Short-nosed Fruit Bat, East Asian Porcupine, Leopard Cat, Pallas's Squirrel, Red Muntjac, Small Indian Civet and Motschulsky's Starworm) <u>Literature Review</u> Five flora species of conservation importance (i.e. *Aralia chinensis*, Emarginate-leaved Ormosia, Eurya-leaved Camellia, Incense Tree and Lamb of Tartary) and 35 fauna species of conservation importance (i.e. Ashy Drongo, Asian Barred Owlet, Besra, Black Kite, Black-throated Laughingthrush, Chinese Hwamei, Chinese Sparrowhawk, Collared Crow, Collared Scops Owl, Common Emerald Dove, Crested Serpent Eagle, Greater Coucal, Grey Heron, Grey-chinned Minivet, Grey-headed Canary-flycatcher, Lesser Shortwing, Pygmy Wren-babbler, Rufous-capped Babbler, Silver-eared Leiothrix, Slaty-legged Crake, Speckled Piculet, White-bellied Erpornis, White-shouldered Starling, Wood Sandpiper, East Asian Porcupine, Japanese Pipistrelle, Leopard Cat,	Moderate to high diversity of flora (includes ruderal and exotic species) Moderate diversity of fauna Moderate abundance of wildlife <u>Current Survey</u> Three flora species of conservation importance (i.e. Incense Tree, *Aralia chinensis* and Lamb of Tartary) and 20 fauna species of conservation importance (i.e. Chinese Pond Heron, Collared Scops Owl, Common Emerald Dove, Greater Coucal, Grey Treepie, Indian Cuckoo, Lesser Coucal, Oriental Magpie, Rufous-capped Babbler, Chinese Noctule, Japanese Pipistrelle, Lesser Yellow Bat, Unknown Vespertilionidae species 1, East Asian Porcupine, Leopard Cat, Pallas's Squirrel, Red Muntjac, Rhesus Macaque, Small Indian Civet and Blackvein Sergeant) <u>Literature Review</u> Two flora species of conservation importance (i.e. *Aralia chinensis* and Incense Tree) and 25 fauna species of conservation importance (i.e. Asian Barred Owlet, Black-throated Laughingthrush, Chinese Hwamei, Collared Crow, Collared Scops Owl, Common Kestrel, Crested Goshawk, Crested Serpent Eagle, Greater Coucal, Grey-chinned Minivet, Rufous-capped Babbler, Speckled Piculet, White-bellied Erpornis, Chinese Noctule, Himalayan Leaf-nosed Bat, Japanese Pipistrelle, Leopard Cat, Lesser Bamboo Bat, Pallas's Squirrel, Red Muntjac, Unknown Vespertilionidae species 1, Forget-me-not, Dingy Dusk-hawker and Chinese Cobra)

Criterion	Woodland	Mixed Woodland
	Masked Palm Civet, Pallas's Squirrel, Red Muntjac, Short-nosed Fruit Bat, Small Asian Mongoose, Small Indian Civet, Unknown Vespertilionidae species 1 and Spotted Royal)	
Size/Abundance	10.35 ha would be lost due to the proposed alignment 6.28 ha would be impacted by temporary works	3.74 ha would be lost due to the proposed alignment 9.02 ha would be impacted by temporary works
Duration	Direct impact (habitat loss) on 10.35 ha would be permanent Direct impact (habitat loss) on 6.28 ha would be temporary Indirect impact (noise, dust and glare) during construction phase would be temporary Indirect impact (noise, dust and glare) during operation phase would be permanent	Direct impact (habitat loss) on 3.74 ha would be permanent Direct impact (habitat loss) on 9.02 ha would be temporary Indirect impact (noise, dust and glare) during construction phase would be temporary Indirect impact (noise, dust and glare) during operation phase would be permanent
Reversibility	Direct impact (habitat loss) on 10.35 ha during construction phase would be irreversible Direct impact (habitat loss) on 6.28 ha during construction phase would be reversible Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be irreversible	Direct impact (habitat loss) on 3.74 ha during construction phase would be irreversible Direct impact (habitat loss) on 9.02 ha during construction phase would be reversible Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be irreversible
Magnitude	Low to moderate	Low to moderate
Regional significance	Low regional significance as the habitat is common regionally; supports widespread flora and fauna; and the loss does not affect the regional ecological connectivity between habitats and species population	Low regional significance as the habitat is common regionally; supports mostly common flora and widespread fauna; and the loss does not affect the regional ecological connectivity between habitats and species population
Overall Impact Significance	**Low to moderate**	**Low**

Table 8.16 Evaluation of Potential Ecological Impacts to Shrubland and Grassland

Criterion	Shrubland	Grassland
Habitat Quality	Low to moderate	Low to moderate for hillside grassland Low for low-lying grassland
Species/Ecological Resources	Moderate diversity of flora Low to moderate diversity of fauna Low abundance of wildlife <u>Current Survey</u> One flora species of conservation importance (i.e. Incense Tree) and 11 fauna species of conservation importance (i.e. Black-throated Laughingthrush, Collared Scops Owl, Greater Coucal, Grey Treepie, Long-tailed Shrike, Rufous-capped Babbler, East Asian Porcupine, Pallas's Squirrel, Red Muntjac, Small Indian Civet and Small Three-ring) <u>Literature Review</u> Two flora species of conservation importance (i.e. Cycad-fern and Incense Tree) and 24 fauna species of conservation importance (i.e. Black-throated Laughingthrush, Chinese Hwamei, Crested Serpent Eagle, Greater Coucal, Grey Heron, Rufous-capped Babbler, Siberian Rubythroat, Speckled Piculet, East Asian Porcupine, Himalayan Leaf-nosed Bat, Japanese Pipistrelle, Leopard Cat, Lesser Bamboo Bat, Pallas's Squirrel, Red Muntjac, Small Indian Civet, Broad Spark, Dark Brown Ace, Malayan, Rounded 6-line Blue, Blue-spotted Dusk-hawker, Orange-backed Threadtail, Ruby Darter and King Cobra)	Moderate diversity of flora and fauna Moderate abundance of wildlife <u>Current Survey</u> Two flora species of conservation importance (i.e. Cycad-fern and Chinese Frienge Lily) and 19 fauna species of conservation importance (i.e. Asian Barred Owlet, Black-faced Bunting, Black-throated Laughingthrush, Chinese Francolin, Chinese Hwamei, Crested Serpent Eagle, Greater Coucal, Sooty-headed Bulbul, White-throated Kingfisher, Zitting Cisticola, Japanese Pipistrelle, East Asian Porcupine, Red Muntjac, Small Indian Civet, Danaid Eggfly, Silver Royal, Small Three-ring, Burmese Python and Chinese Cobra) <u>Literature Review</u> Three flora species of conservation importance (i.e. Chinese Frienge Lily, Cycad-fern and Incense Tree) and 22 fauna species of conservation importance (i.e. Amur Falcon, Black Kite, Chinese Francolin, Chinese Hwamei, Collared Crow, Collared Scops Owl, Crested Serpent Eagle, Greater Coucal, Little Egret, Chinese Noctule, Himalayan Leaf-nosed Bat, Japanese Pipistrelle, Pallas's Squirrel, Short-nosed Fruit Bat, Unknown Vespertilionidae species 2, Danaid Eggfly, Forget-me-not, Small Three-ring, Spotted Angle, Tiny Grass Blue, Scarlet Basker and King Cobra)
Size/Abundance	8.68 ha would be lost due to the proposed alignment 8.26 ha would be impacted by temporary works	14.81 ha would be lost due to the proposed alignment 10.17 ha would be impacted by temporary works

Criterion	Shrubland	Grassland
Duration	Direct impact (habitat loss) on 8.68 ha would be permanent Direct impact (habitat loss) on 8.26 ha would be temporary Indirect impact (noise, dust and glare) during construction phase would be temporary Indirect impact (noise, dust and glare) during operation phase would be permanent	Direct impact (habitat loss) on 14.81 ha would be permanent Direct impact (habitat loss) on 10.17 ha would be temporary Indirect impact (noise, dust and glare) during construction phase would be temporary Indirect impact (noise, dust and glare) during operation phase would be permanent
Reversibility	Direct impact (habitat loss) on 8.68 ha during construction phase would be irreversible Direct impact (habitat loss) on 8.26 ha during construction phase would be reversible Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be irreversible	Direct impact (habitat loss) on 14.81 ha during construction phase would be irreversible Direct impact (habitat loss) on 10.17 ha during construction phase would be reversible Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be irreversible
Magnitude	Low	Low to moderate
Regional significance	Low regional significance as the habitat is common regionally; supports mostly common flora and widespread fauna; and the loss does not affect the regional ecological connectivity between habitats and species population	Low regional significance as the habitat is common regionally; supports mostly common flora and widespread fauna; and the loss does not affect the regional ecological connectivity between habitats and species population
Overall Impact Significance	**Low**	**Low**

Table 8.17 Evaluation of Potential Ecological Impacts to Plantation

Criterion	Plantation
Habitat Quality	Low to moderate for hillside plantation Low for roadside plantation
Species/Ecological Resources	Moderate to high diversity of flora (includes ruderal and exotic species) and fauna Moderate abundance of wildlife Presence of Ko Po Road Night Roost <u>Current Survey</u> Two flora species of conservation importance (i.e. Incense Tree and Cycad-fern) and 32 fauna species of conservation importance (i.e. Asian Barred Owlet, Black-throated Laughingthrush, Chinese

Criterion	Plantation
	Hwamei, Collared Scops Owl, Common Emerald Dove, Common Rosefinch, Crested Serpent Eagle, Greater Coucal, Grey Nightjar, Lesser Coucal, Rufous-capped Babbler, Chinese Noctule, Himalayan Leaf-nosed Bat, Japanese Pipistrelle, Lesser Bamboo Bat, Lesser Bent-winged Bat, Short-nosed Fruit Bat, Unknown Vespertilionidae species 1, East Asian Porcupine, Leopard Cat, Masked Palm Civet, Pallas's Squirrel, Red Muntjac, Rhesus Macaque, Small Asian Mongoose, Small Indian Civet, Small-toothed Ferret Badger, Danaid Eggfly, Little Branded Swift, Small Three-ring, Dusk-hawker and Four-clawed Gecko) <u>Literature Review</u> Three flora species of conservation importance (i.e. *Aralia chinensis*, Cycad-fern and Incense Tree) and 35 fauna species of conservation importance (i.e. Besra, Black Kite, Black-throated Laughingthrush, Collared Scops Owl, Common Emerald Dove, Crested Goshawk, Crested Serpent Eagle, Greater Coucal, Rufous-capped Babbler, Slaty-legged Crake, Speckled Piculet, White-bellied Erpornis, White-throated Kingfisher, Chinese Noctule, East Asian Porcupine, Himalayan Leaf-nosed Bat, Intermediate Horseshoe Bat, Japanese Pipistrelle, Leopard Cat, Lesser Bamboo Bat, Lesser Yellow Bat, Masked Palm Civet, Red Muntjac, Small Asian Mongoose, Small Indian Civet, Unknown Vespertilionidae species 1, Unknown Vespertilionidae species 2, Banded Demon, Baron, Common Dart, Danaid Eggfly, Forget-me-not, Green Skirt Baron, Tiny Grass Blue and Chinese Water Dragon)
Size/Abundance	3.16 ha would be lost due to the proposed alignment 12.89 ha would be impacted by temporary works No direct impact on Ko Po Road Night Roost
Duration	Direct impact (habitat loss) on 3.16 ha would be permanent Direct impact (habitat loss) on 12.89 ha would be temporary Indirect impact (noise, dust and glare) during construction phase would be temporary Indirect impact (noise, dust and glare) during operation phase would be permanent
Reversibility	Direct impact (habitat loss) on 3.16 ha during construction phase would be irreversible Direct impact (habitat loss) on 12.89 ha during construction phase would be reversible Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be irreversible
Magnitude	Low
Regional significance	Low regional significance as the habitat is common regionally; supports mostly common flora and widespread fauna; and the loss does not affect the regional ecological connectivity between habitats and species population

Criterion	Plantation
Overall Impact Significance	**Low**

Table 8.18 Evaluation of Potential Ecological Impacts to Village/Orchard and Developed Area/Wasteland

Criterion	Village/Orchard	Developed Area/Wasteland
Habitat Quality	Low to moderate	Low
Species/Ecological Resources	Moderate to high diversity of flora (consisting of native ruderal and exotic species) Moderate diversity of fauna Moderate abundance of wildlife <u>Current Survey</u> Two flora species of conservation importance (i.e. Incense Tree and Hong Kong Pavetta) and 18 fauna species of conservation importance (i.e. Brown Fish Owl, Collared Scops Owl, Great Egret, Greater Coucal, Little Egret, Chinese Horseshoe Bat, Chinese Noctule, Himalayan Leaf-nosed Bat, Japanese Pipistrelle, Lesser Bamboo Bat, Lesser Yellow Bat, Short-nosed Fruit Bat, Unknown Vespertilionidae species 1, Unknown Vespertilionidae species 2, Pallas's Squirrel, Red Muntjac, Four-clawed Gecko, Motschulsky's Starworm) <u>Literature Review</u> Three flora species of conservation importance (i.e. *Aralia chinensis*, Hong Kong Pavetta and Incense Tree) and 21 fauna species of conservation importance (i.e. Asian Barred Owlet, Besra, Chinese Grosbeak, Collared Scops Owl, Common Emerald Dove, Eastern Buzzard, Greater Coucal, Chinese Noctule, Himalayan Leaf-nosed Bat, Japanese	Moderate to high diversity of flora (consisting of native ruderal and exotic species) Moderate diversity of fauna Moderate abundance of wildlife Presence of Sha Po Night Roost and Day-Roosting of Himalayan Leaf-nosed Bat <u>Current Survey</u> 22 fauna species of conservation importance (i.e. Chinese Pond Heron, Greater Coucal, Little Egret, Oriental Magpie, Rufous-capped Babbler, White-throated Kingfisher, Chinese Noctule, Himalayan Leaf-nosed Bat, Japanese Pipistrelle, Lesser Bamboo Bat, Lesser Bent-winged Bat, Lesser Yellow Bat, Unknown Vespertilionidae species 1, Unknown Vespertilionidae species 2, Pallas's Squirrel, Emerald Cascader, Chinese Bullfrog, Banded Krait, Four-clawed Gecko, Reeves' Turtle, *Nanhaipotamon hongkongense* and Motschulsky's Starworm) <u>Literature Review</u> Two flora species of conservation importance (i.e. *Aralia chinensis* and Incense Tree) and 25 fauna species of conservation importance (i.e. Black Kite, Chinese Pond Heron, Collared Crow, Crested Serpent Eagle, Great Egret, Greater Coucal, Grey Heron, Little Egret, Red-billed Starling, Chinese Horseshoe Bat, Chinese Noctule, Himalayan Leaf-nosed Bat, Japanese Pipistrelle, Lesser Bamboo

Criterion	Village/Orchard	Developed Area/Wasteland
	Pipistrelle, Lesser Yellow Bat, Pallas's Squirrel, Unknown Vespertilionidae species 1, Unknown Vespertilionidae species 2, Common Birdwing, Common Rose, Pale Palm Dart, Peacock Royal, Chinese Bullfrog, Common Rat Snake and Copperhead Racer)	Bat, Pallas's Squirrel, Unknown Myotis sp., Unknown Vespertilionidae species 1, Unknown Vespertilionidae species 2, Danaid Eggfly, Grass Demon, Tiny Grass Blue, Chinese Bullfrog, Buff-striped Keelback, Chinese Cobra and Indo-chinese Rat Snake)
Size/Abundance	7.86 ha would be lost due to the proposed alignment 18.60 ha would be impacted by temporary works	10.93 ha would be lost due to the proposed alignment 49.03 ha would be impacted by temporary works No direct impact on Sha Po Night Roost and Day-Roosting of Himalayan Leaf-nosed Bat
Duration	Direct impact (habitat loss) on 7.86 ha will be permanent Direct impact (habitat loss) on 18.60 ha will be temporary Indirect impact (noise, dust and glare) would be temporary Indirect impact (noise, dust and glare) during operation phase would be permanent	Direct impact (habitat loss) on 10.93 ha will be permanent Direct impact (habitat loss) on 49.03 ha will be temporary Indirect impact (noise, dust and glare) would be temporary Indirect impact (noise, dust and glare) during operation phase would be permanent
Reversibility	Direct impact (habitat loss) on 7.86 ha during construction phase would be irreversible Direct impact (habitat loss) on 18.60 ha during construction phase would be reversible Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be irreversible	Direct impact (habitat loss) on 10.93 ha during construction phase would be irreversible Direct impact (habitat loss) on 49.03 ha during construction phase would be reversible Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be irreversible
Magnitude	Low	Low
Regional significance	Low regional significance as the habitat is artificial and common regionally; supports mostly exotic flora and widespread fauna; and the loss does not affect the regional ecological connectivity between habitats and species population	Low regional significance as the habitat is artificial and common regionally; supports mostly exotic flora and widespread fauna; and the loss does not affect the regional ecological connectivity between habitats and species population

Criterion	Village/Orchard	Developed Area/Wasteland
Overall Impact Significance	**Low**	**Low to moderate** for Sha Po Night Roost and Day-Roosting of Himalayan Leaf-nosed Bat **Low** for remaining developed area/wasteland habitat

Table 8.19 Evaluation of Potential Ecological Impacts to Agricultural Land

Criterion	Agricultural Land
Habitat Quality	Low
Species/Ecological Resources	Low diversity of flora and fauna Low abundance of wildlife <u>Current Survey</u> Five fauna species of conservation importance (i.e. Black-throated Laughingthrush, Chinese Pond Heron, Common Snipe, Green Sandpiper and Japanese Pipistrelle) <u>Literature Review</u> Six fauna species of conservation importance (i.e. Black Kite, Chinese Pond Heron, Greater Coucal, Lesser Bamboo Bat, Unknown Vespertilionidae species 2 and Chinese Bullfrog)
Size/Abundance	0.04 ha would be lost due to the proposed alignment 0.72 ha would be impacted by temporary works
Duration	Direct impact (habitat loss) on 0.04 ha will be permanent Direct impact (habitat loss) on 0.72 ha will be temporary Indirect impact (noise, dust and glare) during construction phase would be temporary Indirect impact (noise, dust and glare) during operation phase would be permanent
Reversibility	Direct impact (habitat loss) on 0.04 ha during construction phase would be irreversible Direct impact (habitat loss) on 0.72 ha during construction phase would be reversible Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be irreversible
Magnitude	Low
Regional significance	Low regional significance as the affected area is small and scattered and the habitat is artificial and common regionally; supports mostly cultivated flora and widespread fauna; and the loss does not affect the regional ecological connectivity between habitats and species population
Overall Impact Significance	**Low**

Table 8.20 Evaluation of Potential Ecological Impacts to Seasonally Wet Grassland and Marsh/Reed

Criterion	Seasonally Wet Grassland	Marsh/Reed
Habitat Quality	Low	Moderate to high for Sha Po Marsh Low to moderate for other marsh/reed
Species/Ecological Resources	Low diversity of flora and fauna Low abundance of wildlife <u>Current Survey</u> One fauna species of conservation importance (i.e. Greater Coucal) <u>Literature Review</u> One fauna species of conservation importance (i.e. Chinese Pond Heron)	Low to moderate diversity of flora and fauna Moderate abundance of wildlife for Sha Po Marsh; Low to moderate abundance of wildlife in other marsh/reed <u>Current Survey</u> Seven fauna species of conservation importance recorded (i.e. Chinese Pond Heron, Greater Coucal, Grey Heron, Little Egret, Japanese Pipistrelle, Lesser Bamboo Bat, Lesser Yellow Bat and Unknown Vespertilionidae species 1) <u>Literature Review</u> Two flora species of conservation importance (i.e. Coastal Bulrush and Prince's Feather) and 20 fauna species of conservation importance recorded (i.e. Chinese Pond Heron, Great Egret, Greater Coucal, Grey Heron, Little Egret, Purple Heron, Siberian Rubythroat, Von Schrenck's Bittern, Zitting Cisticola, Chinese Horseshoe Bat, Intermediate Horseshoe Bat, Japanese Pipistrelle, Lesser Bamboo Bat, Small Asian Mongoose, Unknown Vespertilionidae species 1, Unknown Vespertilionidae species 2, Danaid Eggfly, Tiny Grass Blue, Four-spot Midget and Mangrove Skimmer)
Size/Abundance	No direct loss	0.06 ha would be lost due to the proposed alignment 1.07 ha would be impacted by temporary works No direct loss on Sha Po Marsh
Duration	No direct impact Indirect impact (noise, dust and glare) during construction phase would be temporary	Direct impact (habitat loss) on 0.06 ha will be permanent Direct impact (habitat loss) on 1.07 ha will be temporary

Criterion	Seasonally Wet Grassland	Marsh/Reed
	Indirect impact (noise, dust and glare) during operation phase would be permanent	Indirect impact (noise, dust and glare) during construction phase would be temporary Indirect impact (noise, dust and glare) during operation phase would be permanent
Reversibility	No direct impact Indirect impact (disturbance) during both construction and operation phase would be reversible	Direct impact (habitat loss) on 0.06 ha during construction phase would be irreversible Direct impact (habitat loss) on 1.07 ha during construction phase would be reversible Indirect impact (disturbance) during both construction and operation phase would be reversible
Magnitude	Low	Low
Regional significance	Low to moderate regional significance as the habitat is scarce regionally, however, small and scattered within the Project Boundary; supports mostly common flora and widespread fauna; and the loss does not affect the regional ecological connectivity between habitats and species population	Low to moderate regional significance as the habitat is scarce regionally, however, small and scattered within the Project Boundary; supports mostly common flora and widespread fauna; and the loss does not affect the regional ecological connectivity between habitats and species population
Overall Impact Significance	**Low**	**Low to moderate**

Table 8.21 Evaluation of Potential Ecological Impacts to Pond

Criterion	Pond within WBA and WCA	Pond near NTMWTW	Other Pond
Habitat Quality	Moderate	Low to moderate	Low to moderate for Kwu Tung Reservoir Low for remaining ponds
Species/Ecological Resources	Low to moderate diversity of flora and fauna Moderate abundance of wildlife <u>Current Survey</u> Eight fauna species of conservation importance recorded (i.e. Black Kite, Chinese Pond Heron,	Low to moderate diversity of flora and fauna Low to moderate abundance of wildlife <u>Current Survey</u> Eight fauna species of conservation importance recorded (i.e. Asian Barred Owlet, Chinese Pond	Low to moderate diversity of flora and fauna Low to moderate abundance of wildlife <u>Current Survey</u> Seven fauna species of conservation importance recorded (i.e. Great Cormorant, Grey Heron, Green

Criterion	Pond within WBA and WCA	Pond near NTMWTW	Other Pond
	Great Cormorant, Great Egret, Grey Heron, Little Egret, Little Grebe and Unknown Vespertilionidae species 1) <u>Literature Review</u> 15 fauna species of conservation importance recorded (i.e. Black Kite, Chinese Pond Heron, Collared Crow, Great Cormorant, Great Egret, Greater Coucal, Grey Heron, Kentish Plover, Little Egret, Little Grebe, Purple Heron, Red-billed Starling, White-throated Kingfisher, Japanese Pipistrelle and Unknown Vespertilionidae species 1)	Heron, Great Egret, Greater Coucal, Grey Heron, Little Egret, Small Indian Civet and Blackvein Sergeant) <u>Literature Review</u> 11 fauna species of conservation importance recorded (i.e. Black-winged Stilt, Chinese Pond Heron, Great Egret, Greater Coucal, Grey Heron, Little Egret, White-throated Kingfisher, Japanese Pipistrelle, Unknown Vespertilionidae species 1, Dingy Dusk-hawker and Chinese Bullfrog)	Sandpiper, Chinese Pond Heron, Great Egret, White-throated Kingfisher and Chinese Bullfrog) <u>Literature Review</u> Nine fauna species of conservation importance recorded (i.e. Chinese Pond Heron, Little Egret, Purple Heron, Japanese Pipistrelle, Lesser Bamboo Bat, Lesser Yellow Bat, Unknown Vespertilionidae species 1, Coastal Glider and Chinese Bullfrog)
Size/Abundance	No direct loss	0.05 ha would be lost due to the proposed alignment 0.75 ha would be impacted by temporary works	0.02 ha would be lost due to the proposed alignment 0.36 ha would be impacted by temporary works
Duration	No direct impact Indirect impact (noise, dust and glare) during construction phase would be temporary Indirect impact (noise, dust and glare) during operation phase would be permanent	Direct impact (habitat loss) on 0.05 ha will be permanent Direct impact (habitat loss) on 0.75 ha will be temporary Indirect impact (noise, dust and glare) during construction phase would be temporary Indirect impact (noise, dust and glare) during operation phase would be permanent	Direct impact (habitat loss) on 0.02 ha will be permanent Direct impact (habitat loss) on 0.36 ha will be temporary Indirect impact (noise, dust and glare) during construction phase would be temporary Indirect impact (noise, dust and glare) during operation phase would be permanent

Criterion	Pond within WBA and WCA	Pond near NTMWTW	Other Pond
Reversibility	No direct impact Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be irreversible	Direct impact (habitat loss) on 0.05 ha during construction phase would be irreversible Direct impact (habitat loss) on 0.75 ha during construction phase would be reversible Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be irreversible	Direct impact (habitat loss) on 0.02 ha during construction phase would be irreversible Direct impact (habitat loss) on 0.36 ha during construction phase would be reversible Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be irreversible
Magnitude	Low	Low to moderate	Low
Regional significance	Moderate regional significance as the habitat is uncommon regionally which is mainly restricted to north-western New Territories; supports mostly common flora and widespread fauna; and the loss may affect the regional ecological connectivity between habitats and species population	Low to moderate regional significance as the habitat is uncommon regionally and relatively concentrated which is mainly restricted to north-western New Territories; supports mostly common flora and widespread fauna; and the loss may affect the regional ecological connectivity between habitats and species population	Low to moderate regional significance as the habitat is artificial and relatively scattered; supports mostly common flora and widespread fauna; and the loss does not affect the regional ecological connectivity between habitats and species population
Overall Impact Significance	**Low**	**Low to moderate** for ponds near NTMWTW	**Low**

Table 8.22 Evaluation of Potential Ecological Impacts to Natural Watercourse, Semi-Natural Watercourse and Modified Watercourse

Criterion	Natural Watercourse	Semi-Natural Watercourse	Modified Watercourse
Habitat Quality	Low to moderate	Low to moderate	Moderate for KTMDC Low to moderate for ST-S3, ST-S5, modified watercourse in Ngau Tam Mei

Criterion	Natural Watercourse	Semi-Natural Watercourse	Modified Watercourse
			Low for the remaining modified watercourse
Species/Ecological Resources	Low diversity and abundance of flora and fauna <u>Current Survey</u> Five fauna species of conservation importance (i.e. Eurasian Coot, Grey Heron, Little Egret, Lesser Yellow Bat and *Cryptopotamon anacoluthon)* <u>Literature Review</u> One flora species of conservation importance (i.e. Lamb of Tartary) and 12 fauna species of conservation importance (i.e. Chinese Pond Heron, Greater Coucal, Grey	Low to moderate diversity of flora Moderate diversity of fauna Low to moderate abundance of wildlife <u>Current Survey</u> Nine fauna species of conservation importance (i.e. Chinese Pond Heron, Common Emerald Dove, Greater Coucal, Little Egret, Japanese Pipistrelle, Leopard Cat, Chinese Soft-shelled Turtle, Blotched Snakehead and Small Snakehead) <u>Literature Review</u> Nine fauna species of conservation importance (i.e. Chinese Pond Heron, Greater Coucal, Grey Heron, Little Egret, Chinese Noctule, Japanese Pipistrelle, Mangrove Skimmer,	Low to moderate diversity of flora Moderate diversity of fauna Moderate abundance of wildlife <u>Current Survey</u> 32 fauna species of conservation importance (i.e. Black Kite, Black-faced Spoonbill, Black-winged Stilt, Chinese Pond Heron, Collared Crow, Common Greenshank, Common Redshank, Eurasian Curlew, Eurasian Teal, Eurasian Wigeon, Great Cormorant, Great Egret, Greater Coucal, Green Sandpiper, Grey Heron, Indian Cuckoo, Little Egret, Northern Pintail, Northern Shoveler, Oriental Magpie, Pied Avocet, Spotted Redshank, White-throated Kingfisher, Wood Sandpiper, Chinese Noctule, Japanese Pipistrelle, Lesser Bamboo Bat, Unknown Myotis Species, Unknown Vespertilionidae species 1, Unknown Vespertilionidae species 2, Leopard Cat and Great Blue Spotted Mudskipper) <u>Literature Review</u> 23 fauna species of conservation importance (i.e. Black-winged Stilt, Chinese Pond Heron, Collared Crow, Common Greenshank, Common Redshank,

Criterion	Natural Watercourse	Semi-Natural Watercourse	Modified Watercourse
	Heron, Little Egret, Japanese Pipistrelle, White-banded Shadowdamsel, Brown Wood Frog, Chinese Bullfrog, Lesser Spiny Frog, *Cryptopotamon anacoluthon, Nanhaipotamon hongkongense* and *Somanniathelphusa zanklon*)	Chinese Bullfrog and Small Snakehead)	Eurasian Teal, Eurasian Wigeon, Great Cormorant, Great Egret, Greater Coucal, Grey Heron, Grey-headed Lapwing, Little Egret, Little Grebe, Marsh Sandpiper, Northern Shoveler, Pied Avocet, Wood Sandpiper, Japanese Pipistrelle, Unknown Vespertilionidae species 2, Chinese Bullfrog, *Gobiopterus macrolepis* and *Somanniathelphusa zanklon*)
Size/Abundance	0.03 ha would be lost due to the proposed alignment Negligible area would be impacted by temporary works	0.11 ha would be lost due to the proposed alignment 0.59 ha would be impacted by temporary works	0.03 ha would be lost due to the proposed alignment 2.43 ha would be impacted by temporary works
Duration	Direct impact (habitat loss) on 0.03 ha will be permanent Indirect impact (noise, dust and glare) during construction phase would be temporary Indirect impact (noise, dust and glare) during operation phase would be permanent	Direct impact (habitat loss) on 0.11 ha will be permanent Direct impact (habitat loss) on 0.59 ha will be temporary Indirect impact (noise, dust and glare) during construction phase would be temporary Indirect impact (noise, dust and glare) during operation phase would be permanent	Direct impact (habitat loss) on 0.03 ha will be permanent Direct impact (habitat loss) on 2.43 ha will be temporary Indirect impact (noise, dust and glare) during construction phase would be temporary Indirect impact (noise, dust and glare) during operation phase would be permanent
Reversibility	Direct impact (habitat loss) on 0.03 ha during construction phase would be irreversible Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during	Direct impact (habitat loss) on 0.11 ha during construction phase would be irreversible Direct impact (habitat loss) on 0.59 ha during construction phase would be reversible Indirect impact (disturbance) during	Direct impact (habitat loss) on 0.03 ha during construction phase would be irreversible Direct impact (habitat loss) on 2.43 ha during construction phase would be reversible Indirect impact (disturbance) during

Criterion	Natural Watercourse	Semi-Natural Watercourse	Modified Watercourse
	operation phase would be irreversible	construction phase would be reversible Indirect impact (disturbance) during operation phase would be irreversible	construction phase would be reversible Indirect impact (disturbance) during operation phase would be irreversible
Magnitude	Low	Low	Low to moderate
Regional significance	Low to moderate regional significance as the habitat is scarce regionally; supports mostly common flora and widespread fauna; and the loss does not affect the regional ecological connectivity between habitats and species population	Low to moderate regional significance as the habitat is scarce regionally; supports mostly common flora and widespread fauna; and the loss does not affect the regional ecological connectivity between habitats and species population	Low to moderate for KTMDC as it supports relatively high abundance of fauna. Low regional significance for other modified watercourses as the habitat is artificial and common regionally; supports mostly common flora and widespread fauna; and the loss may affect the regional ecological connectivity between habitats and species population
Overall Impact Significance	**Low**	**Low**	**Low to moderate** for KTMDC **Low** for others

Table 8.23 Evaluation of Potential Ecological Impacts to Mitigation Wetland

Criterion	Mitigation Wetland
Habitat Quality	Moderate for YLBFEW Low to moderate for West Rail Compensatory Wetland
Species/Ecological Resources	Low to moderate diversity of flora and fauna for YLBFEW Low to moderate diversity of flora and low diversity of fauna for West Rail Compensatory Wetland Low to moderate abundance of wildlife for YLBFEW Low abundance of wildlife for West Rail Compensatory Wetland <u>Current Survey</u> 17 fauna species of conservation species (i.e. Black Kite, Chinese Pond Heron, Common Greenshank, Collared Crow, Eurasian Curlew, Eurasian Teal, Great Cormorant, Great Egret, Grey Heron, Little Egret, Little Grebe, Northern Shoveler, Oriental Magpie, White-throated Kingfisher, Japanese Pipistrelle, Ruby Darter and *Nanhaipotamon hongkongense*) <u>Literature Review</u> 24 fauna species of conservation importance (i.e. Black Kite, Black-winged Stilt, Chinese Pond Heron, Cinnamon Bittern, Collared Crow, Crested Serpent Eagle, Eastern Marsh Harrier, Eurasian Coot, Eurasian

Criterion	Mitigation Wetland
	Curlew, Eurasian Teal, Great Cormorant, Great Egret, Grey Heron, Intermediate Egret, Little Egret, Little Grebe, Northern Shoveler, Pied Kingfisher, Tufted Duck, Western Osprey, White-shouldered Starling, White-throated Kingfisher, Unknown Vespertilionidae species 1 and Common Rat Snake)
Size/Abundance	No direct loss
Duration	No direct impact Indirect impact (noise, dust and glare) during construction phase would be temporary Indirect impact (noise, dust and glare) during operation phase would be permanent
Reversibility	No direct impact Indirect impact (disturbance) during construction phase would be reversible Indirect impact (disturbance) during operation phase would be irreversible
Magnitude	Low
Regional significance	Low to moderate regional significance as the habitat is artificial and scarce regionally; supports mostly common flora and widespread fauna; and the loss does not affect the regional ecological connectivity between habitats and species population
Overall Impact Significance	**Low to moderate** for YLBFEW **Low** for West Rail Compensatory Wetland

Table 8.24 Summary of Potential Ecological Impacts

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
Construction Phase – Direct Impact						
Direct impact on CA (Kai Kung Leng / Ngau Tam Shan / Hadden Hill)	Construction of permanent structures (At-grade roads, pile and pier structures, slope works, tunnel portals); Temporary works area	CA zones within Project Boundary	Small areas of habitat loss (portal, slip road, associated temporary works) and local disturbance; alignment largely by tunnel in Ngau Tam Shan / Hadden Hill reducing aboveground loss	Low	No	Nil
Direct impact on WCA / WBA / Priority Site (Deep Bay Wetland outside Ramsar Site) / IBA	Construction of permanent structures (pile and pier structures); Temporary works area	WCA / WBA / Priority Site / IBA within Project Boundary	Small area of habitat loss, mainly man-made habitats (developed area/wasteland, roadside plantation) and small area of modified watercourse (e.g. culverts); no loss of natural habitats anticipated	Low	No	Nil
Direct impact on other ecologically sensitive resources	Construction of permanent structures (pile and pier structures); Temporary works area	KTMDC	<u>Impacted area by permanent structure</u> Modified watercourse (KTMDC): 0.03 ha <u>Impacted area by temporary works</u> Modified watercourse (KTMDC): 0.85 ha	Low to moderate	Yes (Careful planning during detailed design stage to minimize works footprint; Phasing of construction works; Wetland compensation)	
Loss of wooded and vegetated habitats	Construction of permanent structures (At-grade roads; pile and pier structures, slope works, tunnel portals); Temporary works area	Woodland	<u>Impacted area by proposed alignment</u> Woodland: 10.35 ha <u>Impacted area by temporary works</u> Woodland: 6.28 ha	Woodland: Low to moderate	Yes (Careful planning during detailed design stage to minimise loss; Woodland compensation)	Nil
		Mixed woodland Shrubland Grassland Plantation	<u>Impacted area by proposed alignment</u> Mixed Woodland: 3.74 ha Shrubland: 8.68 ha Grassland: 14.81 ha	Mixed Woodland: Low Shrubland: Low Grassland: Low Plantation: Low	No	Nil

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
			Plantation: 3.16 ha Impacted area by temporary works Mixed Woodland: 9.02 ha Shrubland: 8.26 ha Grassland: 10.17 ha Plantation: 12.89 ha			
Loss of wetland habitat and watercourse	Construction of permanent structures (At-grade works; pile and pier structures); Temporary works area (conservative demarcation)	Marsh/reed	Impacted area by permanent structure Marsh/reed: 0.06 ha Impacted area by temporary works Marsh/reed: 1.07 ha	Low to moderate	Yes (Wetland compensation)	Nil
		Pond	Impacted area by permanent structure Pond near NTMWTW: 0.05 ha Other ponds: 0.02 ha Impacted area by temporary works Pond near NTMWTW: 0.75 ha Other ponds: 0.36 ha	Low to moderate for ponds near NTMWTW Low for others	Yes (Wetland compensation)	Nil
		Natural watercourse, and semi-natural watercourse	Impacted area by permanent structure Natural watercourse: 0.03 ha Semi-natural watercourse: 0.11 ha Impacted area by temporary works Natural watercourse: Negligible Semi-natural watercourse: 0.59 ha	Natural watercourse: Low Semi-natural watercourse: Low	No	Nil
		Modified watercourse	Impacted area by permanent structure Modified watercourse (KTMDC): 0.03 ha Modified watercourse (others): Negligible Impacted area by temporary works	Low to moderate for KTMDC Low for others	KTMDC: Yes (Careful planning during detailed design stage to minimize works footprint; Phasing of construction works; Wetland compensation)	Nil

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
			Modified watercourse (KTMDC): 0.85 ha Modified watercourse (others): 0.66 ha		Others: No	
Loss of man-made habitat	Construction of permanent structures (At-grade works; pile and pier structures); Temporary works area	Man-made habitats (Agricultural land, village/orchard and developed area/wasteland)	<u>Impacted area by permanent structure</u> Agricultural land: 0.04 ha Village/orchard: 7.86 ha Developed area/wasteland: 10.93 ha <u>Impacted area by temporary works</u> Agricultural land: 0.72 ha Village/orchard: 18.60 ha Developed area/wasteland: 49.03 ha	Agricultural land: Low Village/orchard: Low Developed area/wasteland: Low	No	Nil
Fragmentation and Isolation of vegetated habitats	Construction of permanent structures (At-grade works; pile and pier structures; and associated slope works)	Vegetated habitats at Kai Kung Leng	Potential barrier effect on non-flying mammals/herpetofauna movement; small scale and low impact relative to overall wooded habitat	Low	No	Nil
Direct impact on flora species of conservation importance	Vegetation clearance at at-grade works area	*Aralia chinensis*, Cycad-fern, Incense Tree, Hong Kong Pavetta	Potential mortality and loss of individuals in absence of mitigation.	Low to moderate	Yes (Pre-construction survey and transplantation / protection measures)	Nil
Direct impact on fauna species of conservation importance	Construction activities for permanent structures (At-grade roads, pile and pier structures; and slope works); Temporary works area	Avifauna species of conservation importance	Loss of habitat and foraging ground, and potential injury / mortality	Low	No	Nil
		Flying mammals (bats) and non-flying mammal species of conservation importance	Loss of habitat and foraging ground, and potential injury / mortality	Low	No	Nil
		Butterfly species of conservation importance	Loss of habitat and foraging ground, and potential injury / mortality	Low	No	Nil
		Odonate species of conservation importance	Loss of habitat and foraging ground, and potential injury / mortality	Low	No	Nil

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
		Herpetofauna species of conservation importance	Loss of habitat and foraging ground, and potential injury / mortality	Low	No	Nil
		Firefly species of conservation importance (Motschulsky's Starworm)	Loss of habitat and foraging ground, and potential injury / mortality	Low to moderate	Yes (Pre-construction survey and translocation)	Nil
		Aquatic species of conservation importance (Small Snakehead)	Loss of habitat and foraging ground, and potential injury / mortality	Low to moderate	Yes (Pre-construction survey and translocation)	Nil
Direct impact on roosting sites	Construction activities for permanent structures; Temporary works area	Sha Po Night Roost	Ecological resources located within Project Boundary, but direct encroachment has been avoided	Nil	No	Nil
		Planned bat shelter near Pok Wai bat roost		Nil	No	Nil
Bird collision	Cranes/heavy machinery; temporary structures; transparent/reflective materials	Ardeids and waterbirds using KTMDC flight corridor; general avifauna	Collision risk particularly at viaduct / pier works along Tsing Long Highway crossing KTMDC.	Low to moderate	Yes (Use of non-transparent/non-glaring materials)	Nil
Construction Phase – Indirect Impact						
Construction disturbance to Priority Site	Disturbance (Noise, glare, dust, vibration, human activities) from construction activities	Deep Bay Wetland outside Ramsar Site Priority Site for Enhanced Conservation	Some additional disturbance from construction activities, but not expected to be significant over existing disturbance Small-scale and largely temporary works	Low	No	Nil
Construction disturbance to LTCP and CA	Disturbance from construction activities	LTCP and CA, and associated natural habitats	Potential disturbance to woodland of moderate value near Kai Kung Leng; tunnel sections reduce disturbance elsewhere; existing disturbance (e.g., tomb sweeping) in some CA areas.	LTCP: Low to moderate; CA: Low	Yes (Careful control of site boundary and access)	Nil
Construction disturbance to WCA and WBA	Disturbance from construction activities	WCA and WBA in Nam Sang Wai and near San Tin Highway	Some additional disturbance from construction activities, but not	Low	No	Nil

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
			expected to be significant over existing disturbance Small-scale and largely temporary works			
Disturbance to IBA	Disturbance from construction activities	IBA	Construction disturbance including noise, glare, dust, vibration and other human activities	Very low	No	Nil
Disturbance to night roosts	Disturbance from construction activities (esp. evening / night-time disturbance, if any)	Sha Po Night Roost (~25 m from alignment); Ko Po Road Night Roost (~360 m from boundary)	Construction disturbance including noise, glare, dust, vibration and other human activities, potentially reduce the usage by roosting ardeids during overwintering season	Low to moderate for Sha Po Night Roost; Low for Ko Po Road Night Roost	Yes (Minimise disturbance with buffer area and timing control of construction activities for Sha Po Night Roost)	Nil
Disturbance to Sha Po Marsh	Disturbance from construction activities adjacent to marsh	Sha Po Marsh	Construction disturbance including noise, glare, dust, vibration and other human activities "No work zone" is incorporated into the Project design between Sha Po Marsh and proposed alignment	Low	No	Nil
Disturbance to YLBFEW	Disturbance from construction activities	Yuen Long Bypass Floodway Engineered Wetland (~40 m away)	Construction disturbance including noise, glare, dust, vibration and other human activities	Low to moderate	Yes (Good site practices; other noise, air and water quality mitigation measures; provision of screening)	Nil
Disturbance to KTMDC and waterbirds	Disturbance from construction activities	KTMDC	Construction disturbance including noise, glare, dust, vibration and other human activities may deter/obstruct fauna using KTMDC "No work zone" is incorporated into the Project design along Project Boundary across KTMDC	Low to moderate	Yes (No-work zones; phasing of construction activities to minimise impacted area; good site practices; adoption of non-dredged method for construction works; piling works would	Nil

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
					be scheduled to conduct during high tidal level (i.e. above 1.0 mCD) during dry season)	
Disturbance to West Rail Compensatory Wetland	Disturbance from construction activities	West Rail Compensatory Wetland	Construction disturbance including noise, glare, dust, vibration and other human activities	Low	No	Nil
Disturbance to planned bat shelter in Pok Wai	Disturbance from construction activities	Planned bat shelter near Pok Wai bat roost	Construction disturbance including noise, glare, dust, vibration and other human activities, with proper implementation of site control to minimise disturbance Potential barrier effect and impact on microclimate	Low to moderate for other disturbance impacts; Low for potential barrier effect and impact on microclimate	Yes (Provision of screening; use of directional lighting; good site practices; other noise, air and water quality mitigation measures)	Nil
Disturbance to planned compensation wetland for Ngau Tam Mei (NTM) Development Area	Disturbance from construction activities	Planned compensation wetland for Ngau Tam Mei (NTM) Development Area	Construction disturbance including noise, glare, dust, vibration and other human activities, with proper implementation of site control to minimise disturbance	Low to moderate	Yes (Good site practices; other noise, air and water quality mitigation measures; provision of screening)	Nil
Disturbance to planned Wetland compensation site in Kam Tin	Disturbance from construction activities	Planned Wetland compensation site in Kam Tin	Construction disturbance including noise, glare, dust, vibration and other human activities	Low	No	Nil
Disturbance to proposed Nam Sang Wai Wetland Conservation Park	Disturbance from construction activities	Proposed Nam Sang Wai Wetland Conservation Park	Construction disturbance including noise, glare, dust, vibration and other human activities	Low to moderate	Yes (Good site practices; other noise, air and water quality mitigation measures; provision of screening)	Nil

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
Disturbance to watercourses	Disturbance from construction activities; Potential water quality impacts (runoff/pollutants)	NTM-S3 (natural); NTM-S2/S4/S4a/S5 and upstream of YL-S6 (semi-natural); other watercourses	Construction disturbance including runoff, noise, glare, dust, vibration and other human activities	Low	No	Nil
Disturbance to other wetland habitats	Disturbance from construction activities; Potential water quality impacts (runoff/pollutants)	Marsh/reed and ponds adjacent to boundary (NTMWTW area; Nam Sang Wai)	Construction disturbance including noise, glare, dust, vibration and other human activities	Low to moderate	Yes (Good site practices; other noise, air and water quality mitigation measures such as runoff control and pollution prevention)	Nil
Disturbance to adjacent non-wetland habitats	Disturbance from construction activities (e.g. dust and human activities)	Mixed woodland (Tit Hang, Ki Lun Tsuen); Hillside wooded areas near Kai Kung Leng / LTCP / CA; Urbanised habitats	Construction disturbance including noise, glare, dust, vibration and other human activities	Low to moderate (for hillside wooded areas near LTCP / CA); Low (for others)	Yes (Good site practice; other noise, air and water quality mitigation measures)	Nil
Potential Obstruction and Disturbance to Flight Corridor	Construction works in KTMDC	KTMDC, avifauna transverse along KTMDC	The construction works area may potentially obstruct the flight corridor in KTMDC; Construction disturbance including noise, glare, dust, vibration and other human activities may disturb avifauna transverse along KTMDC and Sha Po Night Roost	Low to moderate	Yes (No-work zones; phasing of construction activities to minimise impacted area; good site practices; adoption of non-dredged method for construction works)	Nil
Potential water quality and water holding/water flow impacts	Runoff/erosion; construction discharge; accidental spillage; pile/pier in water	KTMDC and impacted watercourses (YL-S6, NTM-S2/3/4/4a/5, NTMDC), ponds, downstream wetlands	Potentially deterioration of water quality, increase in suspended solids and potential contaminants	Low to moderate	Yes (Water quality protection measures and drainage controls)	Nil

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
Groundwater drawdown from tunnel construction	Underground tunnel construction	Potential groundwater-dependent habitats / water regime	Potential water drawdown of watercourses in the area	Low	No	Nil
Night-time glare disturbance	Uncontrolled lighting from construction sites	Nocturnal fauna and adjacent habitats (e.g., LTCP and the potential recipient site of Motschulsky's Starworm in Ki Lun Tsuen, etc.)	Potential behavioural disruption in foraging/predation/breeding	Low to moderate	Yes (Minimise and control glare and light spill)	Nil
Ground-borne noise and vibration	Tunnel construction	Hillside plantation and grassland; associated wildlife	Noise/vibration impacts limited by lack of key roost/breeding sites and low abundance of vibration-sensitive fauna.	Low	No	Nil
Operation Phase – Direct Impact						
Direct impact on recognized sites of conservation importance	Operation of road/viaduct/tunnel	CA, WCA, WBA, Priority Site	No further direct impact beyond construction footprint; no additional habitat loss anticipated in operation phase	Nil	No	Nil
Direct impact on ecologically sensitive resources	Operation of road/viaduct/tunnel	KTMDC	No further direct impact beyond construction footprint; no additional habitat loss anticipated in operation phase	Nil	No	Nil
Impacts from Direct Injury/Mortality	At-grade sections of the alignment	Non-flying mammals utilising vegetated areas adjacent to at-grade sections	Potential direct injury or mortality by roadkill	Low to moderate	Yes (Animal barriers)	Nil
Fragmentation and Isolation of vegetated habitats	Permanent structures (pile and pier structures)	Vegetated habitats at Kai Kung Leng	Potential barrier effect on movement of species with lower mobility (e.g. non-flying mammals and herpetofauna); The permanent structures would be located at the margin of the vegetated habitats while	Low	No	Nil

Impact	Source	Receiver	Nature of Impact	Overall Impact Significance in the absence of Mitigation	Mitigation Required?	Residual Impact
			part of the sections would be in form of viaduct sections.			
Bird collision	Operation traffic; permanent structures/noise barriers	Commuting ardeids (Sha Po and Ko Po Road roosts) and general avifauna	Potential collision with reflective/transparent materials	Low	No	Nil
Operation Phase – Indirect Impact						
Disturbance on recognized sites of conservation importance and ecologically sensitive resources	Traffic noise, glare, dust, human activities	Sites within/adjacent boundary (i.e. CA, WCA, WBA, Priority Site, KTMDC, YLBFEW, planned bat shelter, marsh/reed and pond near NTMWTW, the planned wetland compensation site under the Ngau Tam Mei Development and nearby habitats)	Increase in human, noise, air quality, noise, light pollution, traffic and visual disturbance generated from nearby facilities Potential disturbance impacts to planned bat shelter such as barrier effect and impact on microclimate	Low	Yes (directional lighting)	Nil
Disturbance on species of conservation importance, wildlife and habitat	Traffic noise and visual disturbance from viaduct/road operation	Ponds outside proposed piles/viaduct; foraging avifauna	Increase in human, noise, air quality, noise, light pollution, traffic and visual disturbance generated from nearby facilities	Low	No	Nil
Obstruction and disturbance to flight corridor	Traffic noise and visual disturbance from viaduct/road operation	Flight corridor and, ardeids or waterbirds foraging in the KTMDC	Increase in human, noise, air quality, noise, light pollution, traffic and visual disturbance generated from nearby facilities	Low	No	Nil
Water quality / water holding/ water flow impacts	Surface runoff; permanent piers in KTMDC	KTMDC	Potentially deterioration of water quality, increase in suspended solids and potential contaminants	Low	No	Nil

8.10 Cumulative Ecological Impacts

8.10.1.1 The construction of the Project is tentatively scheduled to commence in 2027, with a target completion date of 2036 or earlier. A comprehensive list of concurrent projects is provided in **Section 2.9**, **Table 2.4**, and **Figure 2.3**

8.10.1.2 Potential cumulative impact may arise from the combined effects of concurrent projects within the Assessment Area, leading to various potential ecological impacts, including:

- Kwu Tung North New Development Areas (KTN NDA)
- First Phase Development of the New Territories North – San Tin/Lok Ma Chau Development Node, also known as San Tin Technopole
- Northern Link (NOL) Main Line
- Ngau Tam Mei New Development Area (NTM NDA)
- Ngau Tam Mei Water Treatment Works (NTMWTW) Extension
- The Development of Wetland Conservation Parks (WCP) System, in particular the Proposed Nam Sang Wai Wetland Conservation Park
- Other sections under Northern Metropolis Highway (NMH)

8.10.1.3 The northern end of the proposed alignment overlaps with the works sites under the KTN NDA and San Tin Technopole. A large portion of these overlapping areas mainly comprises developed area/wasteland (including San Tin Highway and Fanling Highway) and adjacent roadside plantation habitats, with small areas of hillside wooded habitats such as shrubland and mixed woodland. Considering the relatively low ecological value of these habitats and the existing disturbances from adjacent highways, no significant cumulative impact on habitat loss is expected.

8.10.1.4 The proposed alignment overlaps with the work sites associated with the NOL Main Line, NTM NDA, and the NTM WTW Extension. The overlapping areas with the NOL Main Line and NTM NDA mainly comprise village/orchard habitat of relatively low ecological value, while the NTM WTW Extension mainly encompasses developed area/wasteland habitat. Nonetheless, some wetland habitats (such as pond and marsh/reed) exist between the proposed NTM NDA and the NTM WTW Extension, which would be encroached upon by the proposed alignment. While these wetland habitats may experience some direct impacts, they have been proposed as a wetland compensation site under the Project. Upon implementing wetland compensation and enhancement measures, the scattered wetland habitats with relatively low ecological value in the area are expected to be consolidated and enhanced as a well-connected, actively managed wetland system, improving habitat quality and ecological connectivity, thereby creating environmental benefits. Consequently, no adverse cumulative ecological impacts on the habitats are expected.

8.10.1.5 The southern portion of the proposed alignment, near Kam Tin River, overlaps with a wetland compensation site proposed under the NTM NDA (CEDD, 2025), while the proposed Nam Sang Wai Wetland Conservation Park is situated adjacent to the proposed alignment, with direct encroachment avoided. The details of the Nam Sang Wai Wetland Conservation Park are currently under development but are expected to result in improved and enhanced habitat quality. Information regarding the compensation wetland under NTM NDA is provided in a separate EIA report. Special attention would be given during the detailed design stage to address

interfacing issues with the construction of the compensation wetland to minimise potential indirect impacts. For instance, low-lying habitats such as waterbodies and marsh planting areas with short vegetation or other suitable shade-tolerant species would be arranged in areas that may be shaded under the viaduct of the Project alignment. Considering the current urbanised nature of this developed area/wasteland, along with the expected environmental benefits from the implementation of the compensation wetland, no adverse cumulative ecological impacts are expected.

8.10.1.6 As the details and alignments of other sections under the NMH were unavailable at the time of this EIA study, potential cumulative impacts with these sections will be assessed in the subsequent studies. Nonetheless, considering that the other sections under NMH is expected to extend to Tin Shui Wai to the southwest of this Project and further north to Kwu Tung and the NTN New Town area, the overlapping impacted area with this Project would be limited. Moreover, underground tunnel and viaduct sections would also be adopted for other NMH sections, which would avoid and minimise direct impact on habitats, subject to details under the other sections. No significant or unacceptable disturbance impacts are anticipated to arise from the construction activities associated with this Project. With additional appropriate mitigation measures in place to minimise potential direct and indirect impacts, the overall cumulative ecological impact is expected to be low at the time of this EIA study.

8.11 Mitigation of Adverse Ecological Impacts

8.11.1 General Principle on Mitigations Measures

8.11.1.1 According to *Annex 16* of the *EIAO-TM*, the general policy for mitigating impacts on important habitats and wildlife, in the order of priority, are avoidance, minimisation, and compensation.

8.11.1.2 Under the current Project, key ecological impacts include direct impact on woodland habitat and small areas of wetland habitats (such as ponds and marsh/reed near the NTMWTW, as well as a short section of modified watercourse at the KTMDC). Additionally , there may be potential disturbances to Sha Po Night Roost and impacts on species of conservation importance, as discussed in **Section 8.8** and **Table 8.24**. Mitigation measures are proposed for each of these ecological resources, as detailed in the following sections.

8.11.2 Avoidance Measures

Avoiding Recognised Sites of Conservation Importance

8.11.2.1 Recognised sites of conservation importance, including Lam Tsuen Country Park (LTCP) located adjacent to the Project Boundary (**Figure 8.1** refers). The Project alignment has been carefully designed and refined to avoid direct encroachment, thereby preserving this site of conservation importance and the associated ecological resources. On the other hand, despite a small area of WCA encroaches within the Project Boundary in Kam Tin, 'no work zone' has been designed to avoid direct encroachment of construction works to the WCA area.

Avoiding Other Ecologically Sensitive Resources

8.11.2.2 *Adoption of Viaduct and Tunnel Sections*: The current proposed alignment incorporates sections of viaduct and tunnel instead of a complete at-grade design, with ratio of approximately 1: 1.14: 1.95 for at-grade, tunnel, and viaduct sections, respectively. This design minimises extensive site clearance and site formation works, significantly reducing the area of habitat loss required under the Project (e.g. avoiding extensive site formation over low-lying wetland habitats by adopting viaduct section, and over hillside wooded habitat by adopting tunnel section).

8.11.2.3 *Adoption of "No Work Zone"*: The Project Boundary includes land required for the Project, allowing for further refinement by the Contractor's planning and design during the construction stage. For impact assessment purposes, all habitats within the Project Boundary are considered potentially impacted areas to ensure a robust evaluation of ecological impacts. Nonetheless, considering the large extent of the Project Boundary and the presence of sensitive ecological resources, a "no work zone" has been established within the Project Boundary to ensure the habitats and associated resources are not encroached upon by construction activities, thereby avoiding direct impacts on these habitats. The "no work zone" delineates areas in wetland and woodland habitats, including the KTMDC (which encompasses the modified watercourse habitat within the WCA and the Deep Bay Wetland outside Ramsar Site Priority Site for Enhanced Conservation), the planned compensation wetland under the Ngau Tam Mei Development, the woodland habitat east of Pok Wai, and the pond habitat near NTMWTW.

8.11.2.4 *Wetland Habitats at Ngau Tam Mei*: Under the original alignment, an interchange (along with associated site office and ancillary facilities) was proposed to the west of NTMWTW, resulting in extensive loss of pond and marsh/reed habitats. The current proposed alignment has been refined, relocating the interchange to the south of Ngau Tam Mei Tsuen. The direct impact on habitats to the west of NTMWTW will be limited to the footprint of the pile and pier-related works, as well as some temporary impacts during the construction phase, thereby avoiding extensive loss of sensitive wetland habitats.

8.11.2.5 *Proposed Nam Sang Wai Wetland Conservation Park*: The proposed Nam Sang Wai Wetland Conservation Park (NSW WCP) is located in close proximity to the Project Boundary (**Figure 8.1** refers). The Project Boundary has been carefully designed and refined to avoid direct encroachment, preserving these sites of conservation importance and their associated ecological resources.

8.11.2.6 *Watercourses along the Proposed Alignment*: A natural watercourse (NTM-S3) has been identified within the Project Boundary, as discussed in **Section 8.8.1.16**. The locations of the proposed pile and pier structures will be fine-tuned, and construction activities within the Project Boundary will avoid direct encroachment onto this natural watercourse to prevent adverse ecological impact.

8.11.3 Minimisation Measures

Phasing of Construction Activities to Minimising Habitat Loss

8.11.3.1 To further minimise potential adverse impacts to the ecologically sensitive habitats (i.e. KTMDC and ponds near NTMWTW), phasing of construction works would be adopted in these areas under the Project. This phased construction in KTMDC and ponds near NTMWTW aims to reduce the scale of disturbance and the total area of construction activities. Specifically, construction activities would be conducted in

multiple phases over a period of at least 2 years. This approach will minimise direct impacts and disturbances to these habitats while allowing for the gradual migration of wildlife to nearby available habitats.

8.11.3.2 As construction activities would be phased over two years or more, the area occupied by actual construction (i.e. pond habitat near NTMWTW and KTMDC) would not exceed half of the designated temporary works area at any given time. The remaining half would be maintained as pond habitat or watercourse. The impacted areas from temporary works will be reinstated to their pre-existing condition as much as practicable before commencing construction on the remaining half. For instance, steel casing would be used to isolate the surrounding riverbed when bored pile construction method is employed in KTMDC. The material within the steel casing will be excavated and confined within the casing for the placement of reinforcement and concrete for the pile, protecting the watercourse area of KTMDC outside the steel casing from being affected. Similarly, for pond habitat near NTMWTW, construction activities would occupy no more than half of the designated temporary works area at any given time. The temporarily affected ponds would be reinstated to their pre-existing conditions before works commence on the remaining ponds, thereby maintaining at least half of the pond habitat throughout the construction phase near NTMWTW.

Minimising Construction Phase Impact on Sha Po Night Roost

8.11.3.3 Given the close proximity of the construction works area to the Sha Po Night Roost, measures will be taken to minimise disturbance to the night roost. Noisy activities, such as pilling and the use of heavy machinery, should be avoided within 100 m of the Sha Po ardeid night roost during the period from one hour before sunset to one hour after sunrise. Sunrise and sunset times will be referenced from the Hong Kong Observatory. Additionally, other appropriate mitigation measures, such as noise barriers and good site practices, as detailed in **Section 8.11.3.15**, will be implemented near the Sha Po Night Roost to further minimise impacts from construction activities.

Minimising Construction Phase Impact on Planned Bat Shelter

8.11.3.4 A planned bat shelter is situated in Pok Wai, adjacent to the viaduct section of the Project. The alignment has been carefully designed to avoid direct impact to this planned bat shelter. To minimise indirect impact on the planned bat shelter, appropriate mitigation measures will be implemented, including the provision of screening, use of directional lighting, noise barriers, good site practices, and controls of air quality, water quality, and glare, as detailed in **Section 8.11.3.15** to **8.11.3.20**.

Minimising Construction Phase Impact on Planned Compensation Wetland

8.11.3.5 A planned wetland compensation site under the Ngau Tam Mei Development is located between Tsing Long Highway and San Tam Road in the Sha Po area. Although this wetland compensation site falls within the Project Boundary, a “no work zone” has been established in this area during the establishment period of the Ngau Tam Mei Development wetland compensation site to avoid direct impact and minimise indirect impacts, such as change in natural light caused by the construction activities. To further protect the planned wetland compensation site and its associated wildlife, measures will include erecting hoardings to clearly

demarcate the construction boundary, as well as implementing good site practices to minimize indirect disturbances, including runoff, noise, dust, and glare. With the proper implementation of the "no work zone" and relevant mitigation measures during the construction phase of the Project, both direct and indirect impacts to the planned wetland compensation site under the Ngau Tam Mei Development is anticipated to be low.

Transplantation of Flora Species of Conservation Importance

8.11.3.6 Several species of conservation importance were recorded within the footprint of the proposed work sites/areas. While no major adverse impacts are expected on fauna species with relatively high mobility, direct impact on certain flora species of conservation importance, such as *Aralia chinensis* found in mixed woodland along Ching Yau Road near Project Boundary, would be avoided during the construction phase.

8.11.3.7 Nonetheless, the proposed construction activities may lead to the direct loss of four floral species of conservation importance, including clusters of Cycad Fern, mature individuals of Incense Tree, *Aralia chinensis* in Kai Kung Leng hillside woodland and Ki Lun Tsuen mixed woodland and Hong Kong Pavetta in village/orchard habitat in Ngau Tam Mei area. These individuals, recorded within the Project footprint, could be directly lost due to the construction activities if not properly mitigated. All unavoidably affected flora individuals of conservation importance should be preserved in-situ or transplanted to nearby suitable habitat(s) prior to the commencement of site clearance as a last resort. A Detailed Vegetation Survey should be undertaken in the identified affected areas (i.e. Kai Kung Leng hillside and Ki Lun Tsuen mixed woodland) by a suitably qualified botanist or ecologist. This survey will identify any potentially affected plant species of conservation importance, ascertain their presence, update their physical conditions, and determine the abundance and locations of the flora species of conservation importance prior to the commencement of any site clearance works. A Protection and Transplantation Proposal, including subsequent monitoring for the affected individuals, should be prepared and conducted by a suitably qualified local ecologist or botanist with at least seven years of relevant experience. The Proposal must be submitted for approval to the relevant government departments (e.g. AFCD and EPD) at least two months before the commencement of transplantation works. If plant preservation or transplantation is deemed impractical by the qualified ecologist or botanist (e.g. due to poor health and low survive rate of the plants), alternative mitigation measures, such as compensation by seedling planting, should be considered.

Translocation of Fauna Species of Conservation Importance

8.11.3.8 Fauna species of conservation importance were recorded within the footprint of Project area. While no adverse direct impacts are expected on fauna groups with relatively high mobility (e.g. avifauna, mammals, butterflies and odonates), there is a potential for direct injury or mortality to slow-moving fauna or aquatic fauna (e.g. freshwater fauna and fireflies) due to the Project. Mitigation measures, including translocation for affected fauna groups should be implemented as discussed below.

8.11.3.9 A population of Small Snakehead has been recorded in the modified watercourse (NTM-S4) within the Project Boundary, which may be affected by nearby construction activities. A pre-construction survey would be conducted in the area of

concern for this species of conservation importance, followed by measures to capture and translocate individuals of Small Snakehead to nearby suitable habitats (e.g. upstream locations of the watercourse) that are free from development pressure prior to the commencement of the relevant construction works. The preparation of Translocation Proposal, including the translocation methodology, subsequent monitoring of the affected individuals, screening and selection of recipient sites, and the capture–release process, should be conducted by qualified ecologist with at least seven years of relevant experience. This Proposal must be submitted for approval to the relevant government departments (e.g. AFCD and EPD) at least two months before the commencement of translocation works.

8.11.3.10 A population of Motschulsky's Starworm was recorded in woodland habitat on the foothills of Hadden Hill in Ki Lun Tsuen within the Project Boundary, which shall be mitigated by translocation to minimise direct impact to this species due to any loss of woodland habitat in this area. The number of Motschulsky's Starworm, including both female and male as well as larvae, should be included in the capture during the survey. The recipient site should be capable of supporting both the existing population and translocated individuals, with minimal light intensity, artificial light, and disturbance. It should be safeguarded against threats and risks, such as foreseeable planned works and developments, incompatible land-use change, fragmentation, human disturbances etc. in order to be considered suitable. During the translocation exercise, suitable habitats (e.g. soil, grass, withered twig and leaves) at the translocation donor site should be searched for both female and male adults and larvae individuals of Motschulsky's Starworm using appropriate capturing methods. Prior to the commencement of construction works, the translocation capturing should be carried out repeatedly throughout the adult flight period, and each session should continue until no more fireflies species of conservation importance are identified. The number of translocated firefly species of conservation importance at the translocation recipient site should be counted during the peak season in the subsequent year, in addition to the first year, to evaluate the survival rate in post-translocation. Potential capture methods for adult and larvae firefly species of conservation importance include sweep netting, hand capture, light trapping, and direct searches on soil, grass, withered twig, and leaves. Preparation of Translocation Proposal including the translocation methodology and the subsequent monitoring for the affected individuals, screening/selection of recipient sites and detailed capture – release process should be conducted by qualified ecologist with at least 7 years relevant experience. The Proposal should be submitted for approval from relevant Government departments (e.g. AFCD and EPD) at least two months before commencement of translocation works.

Minimising Direct Injury or Mortality of Wildlife

8.11.3.11 The majority of the wildlife species recorded within the Assessment Area are widespread and common in Hong Kong. These species are already subject to regular disturbance from nearby developed area/wasteland, including heavy road traffic and highway, open storage, workshops, and residential areas. Species with high mobility are anticipated to disperse and migrate upon the commencement of construction activities. Furthermore, proper screening measures, such as hoarding or barrier, would be provided to restrict and confine the construction activities within the Project Boundary, thereby preventing wildlife from accidentally entering the area.

Minimising Bird Collision

8.11.3.12 The risk of bird collision may increase with the use of transparent or reflective building façades. Considering the presence of avifauna species of conservation importance and the commuting activity of birds in the vicinity, potential bird collision should be mitigated by using low-reflective materials (e.g. tinted glass, low-reflective window film). Non-transparent panels should also be used as noise enclosures, along with non-glaring tinted materials or dark patterns superimposed on the majority of glazing along noise barriers. These measures align with the *Guidelines on Design of Noise Barriers* (EPD & HyD, 2022) and Practice Notes No. BSTR/PN/003 (Revision F) *Noise Barriers with Transparent Panels* (HyD, 2022), to avoid and minimise bird mortality from collisions.

Other Measures to Minimise Impact from Construction Disturbance

Minimising Disturbance to Habitats, Recognised Sites of Conservation Importance and Wildlife

8.11.3.13 While the indirect impacts on recognised sites of conservation importance, ecologically sensitive resources, habitats, and wildlife are considered minor, minimisation measures are recommended to further minimize potential environmental disturbances.

8.11.3.14 Provision of screening (e.g. hoarding) for demarcation of the construction site and good site practices should be adopted in works area near or within recognised sites of conservation importance and other ecologically sensitive resources during the construction phase in order to minimise disturbance impacts to the recognised sites of conservation importance (e.g. WCA, WBA, CA, and LTCP) and other ecologically sensitive resources (i.e. area near planned bat shelter, Sha Po Marsh, and YLBFEW) within the assessment area, while avoiding unintentionally access to the nearby natural habitats.

8.11.3.15 In order to minimize the indirect construction disturbances, including runoff, noise, dust, and glare, to the nearby habitats and the associated wildlife, general good site practices should be implemented, including:

- Restricting construction activities within clearly demarcated works boundaries;
- Using Quality Powered Mechanical Equipment (QPME);
- Using movable noise barrier/noise screening structures;
- Adopting regular water spraying to minimise impacts from dust deposition on adjacent vegetation and habitats during the construction activities;
- Covering trucks or transporting waste in enclosed containers to minimize windblown litter; and
- Using directional lighting, light shields, and optimizing the intensity of light to avoid excessive illumination.

8.11.3.16 While disturbance impacts during operation phase are expected to be lower than that in construction phase, potential glare and noise could still indirectly impact LTCP during operation phase. Mitigation measures such as avoiding/minimizing the installation of strong light sources (i.e. large billboard and strong lighting in

infrastructure) and minimisation of noise (i.e. implementation of noise barriers) near the recognised sites of conservation importance and other ecologically sensitive resources, particularly in the works section near Long Ha, planned bat shelter, and the potential recipient site of Motschulsky's Starworm in Ki Lun Tsuen, are recommended.

8.11.3.17 In addition, to minimise disturbance impacts on KTMDC and the associated wildlife, a non-dredging method for construction works would be adopted in KTMDC. For instance, steel casing would be implemented to isolate the surrounding riverbed when the bored pile construction method is used. The material within the steel casing will be excavated and confined within the casing for placing reinforcement and concrete of the pile. Furthermore, in view of the higher usage by waterbirds during low tide level in KTMDC during the dry season, piling works would be scheduled to conduct during high tidal level (i.e. above 1.0 mCD) when the riverbed of KTMDC is mostly submerged. As a result, disturbance to the watercourse and avifauna due to the construction works would be minimised.

Minimisation of Water Quality and Water Flow Impacts

8.11.3.18 To avoid and minimise any adverse water quality impacts to the surrounding habitats, the good site practices outlined in the *ETWB Technical Circular (Works) No. 5/2005 Protection of Natural Streams/Rivers from Adverse Impacts arising from Construction Works*, and the *ProPECC PN 2/24 Construction Site Drainage*, should also be adopted where applicable.

8.11.3.19 Mitigation measures for water quality impacts during construction and operation phases, where applicable, should include:

- Discharging surface runoff from construction sites into storm drains via adequately designed sand/silt removal facilities such as sand traps, silt traps, and sedimentation basins;
- Providing channels or earth bunds or sandbag barriers on the construction site to properly direct stormwater to these silt removal facilities;
- Installing catchpits or perimeter channels on the construction site to intercept storm runoff from outside the site, preventing it from washing across the site;
- Implementing Best Management Practices (BMPs) for stormwater discharge to reduce stormwater pollution arising from the Project; and
- Providing screening facilities such as standard gully grating and trash grille, with spacing which is capable of screening large substances like fallen leaves and rubbish at the inlet of the drainage system.

Minimisation of Night-time Disturbance

8.11.3.20 Mitigation measures should be implemented to minimise glare impacts on adjacent habitats and associated wildlife arising from construction activities. A balance between lighting for safety and avoiding excessive illumination can be achieved through the use of directional lighting to prevent light spill into sensitive areas (e.g. the planned bat shelter, the potential recipient site of Motschulsky's Starworm in Ki Lun Tsuen, and recognised sites of conservation importance such as LTCP and CAs to the south of the Project Boundary, WBA, and WCA to the west of Tsing Long Highway, which are near construction site). Additionally, controlling of night-time lighting periods, particularly for works site(s) located in proximity to these areas, will

help minimise potential indirect impacts on nocturnal fauna, including nocturnal avifauna, bats, non-flying mammals, herpetofauna, and firefly species in the vicinity.

8.11.4 Compensation Measures

Habitat Compensation Requirements

8.11.4.1 In accordance with Annex 16 of the EIAO-TM, the ecological values of the affected habitats were reviewed, and potential ecological impacts during construction and operation phases were identified and evaluated, taking in account existing, committed, and planned projects. According to Annex 16 of the EIAO-TM, the general policy for mitigating impacts on important habitats and wildlife prioritizes avoidance, minimisation, and compensation. Compensation would only be proposed when no other practicable mitigation measures are available to address the direct and indirect impacts. Furthermore, "from an ecological point of view, mitigation measures for ecological impact shall preferably be carried out on-site, and well in advance of the works, rather than off-site and after the completion of the works". The need for, as well as the type and scope of, off-site ecological mitigation measures to be adopted for a particular project shall be determined by the guidelines listed in **Section 5.4.5** of the Annex 16 of the EIAO-TM.

8.11.4.2 As described in **Section 8.8** and **Table 8.24**, low to moderate ecological impacts may arise from the loss of woodland, marsh/reed, pond (ponds near NTMWTW), and modified watercourse (KTMDC). To address temporary impacts on wetland habitats in ponds near NTMWTW and KTMDC during the construction phase, construction activities will be phased over two or more years, thereby reducing the actual impacted area at any one time (detailed in **Section 8.11.3.1**). Habitat compensation based on a "like-for-like" basis for the direct loss of aforementioned habitats is recommended under the Project and is summarised in **Table 8.25**.

Table 8.25 Summary of Compensation Requirement

Habitat	Direct Loss during Construction Phase	Total (ha)
	Area (ha)[(1)]	
Woodland Compensation		
Woodland	16.63	**16.63**
Wetland Compensation		
Marsh/Reed	1.13	**2.81**
Pond near NTMWTW)	0.80	
Modified Watercourse (KTMDC)	0.88	

Remark:
(1) Within the Project Boundary, approximately 1 ha of wetland habitats (including marsh/reed, pond, and natural watercourse habitats) and 0.05 ha of woodland habitat are overlapping with the concurrent Development at Ngau Tam Mei project. The loss of these habitats (i.e. marsh/reed, pond, natural watercourse and woodland habitats) has already been assessed and addressed under the approved EIA for the concurrent project. Thus, are excluded from the present calculation of habitat loss for this Project.

Compensation of Woodland Habitats

8.11.4.3 The Project will result in a direct loss of 16.63 ha of woodland during the construction phase. To compensate for this unavoidable loss, woodland

compensation would be provided on a "no net loss", "like for like" basis. Compensatory planting would be undertaken for the loss of the 16.63 ha woodland of moderate ecological value. Suitable sites have been identified mainly based on their present ecological value and connectivity with existing wooded areas. Other considerations included topography, soil conditions, and hill fire risk.

8.11.4.4 Due to the limited availability of suitable on-site area within the Project Boundary, off-site woodland compensation is proposed after all feasible design refinements and practicable on-site measures have been exhausted. A "no work zone" has been incorporated into the Project design to delineate areas where no construction activities will occur, minimising loss or damage to the ecological habitats and associated wildlife as far as practicable. However, further avoidance of direct impacts on existing woodland is not feasible due to engineering, environmental, and site constraints. In addition, ongoing and planned developments in the adjacent areas make it unfeasible to identify a contiguous block of 16.63 ha compensation area within the Project Boundary. Off-site compensation offers the advantage of consolidating scattered woodland habitat into a larger, contiguous woodland alongside other planned compensatory woodlands, enhancing ecological function and connectivity with nearby wooded habitats. A suitable area has been identified near the compensatory woodland for the Development of Lok Ma Chau Loop project (CEDD & PlanD, 2013a) and the San Tin / Lok Ma Chau Development Node (CEDD & PlanD, 2024). This compensatory woodland is situated within the Green Belt zone which was originally of grassland/shrubland habitat (ibid.) and falls outside Permitted Burial Grounds. This grassland/shrubland habitat is a common habitat with lower ecological value in the region. In addition, grassland/shrubland habitat could also be found in nearby areas including approximately 53 ha of grassland/shrubland recorded under approved EIA report North East New Territories New Development Areas (AEIAR-175/2013) in area to the west of Crest Hill. Establishing compensatory woodland in this area will increase the overall compensatory woodland area and strengthen ecological connectivity with adjacent wooded areas. A large, coherent compensatory woodland will foster natural succession, enhance habitat quality, and potentially increase the area's overall ecological value. As the compensatory planting will not involve heavy machinery or extensive ground disturbance, the impacts associated with tree planting and subsequent maintenance are expected to be minimal. As the compensatory planting works will not involve heavy machinery or extensive ground disturbance, the impacts associated with tree planting and subsequent maintenance are expected to remain minimal.

8.11.4.5 Native species of different growth forms with high market availability are preferred for compensatory planting. In addition, individuals of Cycad Fern and Incense Tree requiring transplantation could be accommodated in this compensatory woodland if suitable recipient sites near the Project Boundary are unavailable. At maturity, the compensatory woodland would develop multiple structural layers, including canopy, middle layer, and understory, providing habitat complexity that enhances overall ecological value.

Woodland Compensation Plan

8.11.4.6 To facilitate the successful establishment of the compensatory woodland, a detailed Woodland Compensation Plan should be prepared by qualified ecologists / botanist with at least seven years of relevant experience. The Woodland Compensation Plan should include (i) the extent of woodland loss, including site locations and

sizes, and plant species and their numbers to be felled; (ii) plant species selected for compensation with justifications; (iii) the condition and size of planting site(s); (iv) a planting scheme that includes site preparation, size standards for planting, planting spacing, and matrix for each planting site; (v) planting schedule that includes a table listing the implementation party, location, and timing; (vi) fire control measures for planting; and (vii) post-planting monitoring and maintenance programme, detailing implementation requirements, management needs, and monitoring protocols (e.g., methodology, schedule, frequency of monitoring, and monitoring parameters). The Woodland Compensation Plan shall be submitted to the relevant authorities (including AFCD and EPD) for approval at least three months before the commencement of compensatory woodland planting. Site clearance works of the Project affecting the existing woodland shall only start after the commencement of compensatory woodland planting. The compensatory woodland planting shall be completed before the operation of the NMH-San Tin Section. Upon completion of the compensatory woodland planting, a six-year monitoring programme conducted by a qualified ecologist or botanist with at least seven years of relevant experience shall be implemented to ensure the proper establishment of this compensatory woodland. The six-year establishment period (including monitoring) would be conducted by the Project Proponent /or Contractor. The monitoring frequency should be monthly during the first year following the completion of the compensatory planting, and quarterly for the next five years of the monitoring period. Parameters, such as health condition and survival rate of the trees/plants, as well as the presence of weedy species, should be monitored. Maintenance works (e.g. irrigation, weeding, pruning, pest and disease control, replacement planting, and damage repair) shall also be conducted by the Project Proponent and Contractor as necessary during the monitoring period.

Compensation of Wetland Habitats

8.11.4.7 Low to moderate ecological impacts may arise from the direct impact on marsh/reed, ponds near NTMWTW, and KTMDC. As described in **Table 8.25**, the direct loss of these wetland habitats during construction phase would result in a wetland compensation requirement of about 2.81 ha. The compensation for these habitats would be recommended based on the re-provision of similar species/habitats or enhancement of existing habitats basis.

8.11.4.8 Consideration has been given to existing condition, size, connectivity with nearby wetlands, land ownership, and water holding/water flow suitability in identifying potential wetland compensation sites. Priority has been given to potential on-site locations. Based on these factors, a proposed compensation site has been identified immediately west of the NTMWTW, covering about 5 ha, which includes 2.81 ha of village/orchard habitat (**Figure 8.8** refers). This proposed site is currently located on private land but would be resumed under the Project as part of the construction activities for the proposed alignment. The site comprises a mixture of village/orchard and pond habitats with low to moderate ecological values, surrounded by semi-natural watercourses. In addition, the ponds and marsh/reed habitats within the proposed wetland compensation site will be retained as an enhancement measure for the overall NMH Project. Upon completion of construction and establishment of the compensatory wetland habitats, the impacted area from temporary works in pond habitat near NTMWTW and KTMDC would be reinstated to their pre-existing conditions, resulting in a net increase in wetland habitats. Within this proposed wetland compensation site, the existing ponds (primarily consisting of concretised pond bunds) would also be enhanced and

integrate with the newly created wetland, thereby increasing their ecological value through appropriate wetland design under the Habitat Creation and Management Plan. This design may include:

- Consolidating small and fragmented ponds into larger wetland with enhanced connectivity,
- Physical modification of pond structures and pond bunds to allow more gently sloping bank, allowing for shallow areas for foraging avifauna, and
- Creating habitat islands and/or floating platforms, etc.

8.11.4.9 The proposed compensation works at this site maximise environmental benefit and ecological connectivity within existing and surrounding wetland habitats. Converting this area as a wetland compensation site would ensure that the retained wetland habitats are properly managed and maintained upon the completion of construction, rather than reverting to abandoned ponds or marsh/reed areas. A Habitat Creation and Management Plan is recommended to facilitate the proper design of the wetland habitats and their composition within this site, including optimising wetland design beneath the proposed viaduct structure. Given that other sections of the NMH are undergoing separated assessments, the feasibility of integrating the respective wetland compensation areas from these sections of NMH to the area adjacent to the proposed wetland compensation site under San Tin section will be explored in corresponding assessments to form a larger and contiguous wetland, if applicable.

Habitat Creation and Management Plan (HCMP)

8.11.4.10 The implementation details of the proposed compensation wetland, including the design and composition of the proposed wetland habitats, and the associated management and monitoring requirements (e.g. monitoring location, frequency and parameters) will be provided in subsequent HCMP. The HCMP shall be prepared by a qualified ecologist with at least seven years of relevant experience and shall be submitted to the relevant authorities (including AFCD and EPD) for approval at least three months before the commencement of construction of the compensatory wetland. The HCMP should set out design details (including specifications for the target habitats and species), location of the wetland compensation, and an implementation schedule in tabular form that clearly lists the measures to be implemented, the responsible party, the implementation programme, the required environmental performance for each measure, the management strategy, and the ecological monitoring and audit requirements for the wetland compensation area. Construction works at the wetland habitat (i.e. marsh/reed and pond habitat) to the west of NTMWTW shall commence only after the construction of the compensatory wetland has begun. The construction of the compensatory wetland shall be completed before the operation of the NMH-San Tin Section. Upon completion of the compensatory wetland, a 12-month monitoring period shall be undertaken by a qualified ecologist with at least seven years of relevant experience to ensure proper establishment of the compensatory wetland. The 12-month establishment period (including monitoring) would be conducted by the project proponent / its contractor. A Draft HCMP is appended in Appendix 8.6 of the EIA report.

8.12 Evaluation of Residual Impacts

8.12.1.1 Potential ecological impacts arising from the Project have been evaluated in accordance with the criteria under the relevant Annexes 8 and 16of the EIAO-TM,

as presented in **Sections 8.8** and **8.9**. The proposed mitigation measures for these ecological impacts are presented in **Section 8.11**, while a table of potential ecological impacts, corresponding mitigation measures, and potential residual impacts summarised in **Table 8.24**. A summary of the ecological impacts and potential residual impacts are provided below.

8.12.1.2 Direct impacts on habitats from the permanent structures and associated construction activities under the proposed Project alignment are identified and presented in **Section 8.8**. While generally low ecological impacts are anticipated, some habitat loss may lead to low to moderate ecological impacts, particularly affecting woodland, marsh/reed, pond, and modified watercourse habitats. Mitigation measures have been proposed accordingly in the aforementioned sections, including minimisation measures (e.g. phased construction activities), and compensation measures (e.g. woodland and wetland compensation). With the implementation of these proposed mitigation measures, residual ecological impacts from habitat loss are not anticipated. Furthermore, the wetland compensation measure is anticipated to achieve environmental benefit by enhancing overall habitat quality at the compensation site.

8.12.1.3 The construction and operation of the proposed development would result in potential disturbance to Sha Po Night Roost which occurs within the Project Boundary and adjacent to the proposed alignment. Mitigation measures were proposed to minimise disturbance impact to this night roost and the associated flight paths, including the establishment of a 100 m buffer area from the night roost, where noisy works would be subject to timing control. With the proper implementation of mitigation measure, no residual impacts are expected arising from the potential disturbance to the night roost.

8.12.1.4 Species of conservation importance have been recorded within the Project Boundary. In view of the low mobility of flora species and certain fauna species, appropriate mitigation measures have been proposed accordingly (e.g. transplantation of vegetation species of conservation importance, translocation of firefly species of conservation importance). Residual impacts on these species of conservation importance are not anticipated.

8.12.1.5 Given the potential ecological impacts will be mitigated through relevant mitigation measures, and with proper implementation, no residual ecological impacts are anticipated.

8.13 Environmental Monitoring and Audit

8.13.1.1 Key mitigation measures on specific ecological resources are summarised below, including transplantation/translocation of species of conservation importance, woodland compensation and wetland compensation. These measures should be monitored and audited by qualified local ecologist(s)/botanist(s) with relevant experience to ensure proper implementation. Furthermore, regular site audits should be carried out throughout the construction phase to ensure the recommended avoidance and minimisation measures are properly implemented. In event of non-compliance, the contractor should be informed and the proposed measures strengthened accordingly. Details of EM&A requirements are provided in the standalone EM&A Manual.

<u>Transplantation of Flora Species of Conservation Importance</u>

8.13.1.2 All flora species of conservation importance (e.g. Cycad-fern, *Aralia chinensis*, Hong Kong Pavetta and Incense Tree) should be protected and preserved on site as far as practicable. Where unavoidable, affected individuals should be transplanted to nearby suitable habitat(s) prior to the commencement of site clearance as a last resort. A detailed pre-construction vegetation survey should be conducted by a qualified botanist/ecologist with at least seven years of relevant experience to identify and record the affected individuals. A Plant Protection and Transplantation Proposal including subsequent monitoring of the affected individuals, should be prepared by a qualified ecologist/botanist with at least seven years of relevant experience. The Proposal should be submitted to relevant government departments (e.g. AFCD and EPD) for approval at least two months before the commencement of transplantation works.

8.13.1.3 Following transplantation, post-transplantation monitoring should be implemented to assess the health conditions and survival of the transplanted individuals. A three-year establishment period should be provided for flora species of conservation importance. Monitoring should be conducted bi-weekly during the first three months and monthly throughout the remaining establishment period. Details of post-transplantation monitoring are provided in the standalone **EM&A Manual**.

Translocation of Fauna Species of Conservation Importance

8.13.1.4 Fauna species of conservation importance with low mobility should also be preserved as far as practicable, including one firefly species, namely Motschulsky's Starworm, and one freshwater fish species, namely Small Snakehead. The species should be translocated to nearby suitable habitat(s) prior to the commencement of the relevant parts of the construction works. A detailed pre-construction survey should be conducted by a suitably qualified ecologist to identify and record affected individuals prior to the commencement of the relevant construction works. A Translocation Proposal, including the translocation methodology and subsequent monitoring of the affected individuals, should be prepared and implemented by a qualified ecologist with at least seven years of relevant experience. The Proposal should be submitted to the relevant government departments (e.g. AFCD and EPD) for approval at least two months before the commencement of translocation works.

Monitoring of Compensatory Woodland

8.13.1.5 Monitoring of the compensatory woodland should be performed on a regular basis after the first planting to track the survival and establishment of trees, and to document wildlife use. Surveys at each compensatory woodland location will commence after the first planting. Randomly selected individuals of each planted species will be tagged, and their survival rate will be calculated and evaluated. Supplementary planting will be recommended if necessary. Wildlife use of the planted vegetation will also be monitored. Details of the monitoring will be included in the Woodland Compensation Plan to be submitted in the detailed design stage and agreed with the relevant authorities.

Monitoring of Compensatory Wetland

8.13.1.6 A wetland compensation site will be provided to offset the loss of 2.81 ha of wetland habitat due to the Project. A Habitat Creation and Management Plan (HCMP) should be prepared by a qualified ecologist with at least seven years of relevant experience to form the basis of the proposed wetland compensation site and

submitted to relevant government departments (e.g. AFCD and EPD) for approval. The HCMP should cover habitat design, construction methods, and the monitoring protocol, with particular focus on the detailed design and implementation of the proposed wetland compensation site.

8.13.1.7 Upon establishment of the proposed wetland compensation site, monitoring by a qualified ecologist with at least seven years of relevant experience is recommended. Monthly monitoring should be conducted during the 12-month establishment period. Monitoring parameter should focus on the abundance of target species and habitat conditions (e.g. water depth, water quality, and the condition of the wetland vegetation). Management programmes (e.g. water control, structural maintenance, supplemental planting, pest control, repair of damage) should be carried out as necessary in accordance with the approved HCMP.

Other Minimisation Measures

8.13.1.8 As described in **Section 3** and **4**, EM&A programmes were recommended to ensure compliance with the potential air quality and noise impacts (e.g. potential dust emissions during the construction phase and potential construction noise exceedances). Monitoring requirements for construction dust emissions and construction noise are further set out in **Section 3** and **4**, respectively. Regular site environmental audits during the construction phase are also recommended to ensure proper implementation of mitigation measures and good site practices. Details of the EM&A programme are provided in a standalone EM&A Manual.

8.13.1.9 As described in **Section 5**, water quality monitoring and regular site inspections would be undertaken during the construction to ensure that the recommended mitigation measures for water quality are properly implemented. Details on water quality monitoring requirements are further set out in the Water Quality Impact Assessment (**Section 5**) and the standalone EM&A Manual.

8.14 Conclusion

8.14.1.1 The ecological impact assessment for the Project has been conducted in accordance with the EIA Study Brief No. ESB-373/2025, the criteria and guidelines of the EIAO-TM Annexes 8 and 16, as well as relevant EIAO Guidance Notes.

8.14.1.2 The Assessment Area includes areas within a 500 m distance from the San Tin section of the NMH alignment and associated works (i.e. the Project Boundary). Several recognised sites of conservation importance, including Priority Sites for Enhanced Conservation, Lam Tsuen Country Park (LTCP), Conservation Area (CA), Wetland Conservation Area (WCA), Wetland Buffer Area (WBA), and other ecologically sensitive resources such as the proposed Nam Sang Wai Wetland Conservation Park (NSW WCP), were identified within the Assessment Area. The Project Boundary encroaches into small areas of WCA, WBA, and Deep Bay Wetland outside Ramsar Site Priority Sites for Enhanced Conservation, but the encroached habitats are limited to man-made habitats, including developed area/wasteland, roadside plantation, and small section of modified watercourse, all of which are subject to regular disturbance (e.g. traffic and human activities along the existing highway). For CA, impacts may arise from the loss of woodland habitats, which will be compensated with a Woodland Compensation Plan.

8.14.1.3 A total of 15 habitat types were identified during the ecological survey within the 500 m Assessment Area, including woodland, mixed woodland, shrubland, grassland, plantation, village/orchard, developed area/wasteland, agricultural land, seasonally wet grassland, marsh/reed, pond, natural watercourse, semi-natural watercourse, modified watercourse, and mitigation wetland. These habitats are generally considered to have "low" or "low to moderate" ecological values, except for woodland, ponds within WCA and WBA, YLBFEW, and Kam Tin Main Drainage Channel (i.e. YL-S1), which are considered as "moderate" ecological value, as well as Sha Po Marsh (outside of Project Boundary), which is considered to have "moderate to high" value.

8.14.1.4 Direct impacts arising from the Project include direct loss of habitats and vegetation due to the proposed alignment and temporary works. Avoidance and minimisation measures (e.g. refinement of alignment and establishment of "no work zone" during construction activities) have been considered to reduce and minimise the direct and indirect impacts to these habitats. Woodland and wetland compensation measures have also been recommended to mitigate the inevitable loss of ecologically sensitive habitats. Direct impact may also potentially arise from the potential direct injury or mortality of species of conservation importance within the Project Boundary, particularly species with low mobility, including the flora species Cycad-fern, Hong Kong Pavetta and Incense Tree, as well as the firefly species Motschulsky's Starworm and the freshwater fish species Small Snakehead. Pre-construction site checks with associated transplantation and translocation exercises are recommended.

8.14.1.5 On the other hand, indirect impacts such as disturbance impact to adjacent recognised sites of conservation importance and ecologically sensitive resources (such as Sha Po Night Roost and a planned bat shelter), as well as night-time disturbance and potential impacts on water quality, water holding, and water flow in the ponds and watercourses, may be induced by the construction and operation of the Project. Measures such as providing screening, using directional lighting, following good site practice, and implementing other noise, air, and water quality mitigation measures are recommended. Moreover, specific measures for the Sha Po Night Roost are also recommended, including a 100 m Buffer Area for the night roost, where noisy works would be subject to timing control. The use of non-transparent or non-glaring materials is also recommended to avoid and minimise potential direct impacts to the habitats and associated wildlife within and adjacent to the Project Site.

8.14.1.6 With the implementation of the aforementioned mitigation measures, no adverse ecological impacts are anticipated to arise from the Project.

8.15 References

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